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MILITARY STANDARDIZATION HANDBOOK

MAGNESIUM AND MAGNESIUM ALLOYS

95GP

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DEPARTMENT OF DEFENSE

WASHINGTON, D. C.

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Magnesium and Magnesium Alloys
14 April 1972

1. This standardization handbook was developed by the Department of Defense in accordance with established procedures.

2. This publication was approved on 14 April 1972 for printing and inclusion in the military standardization handbook series.

3. This handbook will provide instruction and assistance to designers and engineers concerned with the design of magnesium components for military material. The handbook is not intended to be referenced in purchase specifications except for informational purposes, nor shall it supersede any specification requirements.

4. Every effort has been made to reflect the latest information on magnesium and magnesium alloys. It is the intent to review this handbook periodically to insure its completeness and currency. Users of this document are encouraged to report any errors discovered and any recommendations for changes or inclusions to The Director, Army Materials and Mechanics Research Center, Watertown, Mass. 02172, ATTN: AMXMR-MS.

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Preface

This is one of a group of pamphlets covering the metallic and nonmetallic materials used in the design and construction of military equipment, which as a group constitute the Materials Handbook.

The purpose of the Materials Handbook is to provide, in condensed form, technical information and data of direct usefulness to engineers and designers of military equipment.

The Materials Handbook summarizes the general characteristics of magnesium and magnesium alloys and provides information and data on magnesium alloy properties and characteristics, to assist in selecting an alloy for a given job.

An Appendix provides the specifications for the mechanical properties and chemical composition of each of the standard alloys.

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Magnesium and Magnesium Alloys

GENERAL DISCUSSION

1. General Characteristics. Magnesium, with a specific gravity of only 1.74, is the world's lightest structural metal. Aluminum weighs 1-1/2 times more than magnesium; iron and steel 4 times more; and copper and nickel alloy 5 times more.

Among the physical properties of magnesium, its low density is probably the most outstanding. Due in part to this low density, magnesium has a high strength-to-weight ratio. Increased rigidity is an added benefit when used on an equal weight basis.

Magnesium is a versatile metal and can be shaped and worked by practically all methods known to the metal workers. It can be sand, die and permanent mold cast; extruded into a variety of shapes; and rolled into sheet, plate and strip. Magnesium can be forged and formed into useful shapes by drawing, bending, spinning, impact extrusion and other common methods. Magnesium is considered the easiest of all metals to machine. Machining rates of over 10,000 ft per minute are possible in certain applications. It can be joined by gas, arc and electric resistance welding, as well as by riveting, adhesive bonding, soldering, and other methods.

Magnesium alloys require protective coatings to protect the metal and to insulate contact surfaces to prevent galvanic corrosion. When coatings are properly applied and maintained, magnesium alloys are satisfactory in all variations of natural environments, other than continuous complete or partial immersion in water. There are three main groups of protective treatments for magnesium: (1) chemical treatment; (2) electroplating; (3) painting.

Magnesium resists attack by most alkalis as well as many organic chemicals. It is non-toxic.

Magnesium and its alloys are non-magnetic and possess relatively high thermal and electrical conductivities and good damping characteristics.

Mechanical properties given in table I indicate the minimum standards required in the design of military equipment.

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TABLE I. MECHANICAL PROPERTY RANGES OF MAGNESIUM ALLOYS

Property	Cast	Wrought
Tensile Strength Min, ksi	18 to 35	28 to 48
Yield Strength Min, ksi	10 to 22	14 to 38
Fatigue Strength Typ, ksi	9 to 14	11 to 22
Elongation Min, %	2 to 7	2 to 12
Modulus of Elasticity Typ, psi	6.5 million	

As with other common metals, subjecting magnesium alloys to elevated temperature causes a decrease in tensile strength, yield strength, and hardness, and an increase in elongation. The useful temperature limit of Mg-Al-Zn-Mn and Mg-Zn-Zr alloys is about 300°F(149°C), while Mg-RE alloys are used up to about 500°F (260°C) and Mg-Th alloys up to about 800°F (427°C). Magnesium alloys increase in strength and hardness at sub-zero temperature and decrease in elongation, but there is no sudden loss of ductility.

In common with other metals, this silvery-white element is not used in its pure state for stressed applications, but is alloyed with certain other metals including aluminum, zinc, zirconium, manganese, thorium and the rare earth metals to obtain the strong, light-weight alloys needed for structural uses.

The structural uses of magnesium are many and varied; i.e., transportation equipment, portable equipment, and parts subject to frequent and rapid changes in position.

The main factors limiting the use of magnesium are: the relatively high cost of the metal; the fact that it has to be worked hot; the extra care necessary to prevent fire; corrosion problems; and the necessity of using thick sections to overcome low rigidity.

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2. Economics of Production. By using magnesium rather than other materials that cost less, it is often possible to effect economies in production. The volume of metal used in a part will often be the same regardless of the type of material employed. The specific gravity of steel, for example, is 7.85. Therefore, when comparing the cost of using magnesium alloy to another metal, in this case steel, the price per pound of the magnesium alloy material should be divided by the ratio of the specific gravities involved (about 4.5 to 1).

In the case of castings, somewhat thicker sections may be required for the lighter alloys than for steel in order to compensate for the difference in strength. However, the thickness of sections is often determined by foundry limitations rather than by the mechanical requirements of design. So, the same thickness of section in a magnesium alloy, as in steel, may afford adequate strength for a particular purpose.

The economy arising in machining magnesium alloys may offset an unfavorable price relation with a material which costs less on a volume basis, but is difficult to machine. Also, substantial savings may result from the fact that a die casting can be made of a magnesium alloy and used in place of another material which is not suitable for the die-casting process.

CLASSES OF MAGNESIUM AND MAGNESIUM ALLOYS

3. Types Available. The metal is available in various compositions including the commercially pure metal (99.8 percent minimum), alloys for casting, and alloys for the manufacture of wrought products. Ordinarily, the compositions used for casting are different from those employed for rolling, forging, and other working. Substantially pure magnesium finds practically no use in engineering design or for stressed parts.

Tables 2, 3, and 4 list the magnesium alloys and their applicable specifications.

4. Pure Magnesium. Commercially pure metal (99.8 percent minimum) is rarely used in engineering design or for stressed parts.

5. Casting Alloys. There are four groups of magnesium casting alloys. The first group is based on the Mg-Al binary system and is subdivided into those alloys containing zinc and manganese. The second group is based primarily on the Mg-Zn binary system with zirconium additions. These two groups are valuable for their structural behavior at normal or slightly elevated temperatures of less than 300°F (149°C). The third group is based on the magnesium-rare earth metal binary alloys, and the fourth on the Mg-Th binary system. The last two groups have been specifically designed for their strength at elevated temperatures of 500°F (260°C) to 800°F (427°C). Although some of these casting alloys have outstanding normal-temperature properties they are not competitive with alloys of the first two groups because of the higher cost of the alloying elements.

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To improve the mechanical properties or stabilize dimensions of magnesium alloy sand and permanent mold castings, heat treatment is used; that is, aging, solution heat treatment, and solution treatment followed by aging. (See paragraph 16.)

6. Wrought Alloys. There are many types of wrought products; i.e., rolled sheet or plate, extrusion and forgings. These worked forms have the advantage of lower cost, higher strengths and ductilities, and have greater versatility of mechanical properties than cast forms. Magnesium is well adapted to all three working processes, especially extrusion.

7. Designating System. The ASTM nomenclature system (ASTM B-275) is used exclusively in designating magnesium alloys. In this system the first two letters indicate the principal alloy elements while the numbers indicate the respective percentage. The letters used symbolize the following:

A - aluminum
E - rare earth metals (elements 57-60 or combinations)
H - thorium
K - zirconium
L - lithium
M - manganese
Q - silver
T - tin
Z - zinc

Suffix letters represent variations in composition. The use of X denotes that the alloy is experimental and is not yet adopted by ASTM.

CHOICE OF ALLOY

8. Selection of Magnesium Alloys. When selecting magnesium alloys for a particular application, the designer must consider the intended use, service requirements, mechanical and physical properties of the alloys, form in which manufactured, and the proposed method of production of the part or assembly in question.

After determining mechanical and physical requirements, the choice of an alloy should be made on the basis of which alloy will fulfill both the other technical and economic requirements. In this connection, castability, formability, corrosion resistance, adaptability to joining methods, and necessity for heat treatment as well as cost, should be considered.

Characteristics of magnesium alloys are listed in tables 5 through 8.

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9. Casting Alloys. The choice of a casting composition is dictated largely by certain features of the design and by cost and method of production. For magnesium alloys, the important casting processes are sand, permanent and semi-permanent mold, and die casting. The choice of a casting process depends upon the size and shape of the part, minimum section thickness, tolerances, and types of surface finish, as well as the number of pieces to be produced and the relative cost of finishing parts. (See table 9.)

10. Wrought Alloys. The selection of a wrought magnesium alloy is influenced by the particular application and its requirements, type of wrought products, its mechanical properties and fabricating characteristics.

In specifications used by the Government, the following wrought forms in magnesium alloys are covered: plate and sheet, extrusions (bar, rod, shapes and tube), and forgings.

Magnesium alloy sheet finds numerous applications. It may be readily drawn hot into various shapes. Extrusions have many uses, especially as components in structural assemblies. The extrusion process is unusually adaptable to the production of standard and special shapes in magnesium alloys. In general, forgings are indicated for some applications where greater strength is required than can be obtained with castings, or where the forging process is especially suitable for manufacturing the parts.

FORMS OF ALLOY

11. Forms available. Magnesium alloys are available in all the usual metal forms including: ingots and billets; sand, permanent mold, and die castings; forgings; extruded bars, rods, shapes, and tube; and rolled sheet, plate, and strip.

PHYSICAL PROPERTIES

12. Available Physical Properties. The physical properties of magnesium alloys are listed in tables 10 and 11. These properties provide magnesium and its alloys with many distinctive qualities that have a practical interpretation in many of the uses for which the metal is employed.

The coefficient of thermal expansion of magnesium alloys is approximately 0.0000145/°F in the temperature range from 70°F (21°C) to 212°F (100°C). In applications where magnesium is rigidly connected to other metals and is subject to appreciable variation in temperature, consideration must be given to this relatively high coefficient.

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The dimensional stability of magnesium alloys in service is normally satisfactory. However, some of the magnesium-aluminum alloys are subject to small amounts of permanent growth if exposed for long periods of time to temperature exceeding 200°F (93°C). The growth depends upon the composition and condition of the alloy, and upon the time and temperature of exposure. Growth tends to be greatest in the -T4 temper. It may be mostly eliminated by a stabilizing heat treatment.

MECHANICAL PROPERTIES

13. Property Range. A wide range of mechanical properties are specified for magnesium-alloy products. These properties vary with the composition, condition (cast or wrought), details of fabrication, heat treatment, and other factors. Mechanical properties frequently considered in engineering design include values given in the tension test, in addition to shear strength, compressive yield strength, hardness, impact resistance, and fatigue strength. Values for various mechanical properties are given in tables 12 through 21. Unless otherwise stated, the tensile and compressive yield strengths correspond to 0.2 percent offset. The elongation refers to a gage length of two inches. Brinell hardness numbers are for a 500 kg load and a 100 mm diameter ball. The fatigue strength is based on 100 million cycles of completely reversed stress using the R.R. Moore type of machine and specimen.

Modulus of Elasticity - The accepted value is 6.5 million psi. The modulus of elasticity is affected by elevated temperatures as well as by form and composition of alloys. The decrease in modulus of elasticity with increase in temperature, a characteristic of the Mg-Al-Zn alloys, is considerably less for alloys containing thorium.

Modulus of Rigidity or Shear - The accepted value is 2.4 million psi.

Poisson's Ratio - The value for Poisson's ratio for magnesium alloys is taken as 0.35.

Tensile Properties - Magnesium, in common with other nonferrous metals but in contrast to steel, exhibits no definite yield point. Instead, the metal tends to yield gradually when stressed above the proportional limit. In accordance with ASTM procedure, the yield strength of magnesium alloys is defined as the stress at which the stress-strain curve deviates 0.2 percent from the modulus line.

Compressive Yield Strength - The stress necessary to produce a strain offset of 0.2 percent from the initial elastic modulus line. In the case of magnesium alloys in cast form, the compressive yield strength is essentially the same as tensile strength. The compressive yield strengths of most wrought magnesium alloys, however, will average 60 to 70 percent of the tensile yield strengths.

Shear Strength - Important in the design of connections in magnesium, such as threaded joints and spot welds.

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Bearing Strength - The stress required to produce an offset of 2 percent of the hole diameter from the initial straight portion of the curve. It is particularly important in the design of bolted and riveted joints.

Low Temperature - In general, the yield strength, ultimate strength, and hardness of magnesium alloys increases with decreases in temperature below zero, while the elongation and impact resistance decreases. Fatigue strength is raised appreciably as temperature is lowered, and the modulus of elasticity increases. The individual effect at various sub-zero temperatures vary depending upon the alloy composition, and temper. (See table 22.) Cooling to low temperatures followed by raising the alloys to ordinary temperature generally has no adverse effect on the mechanical properties.

Elevated Temperature - Elevated temperatures have an adverse effect on tensile and yield strengths. There are several considerations when dealing with elevated temperature exposure: (1) the strength as determined by bringing the test specimen up to temperature and testing immediately (short time test); (2) the strength at temperature after prolonged heating at elevated temperature; (3) the effect of heating at elevated temperature for short and long times on the room temperature properties; (4) the deformation produced by prolonged heating under load.

Magnesium alloys melt at relatively low temperature at 1290°F (699°C) and below, and are restricted to use at maximum temperatures from about 600°F (316°C) to 900°F (482°C), depending upon the composition. The upper limit for good strength ranges from about 200°F (93°C) to 800°F (427°C). See tables 23 through 46 for effect of elevated temperature on the mechanical properties of magnesium alloys.

Creep - A measure of strength at elevated temperatures. The values depend upon the composition, temper, grain size, and other factors. Alloys that contain thorium have the greatest resistance to creep at 400°F (204°C) and 600°F (316°C), and the Mg-Al-Zn alloys have the lowest resistance.

Fatigue - The good fatigue properties of magnesium are an asset when the metal is designed into moving parts, and particularly those with a reciprocating motion. High cycles at low stress, show magnesium at its best. The fatigue strength of magnesium can be improved by surface working treatments that induce residual compression. The presence of sharp notches, small radii, fretting or corrosion is more likely to reduce the fatigue life than any variations in chemical composition or heat treatment.

Notch Sensitivity - Broadly defined as the reaction of a material to stress concentrations. Service failures usually occur at stress concentrations associated with general design, joining methods and surface smoothness. In highly stressed areas of magnesium castings, the stress concentration associated with design should be below 1.3. Severe stress concentration may exist near screw threads, bolts, and

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stud holes when service loads are applied. In such areas, where localized stresses may be as high as five times the nominal stress in surrounding materials, the nominal stress is kept low by using thicker sections and bosses. In wrought metal structures, where high stress concentrations may exist at joints, local reinforcements may reduce the nominal stress.

Hardness and Wear Resistance - Magnesium alloys have sufficient hardness for all structural uses, except where subjected to severe abrasion. Although rather wide variations are observed, the resistance of the alloys to abrasion varies by only about 15 to 20 percent. When subjected to wear by rubbing, frequent removal of studs, or heavy bearing loads, magnesium may be protected by inserts of steel, bronze, or nonmetallic materials, attached as sleeves, liners, plates or bushings. Such inserts may be attached mechanically by pressing, shrinking, riveting, bolting or bonding. In castings, inserts may be cast in place.

Magnesium alloys perform satisfactorily as a bearing material where the following conditions are met: (1) loads not in excess of 2000 psi; (2) hardened shafts (350 to 600 Bhn); (3) ample lubrication; (4) low speeds (1000 ft per min max.); (5) operating temperatures not in excess of 200°F (93°C).

Dent Resistance - The dent resistance of magnesium is highly dependent upon conditions of tests. See figures 1 through 4.

14. Casting Alloys. Typical tensile properties for sand and permanent-mold cast alloys, as ordinarily reported, are obtained in tests on standard 1/2-inch diameter test specimens separately cast under standard conditions of solidification. Specimens so cast serve as a control of metal quality, but their properties do not necessarily represent those of commercial castings. The properties obtained may be higher or lower, depending on various factors that influence the rate of solidification in the mold. Likewise, the properties of specimens cut from various locations on a single casting may differ substantially depending on the place from which they are removed.

Specifications may require that the average ultimate tensile strength and average elongation of specimens cut from castings shall not be less than 75 percent and 25 percent, respectively, of the values set for separately cast test bars. Not fewer than three tension-test specimens are machined from a casting, one each from the thinnest, thickest, and an intermediate section.

For small grain size in both bars and castings, the strength values of sections up to 1-1/2 inches thick may be about 90 percent of the minimum test-bar values.

Tensile properties of die castings are based on values obtained from 1/4-inch test specimens, produced in a cold chamber (high pressure) die-casting machine.

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15. Wrought Alloys. The form may have appreciable effect on the mechanical properties of wrought magnesium alloys. Also, the thickness of the section and the direction in which the test specimen is taken in relation to the direction of metal flow during working, are factors that affect strength and other properties.

The tensile properties of the wrought materials are ordinarily based on test data obtained on specimens cut from the actual products (extrusions, forgings, and sheet). When the form and dimensions of the product permit, the standard 1/2-inch diameter test specimen is used. In other cases either a sub-sized round specimen proportioned to the standard specimen, or a standard sheet type specimen is employed. Some products including tube and small bars, may be tested full size. The types of test specimens specified by the Government are illustrated in Fed. Test Method Std. No. 151.

In forgings, as in extrusions, the tensile properties are ordinarily determined on specimens taken in the direction of maximum metal flow. These properties are normally higher than properties determined in other directions of the products.

The properties developed during testing of wrought and cast alloys are represented in tables 47 through 61.

HEAT TREATMENT

16. General. Both cast and wrought magnesium alloys of suitable composition can be heat treated to alter their mechanical properties. The type of treatment used depends upon the nature and extent of the change desired. (See tables 62 and 63.) It is possible to substantially raise the tensile strength and elongation when using one type of treatment, while the use of another treatment may increase the yield strength and hardness of the metal resulting in a marked loss of ductility.

17. Types. Three basic types of thermal treating processes are ordinarily applied to magnesium alloys; solution heat treatment, precipitation (age-hardening), and annealing.

In solution heat treatment, the alloy is treated to a temperature at which certain constituents go into solution, followed by quenching to hold these constituents in solution. Precipitation is carried out by heating the material to moderately elevated temperatures. Annealing consists of heating the alloys at a suitable temperature to effect re-crystallization (softening), to stabilize dimensions, or to remove internal stresses.

A precipitation or aging treatment may be applied to material that has not been given the usual solution heat treatment. Thus, extrusions or forgings in the as-fabricated conditions may be aged to increase the tensile yield and ultimate strength. In their manufacture, these products are heated to temperatures at which substantial amounts of age-hardening constituents are taken into solution. When coming in contact with forming dies or when cooled in air, these products are "more or less" quenched.

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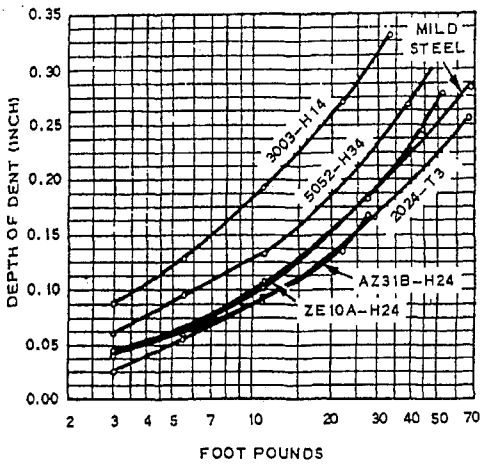


FIGURE 1. Dent Resistance. Comparison of Magnesium and Aluminum Alloys with Mild Steel. All 0.040 inch thickness.

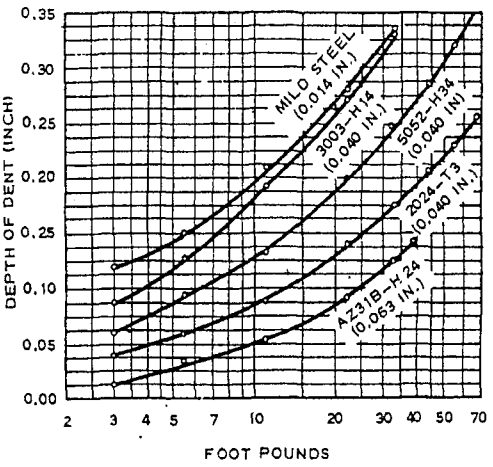


FIGURE 2. Dent Resistance. Equal Weight Comparison of Magnesium and Aluminum Alloys with Mild Steel.

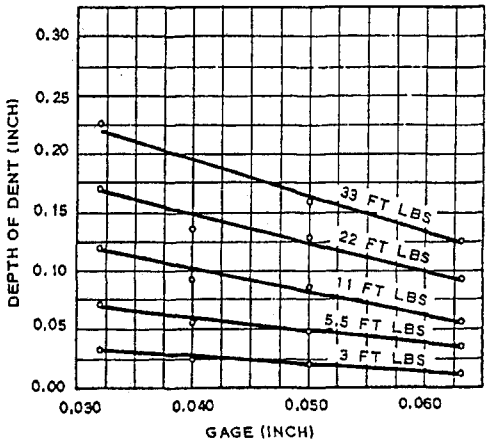


FIGURE 3. Depth of Dent Versus Gage Shown at Various Impact Loads for AZ31B-H24 Magnesium Sheet

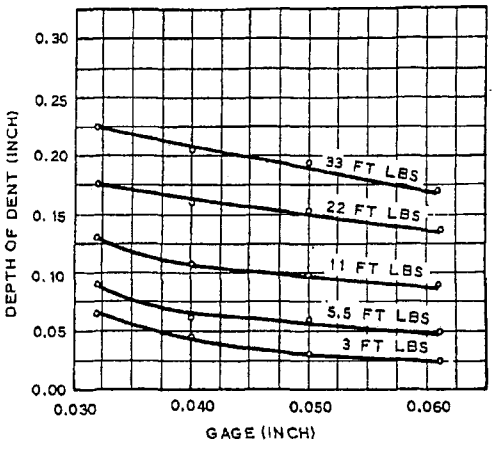


FIGURE 4. Depth of Dent Versus Gage Shown at Various Impact Loads for ZE10A-H24 Magnesium Sheet

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Most magnesium alloys are quenched in still or moving air. Liquid quenching agents are not ordinarily used.

The strain-hardening which results from cold working magnesium alloys may be removed by annealing, that is, by heating to cause recrystallization. For most cold-worked alloys, the annealing temperature is about 650°F (343°C).

Although aging usually gives sufficient freedom from growth of MG-Al castings at elevated temperatures, a stabilization treatment is sometimes used instead. The treatment is normally applied to castings in the -F or -T4 temper. Stabilization treatment has very little effect on the original mechanical properties of the -F temper, but following solution heat treatment it produces properties in the range of the -T6 temper. Stabilization of as-case -F material yields the -T5 temper, and stabilization of the -T4 material yields the -T7 temper.

Magnesium alloys tend to age-harden very slowly at ordinary temperature. Alloys in the -T4 temper are susceptible to room-temperature aging. For example, the aging of cast AZ63A-T4 (QQ-M-56a) for five years at 75°F (24°C) to 90°F (32°C) caused an increase in tensile strength of about 14 percent, in yield strength of 68 percent, and in Brinell hardness of 22 percent. In the same period, the elongation was reduced about 32 percent and the impact resistance 57 percent. The aging of case AZ63A-F alloy for five years at room temperatures increased the yield strength about 20 percent and decreased the elongation 33 percent without materially affecting the other properties. AZ92A-T4 is somewhat more susceptible to room temperature aging than is AZ63A-T4. All heat-treatable alloys in the -T5, -T6, and -T7 tempers are insensitive to room-temperature aging.

18. Temper Designation. The designations for temper are based on the sequence of basic treatments used to produce the temper. The temper designation, which is used for all metal forms except ingot, follows the alloy designation and is separated therefrom by a dash.

Basic temper designations consist of letters. Subdivisons of the basic tempers, where required, are indicated by one or more digits following the letter. These digits designate a specific sequence of basic treatments, but only those operations which are recognized as significantly influencing the characteristics of the product are indicated. Should some other variation of the same sequence of basic operations be applied to the same alloy, resulting in different characteristics, then additional digits are added to the designation.

Temper designations and their subdivisions are defined as follows:

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F As Fabricated — Applies to products that acquire some temper from shaping processes not having special control over the amount of strain-hardening or thermal treatment.

O Annealed, Recrystallized (Wrought Products Only) — Applies to the softest temper of wrought products.

H Strain-Hardened (Wrought Products Only) — Applies to products that have their strength increased by strain-hardening with or without supplementary thermal treatments to produce partial softening. The H is always followed by two or more digits.

Subdivisions of the "H" Temper:

H1 Strain Hardened Only — Applies to products that are strain-hardened to obtain the desired mechanical properties without supplementary thermal treatment. The number following this designation indicates the final degree of strain-hardening.

H2 Strain Hardened and Then Partially Annealed — Applies to products that are strain hardened more than the desired final amount and then reduced in strength tothe desired final amount by partial annealing.

The number following this designation indicates the final degree of strain-hardening remaining after the product has been partially annealed.

H3 Strain Hardened and Then Stabilized — Applies to products that are strain hardened and then stabilized by a low temperature heating to slightly lower their strength and increase ductility. This designation applies only to alloys which, unless stabilized, gradually age soften at room temperature.

The number following this designation indicates the degree of strain-hardening remaining after the product has been strain-hardened a specific amount and then stabilized.

Subdivisions of the "H1", "H2", and "H3" Tempers:

The digit following the designations "H1", "H2", and "H3" indicates the final degree of strain-hardening. Tempers between 0 (annealed) and 8 (full hard) are designated by numerals 1 through 7. Material having a strength about midway between that of the 0 temper and that of the 8 temper is designated by the numeral 4 (half hard): between 0 and 4 by the numeral 2 (quarter hard): between 4 and 8 by the numeral 6 (three-quarter hard): etc.

The third digit, when used, indicates a variation of a two-digit H temper. It is used when the degree of control of temper or the mechanical properties are different from but close to those for the two-digit H temper to which it is added.

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Numerals 1 through 9 may be arbitrarily assigned for an alloy and product to indicate a specific degree of control of temper or specified mechanical property limits.

W Solution Heat-Treated — An unstable temper applicable only to alloys which spontaneously age at room temperature after solution heat-treatment. This designation is specific only when the period of natural aging is indicated: for example, W1/2 hour.

T Thermally Treated to Produce Stable Tempers Other than F, O or H — Applies to products that are thermally treated, with or without supplementary strain-hardening, to produce stable tempers. The T is always followed by one or more digits. Numerals 1 through 10 have been assigned to indicate specific sequence of basic treatments as follows:

Subdivisions of the "T" Tempers:

T1 Cooled From an Elevated Temperature Shaping Process and Naturally Aged to a Substantially Stable Condition — Applies to products for which the rate of cooling from an elevated temperature shaping process, such as casting, or extrusion, is such that their strength is increased by room temperature aging.

T2 Annealed (Cast Products Only) — Applies to a type of annealing treatment used to improve ductility and increase stability of castings.

T3 Solution Heat Treated and Then Cold Worked — Applies to products that are cold worked to improve strength, or in which the effect of cold work in flattening and straightening is recognized in applicable mechanical properties.

T4 Solution Heat Treated and Naturally Aged to a Substantially Stable Condition — Applies to products that are not cold worked after solution heat treatment, or in which the effect of cold work in flattening or straightening may not be recognized in applicable mechanical properties.

T5 Cooled From an Elevated-Temperature Shaping Process and Then Artificially Aged — Applies to products that are cooled from an elevated-temperature shaping process, such as casting or extrusion, and then artificially aged to improve mechanical properties or dimensional stability or both.

T6 Solution Heat Treated and Then Artificially Aged — Applies to products that are not cold worked after solution heat treatment, or in which the effect of cold work in flattening or straightening may not be recognized in applicable mechanical properties.

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T7 Solution Heat Treated and Then Stabilized — Applies to products that are stabilized to carry them beyond the point of maximum strength to provide control of some special characteristics.

T8 Solution Heat Treated, Cold Worked, and Then Artifically Aged — Applies to products that are cold worked to improve strength, or in which the effect of cold work in flattening or straightening is recognized in applicable mechanical properties.

T9 Solution Heat Treated, Artificially Aged, and Then Cold Worked — Applies to products that are cold worked to improve strength.

T10 Cooled From an Elevated Temperature Shaping Process, Artificially Aged, and Then Cold Worked — Applies to products that are artificially aged after cooling from an elevated temperature shaping process, such as casting or extrusion, and then cool worked to further improve strength.

A period of natural aging at room temperature may occur between or after the operations listed for tempers T3 through T10. Control of this period is exercised when it is metallurgically important.

Additional digits may be added to designations T1 through T10 to indicate a variation in treatment that significantly alters the characteristics of the product.

For codification, an alloying element is defined as an element (other than the base metal) having a minimum content greater than zero either directly specified or computed in accordance with the percentages specified for other elements. The amount present is the mean of the range (or the minimum percentage if only that is specified) before rounding off.

The serial letter is arbitrarily assigned in alphabetical sequence starting with A (omitting I and O) and serves to differentiate otherwise identical designations. A serial letter is necessary to complete each designation.

The designation of a casting alloy in ingot form is derived from the composition specified for the corresponding alloy in the form of castings. Thus, a casting ingot designation may consist of an alloy designation having one or more serial letters, one for each product composition, or it may consist of one or more alloy designations.

FORMABILITY

19. Forming. The methods and equipment used for forming magnesium are the same types as commonly employed on other metals (i.e., drawing, pressing, stamping, bending, spinning, dimpling, swaging, forging; stretch, die and rubber forming) with differences in tools and techniques resulting from the fact that magnesium is best formed at elevated temperatures of 350°F (177°C) to 800°F (427°C). Magnesium alloys, in general, have poor cold formability and workharden rapidly at room temperature.

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Formability is affected by alloy composition, temper, type of product (sheet or extrusion), and the kind of fabricating operations used. Most forming operations on sheet are done on hot stock with heated tools, especially those operations requiring deep draws and bends of small radius.

Heating to the temperature required for hot forming (sheet alloys) has almost no adverse effect on the properties of annealed sheet at room temperatures. Depending upon conditions, substantial strengthening of the alloy may result when it is formed. In general, annealed sheet is used where maximum formability is needed. Hard-rolled sheet can also be formed at elevated temperatures, but it cannot be heated to temperatures for maximum drawability without losing the hard-rolled properties. Hard-rolled sheet is used most often for applications requiring maximum strength and on which only a moderate amount of shaping is necessary, since the forming temperature must be limited to prevent excessive loss of mechanical properties.

Data on the forming characteristics of sheet alloys in the soft (—O) and hard (—H2) tempers are given in table 64. These data apply to forming at room temperature and at two elevated temperatures.

Data on the forming characteristics of forging alloys are given in table 65.

20. Bend Radii. It is necessary that a suitable radius be provided for bending, drawing, and other shaping. Proper radius depends upon the bend to be made, the composition and temper of the alloy, and the thickness of the product. Bendability is the main consideration in the selection of material for most sheet-metal work.

The formability of most magnesium alloys at room temperature is limited to bends over large radii and to shallow draws. Heating in the range of 300°F (149°C) to 800°F (427°C) permits severe bending and forming, including single-step draws to a depth of nearly twice the diameter.

Of the extrusion alloys specified in QQ-M-31, AZ31B and ZK60A have the best bending qualities, while AZ80A permits bending only to a limited degree. The mechanical properties of AZ80A-T5 are impaired if the material is heated above 380°F (193°C). Whether heating may be required for bending an extruded shape depends upon alloy composition, temper, cross-section, bending diameter, rate of bending, and bending equipment. (See tables 66 and 67.)

MACHINABILITY

21. Degree of Resistance. The degree of resistance encountered in cutting is usually referred to as "machinability".

22. Machining Characteristics. Magnesium and its alloys have excellent machining characteristics. They can be machined at extremely high speeds using greater depth of cut and higher rate of feed, than can be used in machining other structural metals. For equal volumes of metal removed, the horsepower consumption is lower than with any

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other metal ranging from 0.15 to 0.3 h.p. per cubic inch of metal Magnesium does not drag or tear, and takes a smooth finish. It is not necessary to grind and polish magnesium to obtain an extremely fine finish. Chips machined from magnesium materials readily clear the work and the tool.

In many cases, high-speed steel tools stand up on magnesium as well as carbide tools do on other metals. This results in great savings in tool costs. Because magnesium parts are light, there is less wear on machining equipment, and therefore less maintenance is necessary. In addition, magnesium parts are quickly and easily handled by the operator.

There is little difference in the machinability of various magnesium alloys. All are free cutting; however, M1A alloy does not cut quite as freely as other specified compositions. Magnesium-alloy extrusions make good screw-machine stock. AZ31B alloy, specified in QQ-M-31 is a good general-purpose material for screw machine work.

The relative machinability of magnesium-alloy castings as compared to other cast metals is illustrated by the following approximate ratings:

Cast magnesium alloys	100
Cast aluminum alloys	55
Cast brass	45
Cast iron	30
Cast steel	20

An arbitrary rating is used where the number 100 designates the best machinability. Comparison is based on cutting speed, feed, and power consumption. The ratings represent results of laboratory tests and shop experience. Similar ratings apply to the materials in wrought form.

23. Coolants. Magnesium is usually machined dry at speeds up to 500-700 surface feet per minute. When fine cuts are taken at higher speeds, or in grooving and cut-off operations, fluids (coolants) are used to cool the work and reduce the possibility of distortion and ignition of fine chips. Less heat is generated when machining magnesium than when machining other metals, but higher cutting speeds plus magnesium's high thermal expansion characteristics make it necessary to dissipate the heat developed during machining operations. Correct tooling and machining techniques reduce the amount of heat generated. Coolants should always be mineral oils; animal or vegetable oils are NOT recommended. Water-soluble oils, oil water emulsions, or water solutions of any kind should NOT be used on magnesium. Water reduces the scrap value of magnesium turnings and introduces a potential fire hazard during shipment or storage of machine shop scrap.

24. Safety Practice. Fire may be a safety hazard in maching magnesium alloys. Magnesium alloys must be heated too close to the melting point before ignition can occur. Roughing cuts and medium finishing cuts produce chips of a size not readily ignited during

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machining. Fine finishing cuts, however, produce fine chips which are EASILY ignited. Stopping the feed and letting the tool dwell before disengagement, or letting the tool or tool holder rub on the work, produce extremely fine chips and friction heat. These practices should be avoided.

Factors which tend to increase the chances of chip ignition are: (1) high cutting speeds; (2) extremely fine feeds; (3) dull or chipped tools; (4) improper machining techniques; (5) tool hitting an iron or steel insert. Feeds usually must be less than 0.010 inch per revolution, and cutting speeds must be high before a fire hazard exits. Even under the most adverse conditions, i.e. with dull tools and fine feeds, the chances of chip fires are very slight at speeds below about 700 ft per minute.

Following the rules given below will help reduce the fire hazard:

a. Keep all cutting tools sharp and ground with adequate relieve and clearance angles.

b. Use heavy feeds to produce thick chips.

c. Use mineral oil coolants (4 to 5 gal per min) whenever possible; when not possible, avoid fine cuts and keep speed below 500 ft per min.

d. Remove dust and chips at frequent intervals and store in clean, plainly labeled, covered metal cans. Do not allow chips to accumulate on machines or clothing of operators.

e. Keep an adequate supply of recommended magnesium fire extinguishers within reach of the operators.

If chips should become ignited, extinguish them as follows:

a. Scatter a generous layer of either finely powdered graphite, clean dry cast iron chips, dry sand or powdered talc.

b. Actively burning fires on combustible surfaces like wood floors should be covered with a layer of the powdered graphite; then the entire mass should be shoveled into an iron container or onto a piece of iron plate.

c. Water or any of the common liquid or foamtype extinguishers may intensify a magnesium chip fire and are not generally recommended.

JOINING METHODS

25. Methods. A number of methods may be used to join magnesium alloys. These include mechanical joining, welding, brazing, soldering, and adhesive bonding. The method to be employed depends upon the design, the material to be joined, strength requirements, service conditions, and other factors.

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26. Mechanical Joining. Methods for the mechanical joining of magnesium alloys include riveting and the use of bolts, screws, and similar fasteners.

Riveting — Riveting is probably the most common method used for joining magnesium. Essentially the same procedures employed in riveting other metals are used on magnesium. Standard procedures are used for drilling and countersinking holes. Both dimpling and machine countersinking are used in flush riveting magnesium. Only aluminum alloy 5056 rivets should be used if galvanic incompatibility is to be minimized. Cone-shaped heads are advised for heavy thicknesses and flat bucktail heads for light work. The average ultimate shear strength for 5056 rivets driven cold is 27,000 psi.

Bolts, Screws and Studs — Bolts, screws and similar fasteners are used to join magnesium to dissimilar metals and woods, and are essentially the same types used for other materials. Washers should always be placed under nuts, and they should be used under bolt and stud heads when these parts are to be subjected to tensile loading. Bolts and studs should be made of anodized aluminum alloys and zinc-plated or cadmium-plated steel. Washers should be made of 5052 aluminum alloy and zinc-plated or cadmium-plated steel.

27. Welding. Generally, magnesium alloys cannot be satisfactorily welded to other metals. Ratings of the weldability of various magnesium alloys are given in table 68. Alloy AZ60A has relatively limited weldability. Methods for the welding of magnesium alloys include gas welding, shield-arc welding, electric-resistance welding, spot welding, seam welding, flash welding, and pressure welding.

Gas Welding — Magnesium alloys may be gas welded with oxyacetylene or other mixtures. A suitable flux is required. Gas welding is usually restricted to simple butt and corner joints because the flux is likely to be entrapped in other joints and cause corrosion. The scope of gas welding is relatively limited and has been replaced by shielded-arc welding.

Shielded-Arc Welding — The arc welding process, using either a tungsten or consumable electrode and a shield of inert gas, is broadly applicable to magnesium alloy products. In this process, the inert gas (argon or helium) protects the metal from exposure to air and eliminates the need for a flux; as a result, there is no restriction on the type of joint. Butt, lap, fillet, tee, and edge welds may be made with the shielded arc. Shielded-arc welding gives stronger joints than gas welding. The strength and joint efficiencies of shielded-arc are given in tables 69 and 70.

Electric-Resistance Welding — Good joints can be made in magnesium alloys by electric welding. Using this method, joints may be spot, seam, butt or flash welded.

Spot Welding — Magnesium-alloy sheet can be joined by spot-welding. With available equipment, the maximum thickness that can be spot welded is about 3/16 inch. Sections to be welded should be of approximately the same thickness. If this condition

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cannot be met, the heavier sheet should be more than 2-1/2 times as thick as the lighter one. Different magnesium alloys may be joined by spot welding, but the best results are secured when the same composition (in the same thickness) are welded together. Spot-welded joints are satisfactory for moderately stressed applications not subject to excessive vibration. When loaded in direct tension, spot welds should be subjected only to light stresses.

Seam Welding — Seam welds are made in magnesium in much the same manner used for spot welding. Continuous or intermittent type welds have strength properties comparable to those obtained in spot welds. In seam welding, breaking loads of 1075 pounds per linear inch of welded seam have been obtained in AZ31B alloy of 0.040-inch thickness. The breaking load increases to about 2250 pounds with a thickness of 0.128 inch.

Flash Welding — A limited amount of flash welding has been performed on magnesium alloys. Wrought magnesium alloys can be satisfactorily flash welded. Depending on the composition, size and shape of the welded section, tensile strength of about 26 to 43 ksi with 3 to 8 percent elongation can be obtained.

Pressure Welding — Hot pressure welding of magnesium alloys to each other and to certain aluminum alloys, has been performed successfully on a limited scale. In pressure welding (sometimes referred to as "forge welding"), surfaces to be welded are pressed together in special heated dies. The resultant metal flow forms a weld. Weld strengths in excess of 50 ksi with efficiencies over 98 percent and elongations of 10 to 12 percent have been reported in magnesium ZK60A-T5 alloy.

28. Brazing. Magnesium alloys AZ31B, ZE10A, and K1A can all be flux-dip brazed to themselves or to each other. Because of the rapid heating and relatively large volume of flux involved, much more consistent results and better flow of brazing metal are obtained with flux-dip brazing than with other brazing methods. No brazing process is currently available for the magnesium-thorium and magnesium-zinc-zirconium alloys.

29. Soldering. Soldering is generally used only for filling small surface defects in castings or small dents in sheets, on which a smooth paint job is desired. Soldering is not recommended if the soldered area is required to withstand stresses, and ordinarily is not considered for the joining of magnesium parts or for joining magnesium to other metals because of the low strengths and brittle joints obtained. Because of the marked difference in solution potentials between the solder and the magnesium, soldered joints are unsatisfactory in moist or corrosive atmospheres.

30. Adhesive Bonding. Fatigue characteristics of adhesive-bonded joints are improved over those of other types of joints. Because no drilling is required, the chances for stress concentration failures are minimized. Adhesive bonding permits the use of thinner materials that cannot be riveted effectively. The adhesive fills up the spaces between the contacting surfaces and acts as an insulator between any dissimilar metals in the joint. It also permits smoother surfaces.

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Design of adhesive-bonded joints has been limited almost exclusively to lap joints. The following are a few of the general design factors that should be considered:

a. Joint strengths vary with lap width, metal thickness, direction in which loads are applied, as well as with the type of adhesive used.

b. The joint should be designed so that it provides a sufficiently large bonded area.

c. The adhesive layer should be uniform in thickness.

d. The adhesive layer should be as thin as possible, yet applied in sufficient quantity so that no joints are "starved".

e. Joints should be designed so that pressure and heat can be readily applied.

f. The curing temperatures of the common structural adhesives are below the temperatures where hard rolled magnesium sheet properties are affected, and do not produce a significant reduction in the properties of magnesium alloys in the annealed (—O) temper.

The strength of bonded joints ranges from 1000 to 4000 psi. Table 71 gives various characteristics of adhesives used for bonding magnesium.

CORROSION — ITS CAUSES AND PREVENTION

31. General. Magnesium, because of its position in the electromotive series is not generally regarded as a corrosion resistant metal. Its resistance to corrosion depends to a great degree upon the purity of the metal, the environment to which it is exposed, and its contact with dissimilar metals.

Pure magnesium withstands corrosion better than any of its alloys. The degree of resistance to corrosion of the various alloys differs; i.e., the magnesium-lithium alloys show a greater susceptibility than do the magnesium-thorium combinations.

32. Atmospheric Environments. The resistance of magnesium alloys to corrosion depends to a great degree upon the environment to which they are exposed. A gray film will form on the surface of unprotected magnesium alloys exposed to outdoor atmospheres free from salt spray; such corrosion has a negligible effect on the mechanical properties of the alloy. If chlorides or other foreign materials accumulate on the surface and hold moisture in contact with the magnesium, pitting and general corrosion will occur.

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Magnesium exhibits inherently different corrosion rates depending on the amount and type of alloying elements present, the location, and the humidity. In outdoor urban exposures, the alloys containing aluminum show a decreasing tendency to corrode as the aluminum content increases up to about 9 percent. Relative humidity, or atmospheric moisture, increases corrosion of magnesium alloys.

A film on the surface of magnesium alloys does not protect the metal against further attack. If the film is tight and adherent, however, it decreases the rate of further attack. Unprotected magnesium and magnesium alloys show good corrosion resistance in rural, industrial and mild marine atmospheres, provided they do not contain joints or recesses to entrap water, or are not in contact with dissimilar metals. In marine atmospheres heavily loaded with salt spray, magnesium alloys require protection for prolonged survival.

33. Chemical Environments. Aqueous solutions of oxidizing salts, especially some containing chlorine or sulfur atoms, are more corrosive than nonoxidizing salts; but chromates, vanadates, phosphates and many others form a film and tend to retard corrosion except at elevated temperatures. Magnesium alloys are attacked rapidly by all aqueous solutions of mineral acids except hydrofluoric and chromic acids. Aqueous solutions of organic acids having low molecular weight also attack magnesium alloys but organic acids having high molecular weight will not attack magnesium alloys but may stain them.

In nonaqueous corrosion, magnesium exhibits good resistance in a wide variety of liquid organic compounds. One exception is methanol. Dry chlorine, iodine, bromine, and fluorine also cause little or no attack on magnesium alloys at room temperature.

34. Surface Corrosion. Surface corrosion may occur: (a) as a result of galvanic action at magnesium-to-dissimilar metal contacts; (b) from surface contamination; (c) from the presence of moisture at faying surfaces of magnesium-to-magnesium or nonmetal contacts.

35. Surface Contamination. Certain methods of fabrication, such as sand blasting of castings, or mill scale formed during rolling of sheet, are likely to imbed particles of foreign material in the surface and cause it to corrode in the presence of moisture. The effects of surface contamination can be overcome by in-process pickling and by proper chemical treatment and painting procedures. (See specification MIL-M-3171.)

36. Stress-Corrosion Cracking. Magnesium alloys containing greater than about 2 percent aluminum are susceptible to stress-corrosion cracking. This occurs when the metal is subjected to excessive stress in the presence of a corrosive medium. Strains induced in magnesium-alloy parts by processes such as welding, bending, straightening, etc., and which might cause stress-corrosion cracking may be relieved by suitable heat treatment.

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37. Corrosion at High Temperature. Water vapor in air or in oxygen sharply increases the rate of oxidation at temperature above 212°F, but boron trifluoride and sulfur dioxide are effective in reducing the rate. The presence of boron trifluoride in the ambient atmosphere is particularly effective in suppressing high-temperature oxidation up to and including the temperature at which the alloy normally ignites.

The effect of temperature depends largely on the purity of the metal. Corrosion rate of controlled-purity magnesium and its alloys, immersed in 3 percent NaCl, approximately doubles with increases from room temperature to the boiling point. When cathodic impurities, such as iron and nickel, are present in amounts greater than their tolerance limit, corrosion and pitting increase as the temperature increases.

38. Corrosion Prevention. Magnesium alloys, like other readily corroded metals, require protective finishes of particular design if they are to provide satisfactory service. The type of finish for magnesium depends upon the field of application. Military needs cover a wide range of extremely severe corrosive conditions, and therefore complete protection is necessary. But even the best materials and procedures will not eliminate the problem unless corrosion has been "designed out" of the structure. The basic principles of "designing out" corrosion are briefly as follows:

a. Eliminate "sump" areas where trapped moisture is held in contact with metal:

(1) Design them out in the first place by careful attention to design of structural details.

(2) Provide properly located drain holes; minimum hole size should be 1/8 inch to prevent plugging.

b. Choose non-absorbent, non-wicking materials:

(1) Determine the water absorption qualities of materials to be used.

(2) Use epoxy, vinyl tapes, and coatings of wax and latex for protective barriers.

(3) Avoid, if possible, use of wood, paper, cardboard, open cell foams and sponge rubbers.

c. Protect all faying surfaces:

(1) Use proper sealing materials (tapes, films, sealing compounds) on all faying surfaces. Use primers.

(2) Lengthen the continuous liquid path to prevent forming an electrolytic cell.

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d. Use compatible materials:

(1) For Mg-Al couples, 5000 and 6000 series aluminum alloys are most compatible.

(2) For Mg-steel couples, use tin-plated or cadmium-plated steel.

e. Select proper finishing systems:

(1) Choose chemical treatments, paints or plating on basis of service needs.

(2) Service test the system before setting production run.

(3) Use past experience in similar uses as a guide to choice.

Three basic types of magnesium contacts are found in assembly work. They are illustrated in figures 5 through 11, and treated in the following general manner:

a. Magnesium-to-magnesium contacts (see figure 5) - Give all magnesium faying surfaces one or more coats of chromate pigmented primer.

b. Magnesium-to-dissimilar metals (see figures 6 through 10) - Use moisture impervious films to protect both magnesium and dissimilar metal; use dissimilar metals that are compatible with magnesium; separate metals so that corroding medium cannot complete an electrical circuit; protect the magnesium.

c. Magnesium in contact with wood, paper, etc. (see figure 11) - Seal these absorptive materials with paint, varnish, wax, latex, epoxy or similar coatings, then treat the magnesium faying surface as subparagraph a.

39. Finishing Systems. In nonmarine atmospheres, magnesium and its alloys characteristically form a thin gray film on the bare metal surface. In some atmospheres and uses, this thin film provides adequate protection and may even be satisfactory as a lacquer base. For severe service exposures (military), magnesium parts must be given a suitable finishing system. Such a system should be chosen not only for appearance or protection, but also for durability and the ease of obtaining these qualities. Cleaning, chemical treatment, painting and plating are all to be considered in selecting and applying magnesium finishing systems.

Cleaning and acid pickling methods used on magnesium are given in table 72.

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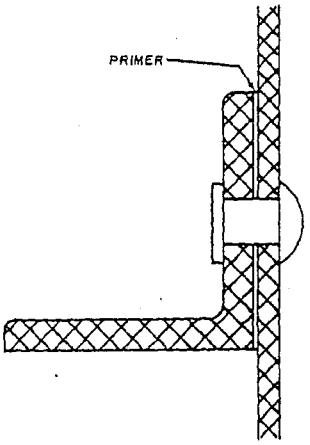


FIGURE 5. Magnesium Riveted to Magnesium

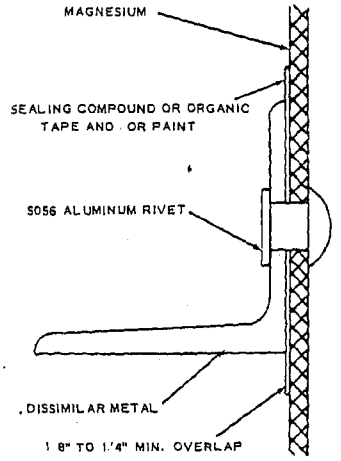


FIGURE 6. Magnesium Riveted to a Dissimilar Metal

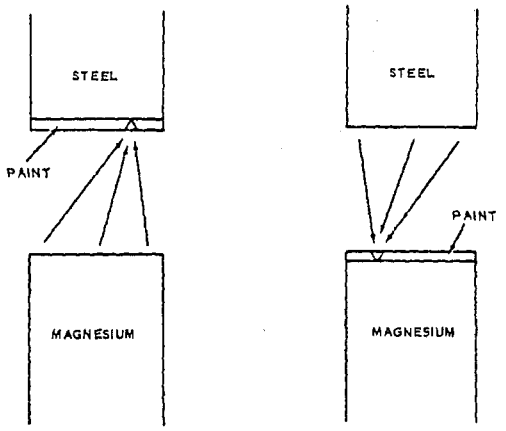


FIGURE 7. Use of Moisture Impervious Films. Proper design requires that both magnesium and dissimilar metals be protected with moisture impervious films (vinyl, etc.)

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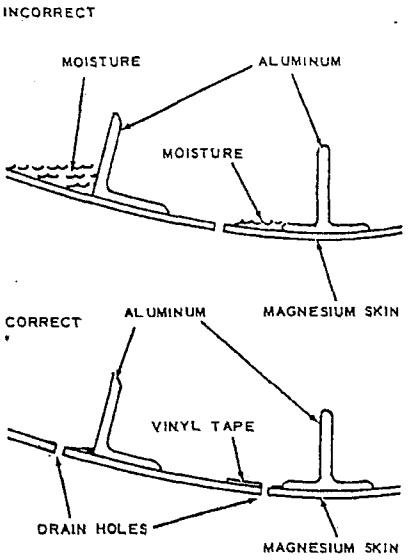


FIGURE 8. Use of Sealing Tapes and Drain Holes. Sealing tapes and drain holes must be properly used so that moisture does not bridge between magnesium and dissimilar metal.

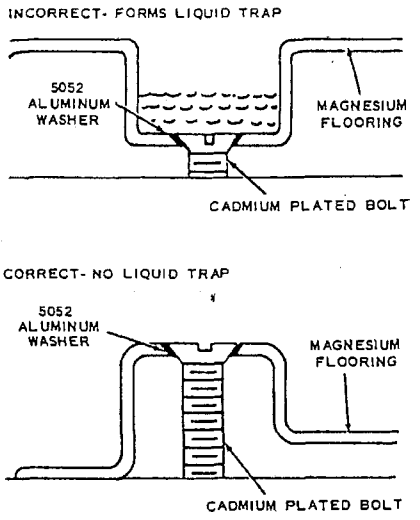


FIGURE 9. Use of 5052 Aluminum Washers. 5052 aluminum washers minimize galvanic corrosion but joint should be designed to prevent washer entrapment.

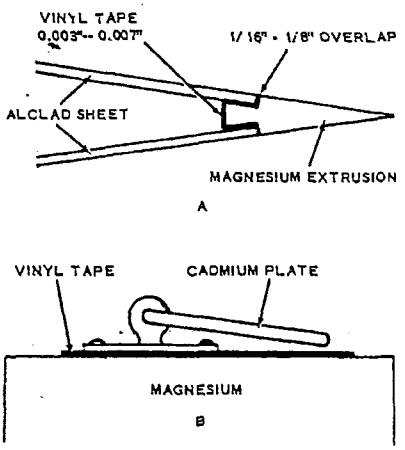


FIGURE 10. Application of Sealing Tapes. Sealing tapes must extend far enough to prevent any possibility of an electrolytic cell being set up.

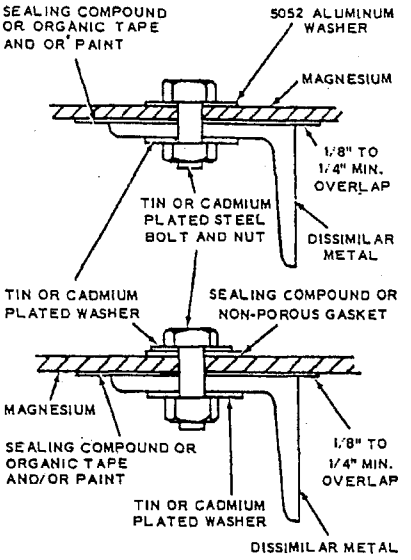


FIGURE 11. Alternate Methods of Protecting Magnesium Bolted to Dissimilar Metal or Wood

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Many chemical treatments for magnesium have been developed over the years. Some are still in general use, some are utilized for special purposes only, and others have been superseded. The treatments now in common use for magnesium appear in table 73.

40. Organic Coatings. Proper surface preparation and the choice of priming materials are two of the major factors that influence the satisfactory painting of magnesium. Most common paint troubles have their beginning at the point of contact between the primer and the metal surface.

Since the application of organic coatings is made primarily for the purpose of protection, the choice of the paint system depends upon the type of exposure to be encountered. The paint schedules most applicable are shown in table 74.

The various magnesium treatments and surface conditions require different paint schedules. Many of these applications are covered by U.S. government specifications (see paragraph 42).

Low temperature porcelain enamels for use as a protective method have been developed, and are reported to offer excellent resistance to abrasion and chemical attack without permitting undercutting at bare or exposed edges. The key to the system is a surface treatment that makes possible the use of standard lead oxide or lithium-phosphate frits. They have good heat-shock resistance; there is no spalling when quenched in water from 800°F (427°C). Porcelain enamels can be used on most magnesium alloys except those with high aluminum content, and those of high total alloy content that contain low-melting eutectics.

41. Plating. Two plating methods are utilized to deposit metallic surfaces on magnesium: conventional electroplating and electroless nickel plating.

Electrodeposits may be applied to magnesium after suitable surface preparation. One commercial process consists in the initial application of a zinc immersion coating, followed by copper striking and electroplating in standard plating baths.

The electroless nickel process utilizes a special chemical etch as the preparatory step, followed by chemical deposition of a nickel coating from an electroless nickel bath. (See table 75.)

SPECIFICATIONS

42. Government Specifications. Following is a list of government specifications covering magnesium-alloy materials.

Source: https://assist.dla.mil -- Downloaded: 202610-26T13:46Z
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Military Specifications

MIL-M-3171	Magnesium Alloy, Processes for Pretreatment and Prevention of Corrosion of
MIL-M-6857	Magnesium Alloy Castings, Heat Treatment of
MIL-W-6858	Welding, Resistance, Aluminum, Magnesium, Non-hardening Steels or Alloys, Nickel Alloys, Heat Resisting Alloys, and Titanium Alloys, Spot and Seam
MIL-R-6944	Rod and Wire, Welding, Magnesium Alloy
MIL-M-8916	Magnesium Alloy, Bar, Rod and Special Shaped Sections, Extruded (HM31A-T5)
MIL-M-8917	Magnesium Alloy, Sheet and Plate (HM21A-T8)
MIL-W-18326	Welding of Magnesium Alloys, Gas and Arc, Manual and Machine Processes for
MIL-M-20161	Magnesium Ingots and Magnesium Sticks
MIL-M-26075	Magnesium Alloy Sheet and Plate (HK31A)
MIL-M-26696	Magnesium Alloy Bars, Rods and Special Shaped Sections Extruded (ZK60B)
MIL-T-38749	Tooling Plate, Magnesium Alloy
MIL-M-45202	Magnesium Alloys, Anodic Treatment of
MIL-M-45207	Magnesium Alloy K1A, Sand Castings
MIL-M-46037	Magnesium Alloy Plate and Sheet ZE10A
MIL-M-46039	Magnesium Alloy Bar, Rod and Special Shaped Sections, Extruded ZK21A
MIL-M-46048	Floor Plate, Raised Pattern, Magnesium Alloy (AZ31B) Rolled
MIL-M-46062	Magnesium Alloy Castings, High Strength

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Military Specifications (Cont.)

MIL-C-46079	Coating, Epoxy, Baking Type for Magnesium Castings
MIL-M-46080	Magnesium Castings, Process for Anodic Cleaning and Surface Sealing of
MIL-M-46143	Magnesium Lithium Alloy Castings (LS141A and LZ145A)

Federal Specifications

QQ-M-31	Magnesium Alloy, Bars, Rods, and Special Shaped Sections, Extruded
QQ-M-38	Magnesium Alloy Die Castings
QQ-M-40	Magnesium Alloy Forgings
QQ-M-44	Magnesium Alloy, Plate and Sheet AZ31
QQ-M-55	Magnesium Alloy, Permanent and Semipermanent mold castings
QQ-M-56	Magnesium Alloy, Sand Castings
WW-T-825	Tube, Magnesium-Alloy, Extruded

43. ASTM Specifications. Following is a list of current ASTM specifications dealing with magnesium alloy materials.

B 80	Magnesium Alloy Sand Castings
B 90	Magnesium Alloy Sheet and Plate
B 91	Magnesium Alloy Forgings
B 92	Magnesium Ingot and Stick for Remelting
B 93	Magnesium Alloys in Ingot Form for Sand Castings, Permanent Mold Castings and Die Castings
B 94	Magnesium Alloy Die Castings
B 107	Magnesium Alloy Extruded Bars, Rods, Shapes Tubes and Wire

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ASTM Specifications (Cont.)

B 199	Magnesium Alloy Permanent Mold Castings
B 275	Codification of Light Metals and Alloys, Cast and Wrought
B 296	Temper Designations of Magnesium Alloys Cast and Wrought
B 403	Magnesium Alloy Investment Castings
B 480	Recommended Practice for Preparation of Magnesium and Magnesium Alloys for Electroplating
E 35	Chemical Analysis of Magnesium and Magnesium Base Alloys
E 155	Reference Radiographs for Inspection of Aluminum and Magnesium Castings

CHECK LIST

44. General. Engineers and designers of military equipment should observe the following when using magnesium alloys.

a. Evaluate both the requirements of the job and the properties of magnesium alloys. Then consider the economic feasibility.

b. Select the alloy which has the best combination of properties, coupled with fabrication advantages.

c. Decide whether the part should be a sand casting, die casting, forging, or fabricated part (welded or formed). The decision should be made on the basis of your requirements and the best combination of properties, with the greatest economy.

d. Keep in mind that strength in tension or compression increases in direct proportion to section area. For most magnesium alloys, compressive strength is not as great as tensile strength.

e. When designing for stiffness, remember that the modulus of elasticity is 6,500,000 psi. Stiffness increases as the cube of the section thickness, or depth.

f. When redesigning (for stiffness) for magnesium, use 4-1/2 times the moment of inertia of a similar steel part and 1-1/2 times the moment of inertia of a similar aluminum part.

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g. The load carrying capacity of a beam is in direct relation to the compressive and tensile yield strengths multiplied by the section modulus. For bending stresses, work to the average of tensile and compressive yield strengths.

h. Strength of columns is proportional to the modulus of elasticity and the compressive yield strength.

i. Design away from stress concentrations by using generous radii and fillets, and by avoiding extreme change of sectional thickness.

j. Avoid conditions where galvanic corrosion may occur. Take precautions against actual contact of dissimilar metals.

k. Observe the following:

(1) Specify 5056 aluminum rivets with magnesium.

(2) Minimum rivet spacing is 4 times the rivet diameter.

(3) Minimum edge distance is 2 times the rivet diameter.

(4) List proper rivet diameter.

l. Allow tolerances as great as possible without affecting the utility of the part. Liberal tolerances cut down scrap allowance, thus decreasing costs.

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TABLE 2. RELATED INDUSTRIAL SPECIFICATIONS FOR MAGNESIUM ALLOYS — CAST AND WROUGHT

Govt. Spec No.	Alloy	Nominal Composition, Elements %							Uses[2]	Industrial Specification Numbers		
		Al	Mn[1]	Zn	Zr	Th	R.E.	Sn		ASTM	SAE	AMS
QQ-M-31	M1A	-	1.2	-	-	-	-	-	E	B107-70	522	-
	AZ31B	3.0	-	1.0	-	-	-	-	E	B107-70	52	-
	AZ61A	6.5	-	1.0	-	-	-	-	E	B107-70	520	4350F
	AZ80A	8.5	-	0.5	-	-	-	-	E	B107-70	523	-
	ZK60A	-	-	5.7	0.55	-	-	-	E	B107-70	524	4352A
QQ-M-38	AZ91A	9.0	-	0.6	-	-	-	-	DC	B94-58	501	4490D
QQ-M-40	M1A	-	1.2	-	-	-	-	-	F	-	533	-
		3.0	-	1.0	-	-	-	-	F	B91-68	-	-
	AZ61A	6.5	-	1.0	-	-	-	-	F	B91-68	531	4358A
	AZ80A	8.5	-	0.5	-	-	-	-	F	B91-68	532	4360
	TA54A	3.5	-	-	-	-	-	5.0	F	B91-68	53	-
	ZK60A	-	-	5.7	0.55	-	-	-	F	B91-68	-	4362
QQ-M-44	AZ31B	3.0	-	1.0	-	-	-	-	P&S	B90 70	510	4375B, 4376A, 4377
QQ-M-55	AM100A	10.0	-	-	-	-	-	-	PMC	B199-70	502	-
	AZ91C	8.7	-	0.7	-	-	-	-	PMC	B199-70	-	-
	AZ92A	9.0	-	2.0	-	-	-	-	PMC	B199-70	503	4484E
	EZ33A	-	-	2.7	0.7	-	3.0	-	PMC	B199-70	-	-
	AZ63A	6.0	-	3.0	-	-	-	-	PMC	-	-	-
	AZ81A	7.6	-	0.7	-	-	-	-	PMC	-	-	-
QQ-M-56	AZ63A	6.0	-	3.0	-	-	-	-	SC	B80-70	50	4420G, 4422H, 4424F
	AZ81A	7.6	-	0.7	-	-	-	-	SC	B80-70	-	-
	AZ91C	8.7	-	0.7	-	-	-	-	SC	B80-70	504	4437
	AZ92A	9.0	-	2.0	-	-	-	-	SC	B80-70	500	4434F
	EZ33A	-	-	2.7	0.7	-	3.0	-	SC	B80-70	-	4442A
	HK31A	-	-	-	0.7	3.0	-	-	SC	B80-70	-	4445
	HZ32A	-	-	2.1	-	3.0	-	-	SC	B80-70	-	4447
	ZE41A	-	-	4.2	0.7	-	1.25	-	SC	B80-70	-	-
	QE22A[3]											
	ZH62A	-	-	5.7	0.7	1.8	-	-	SC	B80-70	-	-
	ZK51A	-	-	4.6	0.7	-	-	-	SC	B80-70	-	4443A
WW-T-825	AZ31B	3.0	-	1.0	-	-	-	-	T	B107-70	52	-
	AZ61A	6.5	-	1.0	-	-	-	-	T	B107-70	520	-
	M1A	-	1.2	-	-	-	-	-	T	B107-70	522	-
	ZK60A	-	-	5.7	0.55	-	-	-	T	B107-70	-	-
MIL-M-26075	HK31A	-	-	-	0.7	3.0	-	-	P&S	B90-70	-	4384A, 4385A

NOTES: 1. Minimum

2. E--extrusions, DC-die castings, F-forgings, P&S--Plate and Sheet, PMC--permanent mold castings, SC--sand castings T--tubes

3. 0.10 Cu, 0.01 Ni max., 2.0 - 3.0 Ag, 1.75 - 2.5 RE, 0.40 - 1.0 Zr, and 0.30 other max.

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TABLE 3. FORMS AND SPECIFICATIONS

Alloy	Sand Cast	Permanent Semi-Perm. Cast	Die Cast	Plate Sheet	Bars - Rods - Wire - Shapes Extruded	Tubing Extruded	Forgings	Welding Rod-Wire
AM100A		MIL-M-46062 QQ-M-55						
AZ31B				QQ-M-44	MIL-F-46048 QQ-M-31	WW-T-825		
AZ61A					QQ-M-31	WW-T-825	QQ-M-40	MIL-R-6944
AZ63A	QQ-M-56	QQ-M-55						MIL-R-6944
AZ80A					QQ-M-31		QQ-M-40	
AZ81A	QQ-M-56	QQ-M-55						
AZ91A			QQ-M-38					
AZ91C	QQ-M-56 MIL-M-46062	QQ-M-55 MIL-M-46062	MIL-M-46062					
AZ92A	QQ-M-56 MIL-M-46062	QQ-M-55 MIL-M-46062	MIL-M-46062					
EZ33A	QQ-M-56	QQ-M-55						
HK31A	QQ-M-56 MIL-M-46062	QQ-M-55 MIL-M-46062	MIL-M-46062	MIL-M-26075				MIL-R-6944
HM21A				MIL-M-8917				
HM31A					MIL-M-8916	-		
HZ32A	QQ-M-56	QQ-M-55						
K1A	MIL-M-45207							
M1A					QQ-M-31	WW-T-825	QQ-M-40	MIL-R-6944
QE22A	QQ-M-56	QQ-M-55 MIL-M-46062	MIL-M-46062					
TA54A							QQ-M-40	
ZH62A	QQ-M-56 MIL-M-46062	MIL-M-46062	MIL-M-46062					
ZK10A				MIL-M-46037				
ZK21A					MIL-M-46039			
ZK51A	QQ-M-56 MIL-M-46062	MIL-M-46062	MIL-M-46062					
ZK60A					QQ-M-31 MIL-M-5354			
ZK60B					MIL-M-26696			
ZK61A	QQ-M-56							

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TABLE 4. STANDARDS CROSS REFERENCE

ALLOY (ASTM B-275)	GOVERNMENT	ASTM	SAE	AMS	H-1	AWS
9980A		B-92			21490	
9990A		B-92			21490	
9990B		B-92			21490	
9995A		B-92			21490	
A3A						
AM80A						
AM100A	QQ-M-55 MIL-M-6857 MIL-M-46062	B80, B93, B199, B403	502	4455, 4483	21327	
AM100B						
AZ10A						
AZ21A						
AZ31A						
AZ31B	QQ-M-40, QQ-M-44 QQ-M-31, WW-T-825 MIL-F-46048	B90, B91, B107	510	4375, 4376, 4377, 4382	21480	
AZ31C		B90, B107				
AZ61A	QQ-M-31, QQ-M-40, WW-T-825, MIL-R-6944	B91, B107	520	4350, 4358	21473	A.5-19-69
AZ63A	QQ-M-55 QQ-M-36 MIL-M-6857	B80, B93	50	4420, 4422, 4424	21467	
AZ80A	QQ-M-31 QQ-M-40	B91, B107	523	4350	21470	
AZ81A	QQ-M-55 QQ-M-56 MIL-M-6857	B80, B93 B403, B199	505		21466	
AZ91A	QQ-M-38 MIL-M-46062	B93, B94	501	4490	21456	
AZ91B		B93, B94	501A		21465	
AZ91C	QQ-M-56 QQ-M-55 MIL-M-46062 MIL-M-6857	B80, B93, B199, B403	504	4437	21465	
AZ92A	QQ-M-55 QQ-M-56 MIL-M-6857 MIL-M-46062	B80, B93, B199, B403	500	4395, 4434, 4453, 4484	21455	A5.8-69
EK30A						
EK41A				4440, 4441	21529	

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TABLE 4 (Continued). STANDARDS CROSS REFERENCE

ALLOY (ASTM B-275)	GOVERNMENT	ASTM	SAE	AMS	H-1	AWS
EZ33A	QQ-M-55 QQ-M-56 MIL-M-6857	B80, B199, B403	506	4396, 4442	21531	A5.19-68
HK31A	QQ-M-55 QQ-M-56 MIL-M-6857 MIL-M-26075 MIL-M-46062	B80, B90, B199, B403	507	4384, 4385 4445	21595 21545	
HM-21A	QQ-M-40 MIL-M-8917	B90		4363 4383 4390 4388 4389		
HM-31A	MIL-M-8916					
HZ32A	QQ-M-55 QQ-M-56 MIL-M-6857	B80		4447	21540	
K1A	MIL-M-45207	B80, B403				
LA141A		B90		4386 4397		
LS141A	MIL-M-46143					
LZ145A	MIL-M-46143					
M1A	QQ-M-31 QQ-M-40 WW-T-825 MIL-R-6944	B107	522		21485	
M1B					21485	
TA54A	QQ-M-40	B91	53		21245	
QE22A	QQ-M-55 QQ-M-56 MIL-M-6857 MIL-M-46062	B80, B199, B403		4418		
ZE10A	MIL-M-46037	B90	534			
ZE41A		B80		4439	21537	
ZH62A	MIL-M-46062 MIL-M-6857 QQ-M-56	B80	508	4438	21538	
ZK21A	MIL-M-46039			4387		
ZK51A	MIL-M-46062 MIL-M-6857 QQ-M-56	B80	509 534	4443	21515	
ZK60A	QQ-M-31 QQ-M-40 WW-T-825	B91, B107	524	4352, 4362	21525	
ZK60B	QQ-M-40 MIL-M-26696					
ZK61A	QQ-M-56 MIL-M-6857 MIL-M-46062	B80, B403	513	4444	21520	

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TABLE 5. CHARACTERISTICS OF MAGNESIUM SAND–CAST ALLOYS

Govt. Spec No.	Alloy	Temper	Characteristics
QQ-M-56	AZ63A	-F	Moderate strength with fair ductility.
		-T4	High tensile strength and ductility.
		-T5	Similar to -F., but with higher creep strength and less growth at elevated Temperatures.
		-T6	High tensile and yield strength with fair elongation.
	AZ81A	-T4	Has high elongation and toughness, with good pressure tightness.
	AZ91C	A11	Similar to some tempers of AZ63A, but has increased pressure tightness.
	AZ92A	-F	Moderate strength with low ductility. Also has good pressure tightness.
		-T4	High tensile strength and ductility.
		-T5	Lower tensile strength and elongation than AZ63A-T5 but with increased pressure tightness.
		-T6	Best combination of high yield strength and moderate elongation with good pressure tightness of Mg-Al-Zn alloys.
	EZ33A	-T5	Good tensile and creep strength up to about 500°F. for long periods of time. Has excellent pressure tightness. More difficult to cast in some designs than Mg-Al-Zn alloys.
	HK31A	-T6	Good tensile and short-time creep strength up to about 700°F. Has excellent pressure tightness. More difficult to cast than EZ33A.
	HZ32A	-T5	Has better long-time creep strength than HK31A-T6 at about 500°F and above. Has excellent pressure tightness. More difficult to cast than EZ33A.
	QE22A	-T6	Has superior tensile yield strength, plus excellent creep resistance and fatigue strength, make QE22A-T6 the best magnesium base alloy for use in sand cast applications operating at temperatures up to approximately 500°F.
	ZH62A	-T5	Has highest tensile and yield strength and good ductility. Has excellent pressure tightness. Castability and weldability increased over ZK51A.
	ZK51A	-T5	Has high tensile strength and elongation and higher yield strength than Mg-Al-Zn alloys. Has good pressure tightness. More difficult to cast than Mg-Al-Zn alloys. Not readily weldable.
	ZK61A	-T6	Provides the highest strength properties for room temperature applications of any magnesium casting alloy.

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TABLE 6. CHARACTERISTICS OF MAGNESIUM SHEET ALLOYS

Govt. Spec No.	Alloy	Temper	Characteristics
QQ-M-44	AZ31B	-O	Has good formability with moderate strength.
		-H24	Has good strength.
--	ZE10A	-O	Has good formability with moderate strength. Does not require stress relief after welding.
		-H24	Has good strength. Does not require stress relief after welding.
QQ-M-44	HK31A	-O	Has good formability and tensile and creep strength up to about 500°F.
		-H24	Has highest tensile and creep strength up to about 500°F.
--	HM21A	-T8	Has highest tensile and creep strength from about 500 to 800°F.

TABLE 7. CHARACTERISTICS OF MAGNESIUM EXTRUSION ALLOYS

Govt. Spec. No.	Alloy	Characteristics
QQ-M-31	AZ31B	Good extrudability. Moderate strength. General purpose extrusion alloy.
	AZ61A	Improved strength over AZ31B.
	AZ80A	High strength. Heat treatable. Not available in hollow shapes or tubing.
	M1A	Lower Strength than AZ31B.
--	ZK21A	Medium strength and good toughness. Does not require stress relief after welding.
QQ-M-31	ZK60A	High strength. Limited weldability. Heat treatable. Tougher than AZ80A.
--	ZK60B	Highest compressive yield strength. Limited weldability. Heat treatable. Tougher than AZ80A.
--	HM31A	Highest tensile and creep strength up to about 800°F.

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TABLE 8. CHARACTERISTICS OF MAGNESIUM FORGING ALLOYS

Govt. Spec No.	Alloy	Characteristics
QQ-M-40	M1A	Low cost, easy formability. Fair strength. Suitable for both press and hammer forging.
	AZ61A	Fairly good formability. Medium mechanical properties. Usually press forged.
	AZ80A	Difficult to form. High mechanical properties. Heat treatable. Press forged only.
	TA54A	General-purpose alloy. Moderate strength with good elongation. Suitable for both press and hammer forging.
	ZK60A	Difficult to form. High mechanical properties. Heat treatable. Tougher than AZ80A. Press forged only.
	AZ31B	Low cost, moderate properties suitable for hammer forging. Good formability and weldability.
	HM21A	Rolled ring and die forgings. Recommended for elevated temperature (650 F and above) applications.

TABLE 9. PHYSICAL PROPERTIES OF MAGNESIUM ALLOYS – CAST AND WROUGHT

Property	Range		Notes
	Casting Alloys	Wrought Alloys	
Specific Gravity	1.74 to 1.86	1.76 to 1.83	About 1/4 that of steel
Weight (pounds per cubic inch)	0.0628 to 0.670	0.0633 to 0.661	Approximately 112 pounds per cubic foot
Electrical Conductivity (International Annealed Copper Standard)	10.0 to 30.3%	11.9 to 34.5%	Values depend upon composition and temper of the alloys. Heat treatment of casting alloys generally tends to lower conductivity as compared with the as-cast condition. The lowest values appear in the -T4 temper for a given composition. The high value for wrought alloys appears in alloy M1A and the low value in AZ80A (both specified in QQ-M-31 and QQ-M-40.
Thermal Conductivity (cgs units at 212°F)	0.10 to 0.29	0.12 to 0.33	Lower than the values for corresponding aluminum-alloy materials.
Thermal Expansion (mean coefficient per degree F. between the range of 68° to 212°F.)	14.5×10^{-6}	14.5×10^{-6}	In applications where magnesium alloys are rigidly connected to other material and the assembly is subject to appreciable variation in temperature, this relatively high coefficient must be given consideration.

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TABLE 10. PHYSICAL PROPERTIES OF MAGNESIUM ALLOYS

Alloy & Temper[1]	Specific Gravity (75°F.)	Density lb/cu. in.	Melting Point °F.	Thermal Conductivity C.G.S. Units (68°F.)	Electrical Resistivity Microhm-centimeters (68°F.)
AM100A-T61	1.81	0.066	1100	0.15	11.6
AZ31B, AZ31C	1.77	0.064	1160	0.18	9.2
AZ61A	1.80	0.065	1145	0.14	12.5
AZ63A-T4	1.83	0.066	1135	0.13	13.9
AZ80A	1.80	0.065	1130	0.12	14.5
AZ81A-T4	1.80	0.066	1115	0.11	15.6
AZ91A	1.81	0.066	1120	0.11	17.0
AZ91C-T4	1.81	0.066	1105	0.11	16.6
AZ91C-T6	1.81	0.066	1105	0.13	13.3
AZ92A-T6	1.82	0.066	1110	0.14	12.2
EK30A-T6	1.79	0.065	1184	0.28	6.0
EK41A-T5	1.81	0.066	1193	0.24	7.0
EK41A-T6	1.81	0.066	1193	0.22	7.5
EZ33A-T5	1.82	0.066	1189	0.26	6.5
HK31A-T6[2]	1.79	0.065	1204	0.22	7.2
HK31A-H24[3]	1.79	0.065	1204	0.25	6.7
HZ32A-T5	1.83	0.066	1198	0.26	6.5
M1A	1.76	0.064	1200	0.33	5.0
ZK60A-F	1.83	0.066	1175	0.28	6.0
ZK60A-T5	1.83	0.066	1175	0.29	5.7

NOTES: 1. All tempers unless otherwise noted. 2. Sand Castings 3. Sheet and Plate

TABLE 11. FABRICATION PROPERTIES OF CASTING ALLOYS

Alloy	Cast Method	Foundry Characteristics[1]				Other Characteristics[1]				
		Fluidity[2]	Pressure Tightness	Micro-Porosity[3]	Heat Treated	Machining[4]	Electro-plating[5]	Surface Treatment[6]	Suitability Braze[7]	Suitability Weld[8]
AZ63A	S	1	3	3	Yes	1	1	1	No	3
AZ81A	S	1	2	2	Yes	1	2	2	No	1
AZ91C	S	1	2	2	Yes	1	2	2	No	2
	PM	1	2	2	Yes	1	1	2	No	2
AZ92A	S	1	2	2	Yes	1	2	2	No	2
	PM	1	3	3	Yes	1	1	2	No	2
EZ33A	S	2	1	1	Yes	1	2	1	No	1
	PM	2	1	1	Yes	1	1	1	No	1
HK31A	S	2	1	1	Yes	1	2	1	*	1
HZ32A	S	2	1	1	Yes	1	*	2	*	2
ZH62A	S	2	2	2	Yes	1	1	1	No	*
ZK51A	S	2	3	2	Yes	1	2	1	No	3

NOTES:
1. Rating of 1 indicates best of group; 3 indicates poorest of group.
2. Ability of liquid alloy to flow readily in mold and fill thin sections.
3. Based on radiographic evidence.
4. Composite rating based on ease of cutting, chip characteristics, quality of finish, and tool life. Ratings, in the case of heat-treatable alloys, based on T6 type temper. Other tempers, particularly the annealed temper, may have lower rating.
5. Ability of casting to take and hold an electroplate applied by present standard methods.
6. Ability of castings to be cleaned in standard pickle solutions and to be conditioned for best paint adhesion.
7. Refers to suitability of alloy to withstand brazing temperature without excessive distortion or melting.
8. Based on ability of material to be fusion-welded with filler rod of same alloy.
* Inexperience with these alloys under wide production conditions makes it undersirable to supply ratings at this time.

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TABLE 12. MECHANICAL REQUIREMENTS OF MAGNESIUM ALLOYS (From Government Specifications).

Govt. Spec. No.	Alloy	Temper	Test Specimen Diameter or Thickness In.	Tensile Strength ksi		Yield Strength at 0.2% Offset or at Extension Indicated ksi				Elong in 2 Inches % Min
								Extension Under Load In. Per In.		
				Min	Avg	Min	Avg	Min	Avg	
QQ-M-31	AZ31B	-F	0.249 and under	35	-	21	-	0.0105	-	7
			0.250 to 1.499	35	-	22	-	0.0108	-	7
			1.500 to 2.499	34	-	22	-	0.0108	-	7
			Hollow Shapes	32	-	16	-	0.0089	-	8
	AZ61A	-F	0.249 and under	38	-	21	-	0.0101	-	8
			0.250 to 2.499	39	-	24	-	0.0114	-	9
			Hollow Shapes	36	-	16	-	0.0089	-	7
	AZ80A	-F	0.249 and under	43	-	28	-	0.0126	-	9
			0.250 to 1.499	43	-	28	-	0.0126	-	8
			1.500 to 2.499	43	-	28	-	0.0126	-	6
		-T5	0.249 and under	47	-	30	-	0.0132	-	4
			0.250 to 1.499	48	-	33	-	0.0142	-	4
			1.500 to 2.499	48	-	33	-	0.0142	-	4
	MI	-F	0.249 and under	30	-	n.r.	-	n.r.	-	2
			0.250 to 1.499	32	-	n.r.	-	n.r.	-	3
			1.500 to 2.499	32	-	n.r.	-	n.r.	-	2
			Hollow Shapes	28	-	n.r.	-	n.r.	-	2
	ZK60A	-F	Up to 5 sq in. cs	43	-	31	-	0.0068	-	5
			Hollow Shapes	40	-	28	-	0.0063	-	5
		-T5	Up to 5 sq in. cs	45	-	36	-	0.0075	-	4
			Hollow Shapes	46	-	38	-	0.0078	-	4
QQ-M-38	AZ91A	-F	-	29	31.5	22	23	-	-	2 - 5
QQ-M-40	AZ31B	-F	-	34	-	19	-	0.0049	-	6
	AZ61A	-F	-	38	-	22	-	0.0054	-	6
	AZ80A	-F	-	42	-	26	-	0.0060	-	5
		-T5	-	42	-	28	-	0.0063	-	2
	HM21A	-T5	-	33	-	25	-	0.0059		3
	M1A	-F	-	30	-	18	-	0.0048	-	3
	TA54A	-F	-	36	-	22	-	0.0054	-	7
	ZK60A	-T5	-	42	-	26	-	0.0060	-	7
		-T6	-	43	-	32	-	0.0069	-	4
QQ-M-44	AZ31B	-O	0.016 to 0.500	32	Note 1	-	-	-	-	12
			0.501 to 2.000	32	Note 1	-	-	-	-	10
			2.001 to 3.000	32	-	-	-	-	-	9
		-H24	0.016 to 0.249	39	-	29	-	0.0129	-	6
			0.250 to 0.374	38	-	26	-	0.0120	-	8
			0.375 to 0.500	37	-	24	-	0.0114	-	8
			0.501 to 1.000	36	-	22	-	0.0108	-	8
			1.001 to 2.000	34	-	20	-	0.0101	-	8
			2.001 to 3.000	34	-	18	-	0.0095	-	8
		-H26	0.250 to 0.374	39	-	27	-	0.0123	-	6
			0.375 to 0.500	38	-	26	-	0.0120	-	6
			0.501 to 0.750	37	-	25	-	0.0117	-	6
			0.751 to 1.000	37	-	23	-	0.0111	-	6
			1.001 to 1.500	35	-	22	-	0.0108	-	6
			1.501 to 2.000	35	-	21	-	0.0105	-	6
			Separately Cast Specimens							
QQ-M-55	AM100A	-F	-	20	-	10	-	0.0035	-	n.r.
		-T4	-	34	-	10	-	0.0035	-	6
		-T6	-	34	-	15	-	0.0043	-	2
		-T61	-	34	-	17	-	0.0046	-	n.r.
	AZ63A	-F	-	26	-	11	-	0.0037	-	4
		-T4	-	34	-	11	-	0.0037	-	7

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TABLE 12 (Continued). MECHANICAL REQUIREMENTS OF MAGNESIUM ALLOYS (From Government Specifications).

Govt. Spec. No.	Alloy	Temper	Test Specimen Diameter or Thickness In.	Tensile Strength ksi		Yield Strength at 0.2% Offset or at Extension Indicated ksi		Extension Under Load In. Per In.		Elong in 2 Inches % Min
				Min	Av.	Min	Avg	Min	Avg	
QQ-M-55	AZ63A	-T5	-	26	-	12	-	0.0038	-	2
		-T6	-	34	-	16	-	0.0045	-	3
	AZ81A	-T4	-	34	-	11	-	0.0037	-	7
	AZ91C	-F	-	23	-	11	-	0.0037	-	n.r.
		-T4	-	34	-	11	-	0.0037	-	7
		-T5	-	23	-	12	-	0.0038	-	2
		-T6	-	34	-	16	-	0.0045	-	3
	AZ92A	-F	-	23	-	11	-	0.0037	-	n.r.
		-T4	-	34	-	11	-	0.0037	-	6
		-T5	-	23	-	12	-	0.0038	-	n.r.
		-T6	-	34	-	18	-	0.0048	-	n.r.
	EZ33A	-T5 Rm Temp	-	20	-	14	-	0.0042	-	2
		-T5 500°F.	-	13	-	8	-	-	-	-
	HK31A	-T6 Rm Temp	-	27	-	13	-	0.0040	-	4
		-T6 500°F	-	21	-	13	-	-	-	-
	HZ32A	-T5 Rm Temp	-	27	-	13	-	0.0040	-	4
		-T5 500°F	-	13	-	8	-	-	-	-
	QE22A	-T6 Rm Temp	-	35	-	25	-	0.0062	-	2
		-T6 600°F	-	13	-	10	-	0.0035	-	-
		- Test Specimens Cut From Castings -								
QQ-M-55	AM100A	-T4	-	17	25.5	9	10	0.0034	0.0035	-
		-T6	-	17	25.5	10	13.5	0.0034	0.0041	-
		-T61	-	17	25.5	11.5	15.5	0.0038	0.0044	-
	AZ63A	-T4	-	17	25.5	9	10	0.0034	0.0035	-
		-T6	-	17	25.5	12	14.5	0.0039	0.0042	-
	AZ81A	-T4	-	17	25.5	9	10	0.0034	0.0035	-
	AZ91C	-T4	-	17	25.5	9	10	0.0034	0.0035	-
		-T6	-	17	25.5	12	14.5	0.0039	0.0042	-
	AZ92A	-T4	-	17	25.5	9	10	0.0034	0.0035	-
		-T6	-	17	25.5	13.5	16	0.0041	0.0045	-
	EZ33A	-T5 Rm Temp	-	13	15	11	12.5	0.0037	0.0039	-
		-T5 500°F	-	10	-	6	-	-	-	-
	HK31A	-T6 Rm Temp	-	19	23	10.5	11.7	0.0036	0.0038	-
		-T6 500°F	-	14	-	10.5	-	-	-	-
	HZ32A	-T5 Rm Temp	-	19	23	10.5	11.7	0.0036	0.0038	-
		-T5 500°F	-	10	-	6	-	-	-	-
	QE22A	-T6 Rm Temp	-	28	32	20	23	0.0051	0.0056	-
		-T6 600°F	-	10	-	8	-	-	-	-
		Separately Cast Specimens								
QQ-M-56	AZ63A	-F	-	26	-	11	-	0.0037	-	4
		-T4	-	34	-	11	-	0.0037	-	7
		-T5	-	26	-	12	-	0.0038	-	2
		-T6	-	34	-	16	-	0.0045	-	3
	AZ81A	-T4	-	34	-	11	-	0.0037	-	7
	AZ91C	-F	-	23	-	11	-	0.0037	-	n.r.
		-T4	-	34	-	11	-	0.0037	-	7

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TABLE 12 (Continued). MECHANICAL REQUIREMENTS OF MAGNESIUM ALLOYS (From Government Specifications).

Govt. Spec. No.	Alloy	Temper	Test Specimen Diameter or Thickness In.	Tensile Strength ksi		Yield Strength at 0.2% Offset or at Extension Indicated				Elong in 2 Inches % Min
						ksi		Extension Under Load In. Per In.		
				Min	Avg	Min	Avg	Min	Avg	
QQ-M-56	AZ91C	-T5	-	23	-	12	-	0.0038	-	2
		-T6	-	34	-	16	-	0.0045	-	3
	AZ92A	-F	-	23	-	11	-	0.0037	-	n. r.
		-T4	-	34	-	11	-	0.0037	-	2
		-T5	-	23	-	12	-	0.0038	-	n. r.
		-T6	-	34	-	18	-	0.0048	-	n. r.
	EZ33A	-T5 Rm Temp	-	20	-	14	-	0.0042	-	2
		-T5 500°F.	-	13	-	8	-	-	-	-
	HK31A	-T6 Rm Temp	-	27	-	13	-	0.0040	-	4
		-T6 500°F	-	21	-	13	-		-	2
	HZ32A	-T5 Room Temp	-	27	-	13	-	0.0040	-	4
		-T5 500°F	-	13	-	8	-	-	-	-
	QE22A	-T6 Room Temp	-	25	-	25	-	0.0058	-	2
		-T6 600°F	-	20	-	25	-	0.0045	-	-
	ZH62A	-T5	-	35	-	22	-	0.0054	-	5
	ZK51A	-T5	-	34	-	20	-	0.0051	-	5
	ZK61A	-T5	-	39	-	26	-	0.0060	-	5
		-T6		39	-	26	-	0.0060	-	5
			- Test Specimens Cut From Casting -							
QQ-M-56	AZ63A	-T4	-	17	25.5	9	10	0.0034	0.0035	-
		-T6	-	17	25.5	12	14.5	0.0038	0.0042	-
	AZ81A	-T4	-	17	25.5	9	10	0.0034	0.0035	-
	AZ91C	-T4	-	17	25.5	9	10	0.0034	0.0035	-
		-T6	-	17	25.5	12	14.5	0.0038	0.0042	-
	AZ92A	-T4	-	17	25.5	9	10	0.0034	0.0035	-
		-T6	-	17	25.5	13.5	16	0.0041	0.0045	-
	EZ33A	-T5 Rm Temp	-	13	15	11	12.5	0.0037	0.0039	-
		-T5 500°F	-	10	-	6	-	-	-	-
	HK31A	-T6 Rm Temp	-	19	23	10.5	11.7	0.0036	0.0038	-
		-T6 500°F	-	10	-	6	-	-	-	-
	HZ32A	-T5 Rm Temp	-	19	23	10.5	11.7	0.0036	0.0038	-
		-T5 500°F	-	10	-	6	-	-	-	-
	ZH62A	-T5	-	28.5	31.5	17.5	19.5	0.0047	0.0050	-
	ZK51A	-T5	-	24	29	14	17	0.0042	0.0046	-
	QE22A	-T6 Room Temp	-	28	32	20	23	0.0051	0.0055	-
		-T6 600°F	-	10	15	8	11	-	-	-
	ZK61A	-T5	-	30	34	21	24	0.0053	0.0057	-
		-T5	-	30	34	21	24	0.0053	0.0057	-
WW-T-825	AZ31A	-F	0.050 to 0.250	32	-	16	-	0.0089	-	-
	AZ61X	-F	0.050 to 0.250	36	-	16	-	0.0089	-	-
	M1	-F	0.050 to 0.250	28	-	-	-	-	-	-

NOTES: 1. Maximum values are 40000.
2. Maximum values are 38000. n. r. denotes not required.

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TABLE 12 (Continued). MECHANICAL REQUIREMENTS OF MAGNESIUM ALLOYS (From Government Specifications)

Govt. Spec. No.	Alloy	Temper	Test Specimen Diameter or Thickness In.	Tensile Strength ksi		Yield Strength at 0.2% Offset or at Extension Indicated				Elong in 2 Inches % Min
						ksi		Extension Under Load In. Per In.		
				Min	Avg	Min	Avg	Min	Avg	
MIL-M-8916	HM31A	-T5	-	37	-	26	-	-	-	4
MIL-M-8917	HM21A	-T8	0.016 to 0.250	33	-	18	-	-	-	6
			0.251 to 0.500	32	-	21	-	-	-	6
			0.501 to 3.000	30	-	21	-	-	-	6
		-T81	0.125 to 0.312	34	-	25	-	-	-	4
MIL-M-26075	HK31A	-H24	0.016 to 0.125	34	-	26	-	-	-	4
			0.126 to 0.250	34	-	24	-	-	-	4
			0.251 to 1.000	34	-	25	-	-	-	4
			1.001 to 3.000	33	-	25	-	-	-	4
		-O	0.016 to 0.250	30 (38 MAX)	-	18	-	-	-	12
			0.251 to 0.500	30	-	16	-	-	-	12
			0.501 to 1.000	30	-	15	-	-	-	12
			1.001 to 3.000	29	-	14	-	-	-	12
MIL-M-26696	ZK60B	-T5	-	45	-	35	-	-	-	4
MIL-M-45207	K1A	-F	-	24	-	6	-	-	-	14
MIL-M-46037	ZE10A	O	0.016 to 0.060	30	-	18	-	-	-	15
			0.061 to 0.250	30	-	15	-	-	-	15
			0.251 to 0.500	29	-	12	-	-	-	12
		-H24	0.016 to 0.125	36	-	25	-	-	-	6
			0.126 to 0.188	34	-	22	-	-	-	6
			0.189 to 0.250	31	-	20	-	-	-	6
MIL-M-46039	ZK21A	-F	-	38	-	28	-	-	-	4
Mechanical Properties, From Designated Areas of Castings										
MIL-M-46062	AM100A*	-T6	-	38	-	20	-	-	-	3
			-	35	-	18	-	-	-	1.5
			-	30	-	16	-	-	-	1
	AZ91C	-T6	-	35	-	18	-	-	-	4
			-	29	-	16	-	-	-	3
			-	27	-	14	-	-	-	2
	AZ92A	-T6	-	40	-	25	-	-	-	3
			-	34	-	20	-	-	-	1
			-	30	-	18	-	-	-	0.75
	HK31A	-T6	-	33	-	16	-	-	-	6
			-	29	-	14	-	-	-	3
			-	25	-	12	-	-	-	1
	ZK51A	-T5	-	36	-	21	-	-	-	6
			-	32	-	19	-	-	-	4
			-	29	-	17	-	-	-	3
	ZK61A	-T6	-	42	-	29	-	-	-	6
			-	37	-	26	-	-	-	4
			-	34	-	23	-	-	-	2
	ZH62A	-T5	-	38	-	23	-	-	-	5
			-	34	-	21	-	-	-	3
			-	31	-	19	-	-	-	2
	QE22A	-T6	-	40	-	28	-	-	-	4
			-	37	-	26	-	-	-	2
			-	33	-	23	-	-	-	2
Mechanical properties, specimens taken from unspecified areas of castings										
MIL-M-46062	AM100A	-T6	-	17	-	10	-	-	-	0.75
	AZ91C	-T6	-	17	-	12	-	-	-	0.75
	AZ92A	-T6	-	17	-	13	-	-	-	0.50
	HK31A	-T6	-	19	-	10	-	-	-	1.0
	ZK51A	-T5	-	24	-	14	-	-	-	1.25
	ZK61A	-T6	-	30	-	21	-	-	-	1.25
	ZH62A	-T5	-	28	-	17	-	-	-	1.25
	QE22A	-T6	-	28	-	20	-	-	-	1.00
MIL-M-46080	LS141A	-T7	-	18.5	-	13.5	-	-	-	18
	LZ145A	-T7	-	28	-	22	-	-	-	4

NOTE* Mechanical properties. - Stress levels of various sections of castings should be carefully considered before specifying the class of mechanical properties for any particular casting section. Since a uniform stress level is seldom required in casting design it would be advantageous from the design and foundry aspect to have higher properties in local designated areas with the remainder of the casting having lower properties. Three classes of mechanical properties are therefore incorporated in the specification for various stress levels and the minimum properties for the various alloys are also given for specimens taken from castings for use in designating stress levels on drawings.

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TABLE 13. TYPICAL MECHANICAL VALUES OF MAGNESIUM ALLOYS

Alloy	Temper	Tensile Strength ksi	Yield Strength ksi	Shear Strength ksi	Elong. %	Hardness, BHN	Hardness, RE	Compressive Yield Strength ksi
AM10	-	-	-	-	-	-	-	-
AM40	-F	24	9	14	6	44	46	9
AM80A	-F	26	14	-	4	46	51	14
	-T4	36.5	12	18	10	48	55	12
	-T6	36.5	13.5	-	6	-	-	15
AM100A	-F	36.5	25	20.5	6	59	70	20.5
	-T4	40	13.	19	10	52	62	13
	-T5	22	16	-	2	58	70	16
	-T6	40	16	22.5	4	69	80	16
	-T7	38	18	-	1	67	78	18
	-T61	40	22	-	1	-	-	22
AM100B	-F	31.5	-	-	2	-	-	-
AM120	-	-	-	-	-	-	-	-
AZ10A	-F	35	22	-	10	-	-	11
AZ31A	-O	40	-	-	12	-	-	-
	-H24	39	29	-	4	-	-	-
AZ31B	-O	37	22.25	21	20	55	66	15
	-F	36-38	24-28	19	13	50	59	12
	-H10	35	20	-	17	-	-	15
	-H11	37	20	-	20	-	-	16
	-H23	42	28	-	15	-	-	25
	-H24	39	27	-	16	73	83	22
	-H26	39	26.5	-	10	-	-	17
AZ31C	-F	35	19	-	14	52	63	11
AZ31X	-	32	16	-	8	-	-	*
AZ51A	-O	40	21	20	19	57	68	-
	-H24	42	25	22	7	72	82	-
AZ61A	-F	41-46	29.5	20	15	55	67	19
AZ61X	-	16	-	-	-	-	-	*
AZ63A	-F	27.3	14	17	5	50	59	14
	-T4	40	16.5	18.5	8	64	74	14
	-T5	31.3	13.6	18	6	55	66	-
	-T6	40	16.3	19	4	73	83	19
	-T7	40	17	20	7	59	71	*
AZ80A	-F	46.8	32	21.3	9	64	74	-
	-T5	50-55	38-40	22.75	7	78	86	34
	-T6	51.6	37	22	7	79	87	-
AZ81A	-T4	40	14	18	12	55	66	14
AZ91A	-F	33	22	20	3	63	73	22
	-T4	40	14	17	12	55	66	22
AZ91B	-F	33	22	20	3	63	73	22
AZ91C	-F	33	14	17	2	52	62	14
	-T4	38	14	17.5	9	55	66	14
	-T5	34	16	-	3	73	83	-
	-T6	38	18	20	4	73	83	19
AZ92A	-F	23.5	14	17.6	2	65	76	14

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TABLE 13 (Continued). TYPICAL MECHANICAL VALUES OF MAGNESIUM ALLOYS

Alloy	Temper	Tensile Strength ksi	Yield Strength ksi	Shear Strength ksi	Elong. %	Hardness BHN	Hardness RE	Compressive Yield Strength ksi
AZ92A	-T4	37	14	17	4	63	75	14
	-T5	28.6	14	18	5	71	81	14
	-T6	38	20.3	21	2	82	89	22
	-T7	40	21	21	3	78	86	21
EK30A	-T6	22	15.3	-	3	45	49	16
EK41A	-T5	21.5	15	-	1	45	49	16
	-T6	23.5	17	-	3	50	59	18
EZ33A	-T5	22	15	-	3	50	59	15
HK31A	-T6	33.7	17.8	21	10	54	65	16
	-H24	36	28	27	9	57	68	24
HM21A	-T8	34	21	19	10	-	-	15
HM21XA	-T8	34	25	-	10	-	-	-
HM31A	-T5	42	33	23	10	-	-	25
HM31XA	-F	42	33	-	10	-	-	-
HZ32A	-T5	29	14.5	-	7	57	68	14
K1A	-F	25	7	8	19	-	-	-
LA141A	-T7	19	14	-	10	-	-	18
LA142	-	-	-	-	-	-	-	-
M1	-	28	-	-	2	-	-	*
M1A	-O	35.5	18	17	14	48	55	-
	-F	31.7	21.5	15.4	8	42	39	-
	-H24	35	23	-	3	-	-	-
M1B	-F	12	-	-	3	48	55	*
MZ10	-	-	-	-	-	-	-	-
QQE22A	-T6	40	30	23	4	77	-	-
TA54A	-F	40	28	16	12	52	62	-
ZE10	-H24	34-48	19-28	-	8-12	-	-	16-26
ZE41	-T5	30	20	22	3.5	62	-	-
ZH62A	-F	40	26	-	8	-	-	-
	-T5	40	26	23	6	62	-	26
ZK51A	-F	33	19	22	8	65	72	-
	-T5	40	24	22	5	65	77	26
	-T6	40	25	-	6	-	-	-
ZK60A	-F	47.5	35.7	26	12	75	84	25
	-T5	50-53	40-44	25	12	82	-	33
ZK60XB	-B	49	38	-	17	-	-	40
ZK61A	-T6	45	28	26	8	70	-	-
Eclipsalloy 130	-T5	32	21	-	9	60	65	-
99.80 Mg	-F	12	3	-	6	39	16	-
	-O	27	14	17	16	40	36	-
Hard Rolled	-	37	27	-	9	50	59	-
General Average	-	35	20	23	8	58	70	18

NOTES:

* Indicates minimum values.

Table 13 represents an attempt to obtain general data on mechanical properties of magnesium alloys from technical publications. Each value listed represents the average of from two to ten published values, including those minimum values given in government specifications.

The following references were used:

American Society for Metals - Handbook and Metal Progress
American Society for Testing Materials - Publications and Handbooks
The Dow Chemical Co. - Magnesium Alloys and Products and other bulletins and literature.
U. S. Federal and Military Specifications.

Published data was not available for all alloys and tempers. Spaces opposite such listed alloys may be completed when information becomes available.

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TABLE 14. NOMINAL COMPOSITION AND TYPICAL MECHANICAL PROPERTIES OF MAGNESIUM ALLOYS AT ROOM TEMPERATURE

Alloy	Composition							Strength				Hardness
	Temper	Al	Mn. min	Zn	Th	Zr	Rare Earths	Tensile Yield ksi	Compressive Yield ksi	Shear ksi	Bearing Yield ksi	Bhn[1]
Sand and Permanent Mold Casting Alloys												
AM100A	-T6	10.0	0.1	-	-	-	-	22	22	-	-	-
AZ63A	-T6	6.0	0.15	3.0	-	-	-	19	19	21	52	73
AZ81A	-T4	7.6	0.13	0.7	-	-	-	12	12	18	44	55
AZ91C	-T6	8.7	0.13	0.7	-	-	-	19	19	21	52	66
AZ92A	-T6	9.0	0.10	2.0	-	-	-	21	21	22	65	84
EK30A	-T6	-	-	-	-	0.3	3.3	16	16	19	34	45
EK41A	-T6	-	-	-	-	0.6	4.0	18	18	20	41	50
EZ33A	-T5	-	-	2.7	-	0.6	3.3	15	15	21	40	50
ZE41A	-T5	-	-	4.2	-	0.7	1.2	20	20	23	51	62
ZK51A	-T5	-	-	4.6	-	0.7	-	24	24	23	47	65
ZK61A	-T5	-	-	6.0	-	0.7	-	27	27	25	-	68
ZK61A	-T6	-	-	6.0	-	0.7	-	28	28	26	-	70
HK31A	-T6	-	-	-	3.3	0.7	-	15	15	21	40	55
HZ32A	-T5	-	-	2.1	3.3	0.7	-	14	14	20	37	57
ZH42	-	-	-	4.0	2.0	0.7	-	-	-	-	-	-
ZH62A	-T5	-	-	5.7	1.8	0.7	-	25	25	24	45	70
Die-Casting Alloys												
AZ91A[2]	-	9.0	0.13	0.7	-	-	-	22	22	-	-	60
AZ91B[2]	-	-	-	-	-	-	-	-	-	-	-	-
Extruded Bars and Shapes												
M1A	-F	-	1.2	-	-	-	-	26	12	18	28	44
AZ31B	-F	3.0	-	1.0	-	-	-	29	14	19	33	49
AZ61A	-F	6.5	-	1.0	-	-	-	33	19	20	41	60
AZ80A	-T5	8.5	-	0.5	-	-	-	40	35	24	-	82
HM31XZ	-F	-	1.2	-	3.0	-	-	33	27	22	50	-
ZK21A	-	-	-	2.3	-	0.45 min	-	34	24	-	-	-
ZK60A	-T5	-	-	5.5	-	0.45 min	-	44	36	26	59	88
ZK60B[3]	-T5	-	-	5.7	-	0.45 min	-	42	42	-	-	-
Sheet and Plate Alloys												
AZ31B	-H24	3.0	-	1.0	-	-	-	32	26	23	47	73
HK31A	-H24	-	-	-	3.0	0.6	-	29	23	20	41	-
HM31A	-	-	0.6	-	2.0	-	-	26	15	18	39	-
ZE10A	-H24	-	-	1.3	-	-	0.17	28	24	-	-	-

NOTES: 1. 500-kg load, 10-min ball.

2. A and B are identical except that residual copper to 0.30% max is allowable in AZ91B.

3. Pelletized.

TABLE 15. STRESS – RUPTURE – TIME

Alloy	Form	Temper	Temperature °F	Stress ksi	Third Stage		Rupture	
					Time to Extension		Time to Extension	
					hrs	%	hrs	%
AZ31B	Extrusion	-F	95	31	570	15	866	30
				32	330	16	418	29
			300	7	390	11	900	55
				8	180	10	410	54
AZ61A	Extrusion	-F	95	42	660	8	1158	25
				44	210	9	299	22
			300	6	360	4	1174	71
				10	114	3	312	44
AZ63A	Cast	-T6	300	9	330	2	728	11
				10	90	1	310	11
AZ92A	Cast	-T6	95	34	480	7	1156	29
				40	115	9	233	27
			250	12	495	3	1080	30
				14	120	6	290	31
			300	8	405	3	1181	33
				10	84	2	288	36
M1A	Extrusion	-F	95	24	1300	17	1420	21
				25	340	14	477	22
			300	12	680	15	1115	38
				14	-	-	232	42

TABLE 16. FATIGUE STRENGTH – ROTATING BEAM ENDURANCE LIMIT

Alloy	Form	Temper	Endurance Limit - ksi			
			Million Cycles To Failure			
			0.1	1	10	100
AZ63A	Cast	-F	16	13	11	10
		-T4	20	18	16	14
		-T6	19	16	14	13
AZ81A		-T4	19	15	13	12
AZ91A		-F	15	13	12	12
AZ91C		-F	16	15	13	12
		-T4	19	17	15	13
		-T6	18	15	13	11
AZ92A		-F	17	15	14	13
		-T4	18	16	14	13
		-T6	18	16	14	13
EZ33A		-T5	14	12	11	10
HK31A		-T6	16	13	11	10
HZ32A		-T5	15	13	11	10
ZK51A		-T5	15	11	10	9
AZ31B	Extruded	-F	26	23	21	19
AZ61A		-F	27	24	22	20
AZ80A		-F	27	25	23	22
M1A		-F	16	14	12	11
ZK60A		-F	30	26	23	20
		-T5	30	26	23	20
AZ61A	Forged	-F	26	23	21	19
AZ80A		-F	29	25	22	19
		-T5	28	24	20	17
ZK60A		-T5	23	20	18	17

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TABLE 17. FATIGUE STRENGTH — DIRECT TENSION ENDURANCE LIMIT (R = 1/4)

Alloy	Form	Temper	Endurance Limit - ksi, Millions of Cycles		
			0.1	1	10
AZ63A	Cast	-F	23	21	19
		-T4	25	22	19
		-T6	27	24	23
AZ92A		-F	24	23	22
		-T4	24	31	30
		-T6	30	28	27
AZ31B	Rolled Sheet	-O	22	20	19
		-H24	24	22	21
HK31A		-H24	23	20	19
AZ31B	Extruded	-F	25	23	22
AZ61A		-F	24	20	19
ZK60A		-F	43	40	40
		-T5	45	43	42

TABLE 18. FATIGUE STRENGTH — REPEATED FLEXURE ENDURANCE LIMIT (Fully Reversed)

Alloy	Form	Temper	Endurance Limit - ksi, Millions of Cycles		
			0.1	1	10
AZ63A	Cast	-F	15	13	11
		-T4	14	12	11
		-T6	13	11	9
AZ91A		-F	10.5	9	8
AZ91C		-F	12	10	8
		-T4	15	13	11
AZ92A		-T6	15	12	10
		-F	12	10	8
		-T4	15	11	9
		-T6	18	15	13
EZ33A		-T5	15	13	11
HK31A		-T6	18	11	9
HZ32A		-T5	15	10	7
ZH62A		-T5	18	13	11
ZK51A		-T5	16	11	9
AZ31B	Rolled Sheet	-O	20	16	15
		-H24	22	18	16
HK31A		-O	17	12	10
		-H24	17	12	10
AZ31B	Extruded	-F	15	12	10
AZ61A		-F	18	13	12
AZ80A		-F	20	15	13
M1A		-F	11	9	8
AZ61A	Forged	-F	16	12	11
M1A		-F	13	9	6
ZK60A		-T5	19	14	12

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TABLE 19. WELD STRENGTH – INERT GAS TUNGSTEN ARC WELDED BUTT JOINTS

Alloy	Form	Temper	Weld Rod	Weld Property		
				Tensile Strength ksi	Elong %	Joint Efficiency %
AZ31B	Sheet-Plate	-H24	AZ61A	37	10	88
	Sheet-Plate	-O	AZ61A	36	12	97
	Extrusions		AZ61A	35	12	92
AZ61A	Extrusions		AZ61A	40	8	89
AZ63A	Casting	-T6	AZ63A	28	3	80
AZ80A	Extrusion	-F	AZ92A	42	4	86
AZ92A	Casting	-T6	AZ92A	28	3	80
EZ33A	Casting	-T5	EX33A	20	2	100
HK31A[1]	Sheet-Plate	-H24	HK31A	30	2	-
	Casting	-T6	HK31A	31	9.5	-
HZ32A[1]	Casting	-T5	HK31A	28	4	-
M1A	Extrusion		M1A	20	2	54

NOTE: 1. Room Temperature Properties. At elevated temperatures (400-600°F) the weld efficiencies of HK31A and HZ32A approach 100%.

TABLE 20. WELD STRENGTH – CONSUMABLE ELECTRODE ARC WELDS

Alloy	Form	Temper	Thickness In.	Weld			
				Wire Dia In.	Tensile Strength ksi	Elong %	Efficiency %
AZ31B	Sheet	-O	0.032	0.050	33.7	8.3	92.7
			0.040	0.050	35.8	7.0	95.5
			-	0.064	36.3	9.3	96.8
			0.050	0.050	35.5	8.3	94.6
			0.064	0.050	35.7	10.6	99.5
			0.250	0.098	35.1	8.8	95.9
AZ31B	Plate	-O	0.375	0.098	36.6	13.4	99.7
			0.500	0.098	36.1	17.5	98.3
AZ31B	Sheet	-H24	0.1875	0.051	40.1	8.8	96.4

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TABLE 21. CREEP STRENGTH

Alloy	Form	Temper	Temp °F	Stress - ksi - To Produce 0.1% Total Extension						0.2% Total Extension					
				10 min	1 hr	10 hr	100 hr	500 hr	1000 hr	10 min	1 hr	10 hr	100 hr	500 hr	1000 hr
AZ31B	Extrusion	-F	200	-	6	5	4	3.5	-	-	10	9	7	6	-
			250	-	5	4	3	2	-	-	9	7	5	4	-
			300	-	4	3	1.5	1	-	-	7	5	3	2	-
			350	-	3	2	-	-	-	-	5	3	-	-	-
	Sheet	-H24	200	-	6	4	2.5	2	-	-	9	7	4	3	-
			250	-	4	2	1	1	-	-	6	4	2	1.5	-
			300	-	2	1	0.5	-	-	-	4	2	1	0.5	-
			350	-	1	0.5	-	-	-	-	2	1	-	-	-
	Sheet	-O	200	-	6	5	4	4	-	-	10	9	8	7	-
			250	-	5	4	2.5	2	-	-	8	7	5	3	-
			300	-	4	2.5	1	0.5	-	-	7	4.5	2	1	-
			350	-	2	1	-	-	-	-	4	2	-	-	-
	Extrusion	-F	200	-	6	5	4	3	-	-	11	10	7	6	-
			250	-	5	4	2	1	-	-	9	7	4	3	-
			300	-	3.5	2	-	-	-	-	7	4.5	1.5	-	-
			350	-	2	1	-	-	-	-	4.5	1.5	-	-	-
AZ61A	Cast	-T6	200	-	8	6	4.5	-	3.5	12	11	9	7.5	-	5.5
			250	-	7	5	3.5	-	2.5	9	9	6.5	4	-	2.5
			300	-	6	4	2.5	-	1.5	6	7.5	4.5	2	-	-
			350	-	4	3	-	-	-	3.5	5	3	-	-	-
			400	-	3	2	-	-	-	-	2.5	1	-	-	-
			500	-	1.5	-	-	-	-	-	-	-	-	-	-
AZ63A	Cast	-T6	200	-	7	6	5	-	4	13	11.5	10	8	-	6.5
			300	-	5	3.5	2.5	-	1.5	9	8	5.5	4	-	2.5
			400	-	4.5	2.5	0.5	-	-	5.5	4.5	3	1.5	-	-
			500	-	2.5	1.5	-	-	-	3.5	2	1	-	-	-
AZ92A	Cast	-T6	200	-	8	6	5	-	4.5	13	11.5	10	8	-	6
			300	-	5.5	3.5	2.5	-	1.5	9	8	5.5	4	-	2
			400	-	3.5	1.5	0.5	-	-	5.5	4.5	3	1.5	-	1
			500	-	2	-	-	-	-	3.5	2	1	-	-	-
EZ33A	Cast	-T5	400	-	6	5.5	5	-	4.5	9.5	9.5	9	8	-	6
			500	-	4.5	3	2	-	1.5	7.5	7.5	5.5	3	-	2
			600	-	2.5	1.5	1	-	0.5	5.5	3.5	2	1	-	1
HK31A	Cast	-T6	400	-	6	6	5.5	-	5.5	10.5	10.5	10	9.5	-	9
			500	-	5	4.5	3.5	-	3	10.5	10	8.5	6.5	-	4
			600	-	4	3	2	-	1	7.5	6	5	3	-	1
			660	-	-	-	-	-	-	6.5	4.5	2.5	1	-	0.5
HZ32A	Sheet	-H24	300	-	6	6	4	-	-	-	6	6	4	-	-
			400	-	6	6	5.5	-	5	8.5	8.5	8	8	-	7.5
	Cast	-T5	400	-	6	6	5	-	3.5	8	8	7	6	-	5
			500	-	5.5	5	4	-	2	6.5	6	5	3.5	-	2
			600	-	4.5	3.5	2.5	-	-	6	5	3.5	2	-	0.5
			660	-	-	-	-	-	-	12	12	12	11	-	8.5
ZH62A	Cast	-T5	200	-	6.5	6	5.5	-	4.5	12	12	9	7.5	-	5.5
			300	-	6.5	6	5.5	-	3.0	10.5	10	7	5	-	2
			400	-	5.5	5.5	4.5	-	7.5	9	9	6	3.5	-	1
			500	-	4.5	3.5	2.5	-	1	7.5	6	3.5	2.5	-	1
ZK51A	Cast	-T5	200	-	6.5	5.5	4.5	-	3.5	12	12	10.5	9.5	-	6.5
			300	-	5.5	4	3	-	2	9	8	8	6.5	-	3.5
			400	-	5	3	2	-	1	7	6	4.5	3	-	2
ZK60A	Extrusion	-T5	200	-	-	-	-	-	-	-	-	-	8	2	-
			250	-	-	-	-	-	-	-	-	-	5	-	-
			300	-	-	-	-	-	-	-	-	-	3	-	-

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TABLE 21 (Continued). CREEP STRENGTH

Alloy	Form	Temper	Temp °F	Stress - ksi - To Produce 0.5% Total Extension						Stress - ksi - To Produce 1% Total Extension					
				10 min	1 hr	10 hr	100 hr	500 hr	1000 hr	10 min	1 hr	10 hr	100 hr	500 hr	1000 hr
AZ31B	Extrusion	-F	200	-	16	14	12	10	-	-	19	17	15	13	-
			250	-	13	11	9	7	-	-	16	14	11	9	-
			300	-	11	8	5	4	-	-	12	12	7	5	-
			350	-	8	6	-	-	-	-	10	8	-	-	-
	Sheet	-H24	200	-	15	11	7	5	-	-	19	14	10	7	-
			250	-	9	6	4	2	-	-	13	8	5	3	-
			300	-	6	3	2	1	-	-	8	5	2	1	-
			350	-	4	1.5	1.5	-	-	-	6	2	-	-	-
	Sheet	-O	200	-	15	14	12	10	-	-	16	15	13	11	-
			250	-	12	11	9	7	-	-	14	12	10	9	-
			300	-	10	7	4	3	-	-	12	9	5	3.5	-
			350	-	7	4	-	-	-	-	8	5	-	-	-
AZ61A	Extrusion	-F	200	-	20	18	14	12	-	-	25	22	19	16	-
			250	-	16	13	8	6	-	-	20	16	11	8	-
			300	-	12	8	3	3	-	-	16	11	5	2	-
			350	-	9	4	-	-	-	-	11	6	-	-	-
AZ63A	Cast	-T6	200	19	17	14.5	12.5	-	10	22.5	20.5	17.5	15	-	12.5
			300	13	12	9.5	7	-	5	16	14.5	11.5	8.5	-	6
			400	8.5	9	5	-	-	-	10	8.5	6	-	-	-
			500	4.5	7	1.5	-	-	-	5.5	4	2.5	-	-	-
AZ92A	Cast	-T6	200	20.5	19	16	13	-	10.5	25	23	20	17	-	14
			300	14.5	12.5	10	7	-	5	17.5	15	12	9	-	6.5
			400	9	7	5	2.5	-	1	11	8.5	6	3.5	-	1.5
			500	5	3.5	2	-	-	-	6.5	5	3	-	-	-
EZ33A	Cast	-T5	400	11	11	11	10	-	8.5	13	13	12.5	11.5	-	9
			500	10.5	9	6.5	4	-	2.5	12.5	10.5	7.5	5	-	3
			600	6.5	4	2.5	1.5	-	1	8	5	3	2	-	1
HK31A	Cast	-T6	400	15	15	15	15	-	14	16	16	16	16	-	16
			500	14.5	14	12.5	9.5	-	7	16	15.5	14.5	12	-	7.5
			600	12	10.5	7	3.5	-	1.5	14	12.5	8.5	4	-	1.5
			660	9.5	6	3	1.5	-	0.5	-	-	-	-	-	-
	Sheet	-H24	300	-	-	20	-	-	-	-	-	-	-	-	-
			400	-	11	9	6	-	-	-	18	16	11	-	-
H232A	Cast	-T5	400	10.5	10	10	8	-	10	11	11	11	11	-	10.5
			500	9.5	9	7	5	-	6.5	10.5	10.5	10	9.5	-	7
			600	8	7.5	6.5	3.5	-	3	9	8.5	7	5.5	-	3
			660	8	6	4	2.5	-	1.5	-	-	-	-	-	-
ZH62A	Cast	-T5	200	21	20	18	16.5	-	13.5	26.5	25	22.5	20	-	17
			300	17.5	16.5	14.5	12.5	-	10.5	20.5	19	17.5	15.5	-	13
			400	13.5	10.5	8	5.5	-	5.5	17	14	10.5	7	-	4
			500	8.5	5.5	3	2	-	3.5	10	6	3.5	2	-	1.5
ZK51A	Cast	-T5	200	19.5	18	14.5	-	12	22	21	19	17	-	-	14.5
			300	14.5	13	9	-	7	17	15.5	13.5	11.5	-	-	9
			400	10.5	9	5.5	-	3.5	12.5	11	9	7	-	-	5
ZK60A	Extrusion	-T5	200	-	15	6	4	-	-	20	14	8	6	-	-
			250	-	10	2.5	1.5	-	-	14	8	4	4	6	-
			300	-	5	1	-	-	-	8	4	1.5	2	-	-

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TABLE 22. LOW–TEMPERATURE PROPERTIES OF MAGNESIUM ALLOYS

Alloy	Temper	Thickness In.	Tensile Strength ksi	Yield Strength ksi	Elongation %
			Transverse Tests at 75°F		
HM21A	-T8	0.250	35.0	24.8	13.7
HK31A	-H24	0.250	35.2	25.9	21.0
	-O	0.250	29.0	18.0	30.5
ZE10A	-H26	0.125	38.9	27.3	20.2
	-H24	0.125	39.2	27.1	22.7
	-O	0.125	34.5	25.2	32.2
			Longitudinal Tests at 75°F		
HM21A	-T8	0.250	32.4	25.1	5.6
Welded[1]		0.250	28.6	18.6	2.7
HK31A	-H24	0.250	34.5	27.3	14.2
Welded[1]		0.250	28.8	21.7	2.4
HK31A	-O	0.250	28.9	17.7	29.7
		0.250	23.4	17.3	3.2
ZE10A	-H26	0.125	39.3	30.6	14.2
Welded[2]		0.125	32.2	19.0	3.5
Welded[3]		0.125	30.8	16.4	4.0
ZE10A	-H24	0.125	39.3	29.5	15.7
Welded[2]		0.125	32.8	21.7	3.0
ZE10A	-O	0.125	35.7	27.5	25.2
		0.125	31.4	17.6	5.7
HK31A	-H24	0.064	36.3	29.0	7.5
HK31A	-O	0.064	29.7	17.9	27.5
HM21A	-T8	0.064	32.2	23.1	7.2
			Longitudinal Tests at -65°F		
HM21A	-T8	0.250	38.4	29.7	4.7
HK31A	-H24	0.250	40.8	33.4	9.0
	-O	0.250	38.3	21.5	18.0
ZE10A	-H26	0.125	44.9	33.3	7.2
	-H24	0.125	45.1	32.1	9.2
	-O	0.125	43.2	29.3	17.2
HK31A	-H24	0.064	43.3	32.0	5.0
	-O	0.064	39.9	21.4	20.7
HM21A	-T8	0.064	39.6	25.6	6.2
HM21XA	-T5	*	39.5	15.8	9.3
			Longitudinal Tests at -98°F		
HM21A	-T8	0.250	40.1	31.3	5.0
Welded[1]		0.250	29.2	17.5	1.5
HK31A	-H24	0.250	42.4	33.8	11.5
Welded[1]		0.250	28.1	23.6	0.5
HK31A	-O	0.250	40.2	21.9	20.2
Welded[1]		0.250	29.4	21.0	2.2
ZE10A	-H26	0.125	45.5	31.5	9.0
Welded[2]		0.125	33.1	19.4	1.5
Welded[3]		0.125	36.9	15.7	3.5
ZE10A	-H24	0.125	45.4	31.8	9.5
Welded[2]		0.125	34.3	19.5	1.5

Note: The values on wrought alloys are averages of two to four tests at room temperature (2-in. gage length). Values of duplicate tests at low temperatures are also averages (1-in. gage length). Values for the cast alloys are averages of two to four tests on separately cast bars.

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TABLE 22 (Continued). LOW–TEMPERATURE PROPERTIES OF MAGNESIUM ALLOYS

Alloy	Temper	Thickness In.	Tensile Strength ksi	Yield Strength ksi	Elongation %
		Longitudinal Tests at -98°F (Cont)			
ZE10A	-O	0.125	44.5	29.5	15.2
Welded[2]		0.125	34.7	20.3	1.5
HK31A	-H24	0.064	42.7	30.6	4.2
HK31A	-O	0.064	41.3	21.1	17.5
HM21A	-T8	0.064	40.8	22.1	4.5
	-T5	*	39.5	18.6	8.0
	-T6	*	46.4	24.1	4.6
	-T5	*	40.0	16.3	8.3
		Longitudinal Tests at -320°F			
HM21A	-T8	0.250	47.3	30.6	4.2
Welded[1]		0.250	33.1	20.9	1.5
HK31A	-H24	0.250	52.9	34.7	8.0
Welded[1]		0.250	33.7	25.9	1.5
HK31A	-O	0.250	47.2	24.7	12.5
Welded[1]		0.250	29.7	21.6	2.2
ZE10A	-H26	0.125	53.6	36.3	4.0
Welded[2]		0.125	37.3	24.5	2.0
Welded[3]		0.125	40.6	19.2	2.0
ZE10A	-H24	0.125	54.4	35.2	6.2
Welded[2]		0.125	40.3	24.1	1.5
ZE10A	-O	0.125	50.1	31.9	7.5
Welded[2]		0.125	39.0	24.1	1.5
	-H26	0.050	57.7	40.7	3.3
		0.050	42.8	35.6	1.0
	-H24	0.080	54.1	40.3	3.7
		0.080	40.7	29.8	0.5
	-H26	0.064	55.2	36.9	6.7
		0.064	36.7	29.6	0.5
HK31A	-H24	0.064	54.0	33.0	6.2
HK31A	-O	0.064	47.9	24.3	12.7
HM21A	-T8	0.064	47.6	24.9	4.0
	-T5	*	45.6	18.9	6.5
	-T6	*	54.0	26.3	5.3
	-T5	*	46.6	18.1	8.0

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TABLE 22 (Continued). LOW–TEMPERATURE PROPERTIES OF MAGNESIUM ALLOYS

Alloy	Temper	Tensile Strength ksi	Yield Strength ksi	Elongation %	Charpy impact ft-lb	
					Unnotched	Notched
		Cast Magnesium Alloys at 75°F				
AZ91C	-T6	41.8	19.2	6.3	5.87	1.0
ZH62A	-T5	39.9	27.9	5.7	11.08	0.75
AZ92A	-T6	41.8	23.4	4.0	5.62	0.50
EZ33A	-T5	27.5	16.9	7.6	5.50	0.62
HK31A	-T6	32.7	16.3	9.5	12.25	2.87
		Cast Magnesium Alloys at -109°F				
AZ91C	-T6	44.3	21.6	5.1	4.62	1.0
ZH62A	-T5	47.6	29.2	2.7	14.00	0.75
AZ92A	-T6	42.7	24.6	2.3	4.75	0.50
EZ33A	-T5	27.6	18.0	3.1	3.56	0.50
HK31A	-T6	43.3	17.5	8.6	12.12	2.37
		Cast Magnesium Alloys at -321°F				
AZ91C	-T6	44.9	26.0	1.7	3.00	0.75
ZH62A	-T5	46.6	34.1	1.0	6.31	0.75
AZ92A	-T6	46.5	28.5	0.8	3.37	0.50
EZ33A	-T5	29.0	20.3	2.2	3.69	0.50
HK31A	-T6	48.1	19.6	6.1	10.12	2.25

NOTES: 1. Welding rod was EZ33; weld bead intact.
2. Welding rod was AZ61; weld bead intact.
3. Weld bead removed.
* Specimen machined from a forging.

TABLE 23. TYPICAL MECHANICAL PROPERTIES AS AFFECTED BY ELEVATED TEMPERATURE – TENSILE STRENGTH

Alloy	Form	Temper	Ultimate Tensile Strength % of 70° Value						
			Temperature °F						
			70	200	300	400	500	600	700
AZ63A	Cast	-T4	100	88	72	38	-	-	-
		-T5	100	97	87	70	57	27	-
		-T6	100	90	60	45	30	20	-
AZ81A		-T4	100	92	70	50	35	-	-
AZ91A		-F	100	95	80	54	33	20	-
AZ91C		-T4	100	85	70	50	-	-	-
		-T6	100	92	68	50	-	-	-
AZ92A		-T5	100	96	92	80	64	36	-
		-T6	100	92	70	42	28	20	-
EZ33A		-T5	100	100	96	91	78	52	-
HK31A		-T6	100	94	87	77	74	64	42
HZ32A		-T5	100	90	76	59	48	41	34
ZH62A		-T5	100	82	65	48	35	-	-
ZK51A		-T5	100	75	58	42	30	20	-
AZ31B	Extruded	-F	100	92	66	39	26	16	-
AZ61A		-F	100	93	73	47	29	18	-
AZ80A		-T5	100	89	62	42	25	18	-
M1A		-F	100	73	57	46	32	24	-
ZK60A		-T5	100	71	50	29	12	-	-
AZ80A	Forged	-T5	100	86	60	44	28	18	-
M1A		-F	100	58	50	42	25	15	-
AZ31B	Sheet	-O	100	79	55	34	21	16	-
		-H24	100	83	54	32	20	15	-
HK31A		-O	100	70	55	45	36	30	-
		-H24	100	82	70	65	54	35	-

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TABLE 24. TYPICAL MECHANICAL PROPERTIES AS AFFECTED BY ELEVATED TEMPERATURE — (0.2% Offset) TENSILE YIELD STRENGTH

Alloy	Form	Temper	Yield Strength - % of 70°F Value						
			Temperature °F						
			70	200	300	400	500	600	700
AZ63A	Cast	-T4	100	93	86	78	-	-	-
		-T5	100	100	87	67	47	27	-
		-T6	100	89	79	63	47	32	-
AZ81A		-T4	100	100	92	85	76	-	-
AZ91A		-F	100	96	74	54	35	16	-
AZ91C		-T4	100	93	93	87	-	-	-
		-T6	100	90	81	67	-	-	-
AZ92A		-T5	100	94	82	65	47	34	-
		-T6	100	91	74	52	35	22	-
EZ33A		-T5	100	94	88	75	62	50	-
HK31A		-T6	100	100	94	88	81	75	50
HZ32A		-T5	100	93	80	67	60	53	47
ZH62A		-T5	100	85	74	56	37	-	-
ZK51A		-T5	100	81	65	50	35	23	-
AZ31B	Extruded	-F	100	76	52	31	17	7	-
AZ61A		-F	100	79	61	42	27	15	-
AZ80A		-T5	100	80	58	40	22	12	-
M1A		-F	100	81	62	46	27	19	-
ZK60A		-T5	100	68	46	24	5	-	-
AZ80A	Forged	-T5	100	79	62	44	24	15	-
M1A		-F	100	74	63	47	32	21	-
AZ31B	Sheet	-O	100	88	58	33	21	8	-
		-H24	100	75	41	25	16	6	-
HK31A		-O	100	81	71	62	52	33	-
		-H24	100	89	79	72	59	24	-

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TABLE 25. TYPICAL MECHANICAL PROPERTIES AS AFFECTED BY ELEVATED TEMPERATURE – ELONGATION

Alloy	Form	Temper	Elongation - %						
			Temperature °F						
			70	200	300	400	500	600	700
AZ63A	Cast	-T4	12	14	15	20	-	-	-
		-T5	4	8	33	45	56	65	-
		-T6	5	11	15	18	15	20	-
AZ81A		-T4	15	22	32	30	25	-	-
AZ91A		-F	4	5	15	17	19	21	-
AZ91C		-T4	14	26	30	30	-	-	-
		-T6	5	24	31	33	-	-	-
AZ92A		-T5	1	2	4	15	32	61	-
		-T6	2	25	35	36	33	49	-
EZ33A		-T5	3	5	10	20	31	50	-
HK31A		-T6	6	8	12	17	19	22	26
HZ32A		-T5	6	15	23	33	39	38	29
ZH62A		-T5	8	20	24	28	30	-	-
ZK51A		-T5	8	12	14	17	16	16	-
AZ31B	Extruded	-F	15	20	38	44	56	75	-
AZ61A		-F	16	21	40	42	52	70	-
AZ80A		-T5	7	20	33	52	80	120	-
M1A		-F	11	15	19	25	60	93	-
ZK60A		-T5	9	41	52	84	177	-	-
AZ80A	Forged	-T5	5	15	30	49	83	123	-
M1A		-F	12	26	31	34	88	140	-
AZ31B	Sheet	-O	23	35	68	82	92	136	-
		-H24	14	35	58	82	92	136	-
HK31A		-O	23	40	50	50	34	60	-
		-H24	8	16	20	21	19	70	-

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TABLE 26. EFFECT OF TEMPERATURE AND NOTCHING ON FATIGUE STRENGTH

Rotating Beam Endurance Limit									
				Endurance Limit - ksi					
				Plain Bars			Notched Bars[1,2]		
				Millions of cycles					
Alloy	Form	Temper	Exposure Temp °F	0.1	10	100	0.1	10	100
AZ63A	Cast	-T6	68	18	16	15	11.5	9.5	9
			250	17	12.5	11	11.5	7	7
			300	-	9	7.5	9.5	6	5
AZ80A	Forged	-T5	68	25	20	18	16.5	9.5	9.5
			300	20.5	11	10.5	10.5	6.5	5
AZ92A	Cast	-T6	68	19.5	16	15	14	9.5	9.5
			250	14.5	11.5	10.5	13.5	8	6
			300	16	11	10	11.5	6.5	6
Axial - Loading Endurance Limit[3]									
AZ63A	Cast	-T6	68	25	23	21	21	18	18
			275	25	24	23	18	16.5	15

NOTES: 1. Stress concentration factor = 2 for rotating beam
2. Stress concentration factor = 2.4 for axial - loading
3. For r = 0.25 = ratio of minimum to maximum stress in each loading cycle.

TABLE 27. EFFECT OF EXPOSURE TO ELEVATED TEMPERATURES ON ELONGATION (Tested at Room Temperature)

				% Elong. After Exposure Time-Hrs				
Alloy	Form	Temper	Exposure Temp °F	0	100	500	2500	5000
AZ31B	Sheet	-H24	200	13.8	14	11	-	-
			250	13.8	14	13.2	-	-
			300	13.8	17	17.5	-	-
			400	13.8	21.6	19	-	-
			500	13.8	22.4	21.2	-	-
			600	13.8	21	21.5	-	-
AZ63A	Cast	-T6	200	3.5	5.5	5.4	5.6	-
			300	3.5	3.8	4	3.7	-
			400	3.5	5.2	4.8	5.2	-
AZ92A	Cast	-T6	200	1.5	1.7	2	3.8	-
			300	1.5	2	1.5	1.7	1.6
			400	1.5	1.8	2.4	2.8	3.1
EK30A	Cast	-T6	400	1.7	2	2	-	-
			500	1.7	3.5	3	-	2.8
			600	1.7	3.5	3	-	3.2
HK31A	Sheet	-H24	300	8	-	-	-	8
			400	8	-	-	-	8
			500	8	-	11	-	16
			600	8	20	21	-	20
ZK60A	Extrusion	-T5	200	8.8	11	9.2	-	-
			250	8.8	10.5	8	-	-
			300	8.8	11	10.2	-	-
			400	8.8	13.8	12.7	-	-
			500	8.8	14.8	14.5	-	-

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TABLE 28. TENSILE PROPERTIES OF SEPARATELY CAST MAGNESIUM – ALLOY TEST BARS AT ELEVATED TEMPERATURES

Govt. Spec No.	Alloy	Temper	Test Temp. °F	Ultimate Tensile Strength ksi	Tensile Yield Strength (0.2% Offset) ksi	Elongation %
			DIE CAST			
QQ-M-38	AZ91A	-F	70	34.4	21.9	4.0
			200	32.7	21.1	5.0
			300	27.6	16.2	14.7
			400	18.5	11.8	16.7
			500	11.5	7.6	18.6
			600	7	3.6	20.7
			SAND CAST			
QQ-M-56	AZ81A	-T4	70	40	13	15
			200	37	13	22
			300	28	12	32
			400	20	11	30
			500	14	10	25
	AZ63A	-T4	70	40	14	12
			200	35	13	14
			300	29	12	15
			400	15	11	20
		-T5	70	30	15	4
			200	29	15	8
			300	26	13	33
			400	21	10	45
			500	17	7	56
			600	8	4	65
		-T6	70	40	19	5
			200	36	17	11
			300	24	15	15
			400	18	12	18
			500	12	9	15
			600	8	6	20
	AZ91C	-T4	70	40	15	14
			200	34	14	26
			300	28	14	30
			400	20	13	30
		-T6	70	40	21	5
			200	37	19	24
			300	27	17	31
			400	20	14	33
	AZ92A	-T5	70	25	17	1
			200	24	16	2
			300	23	14	4
			400	20	11	15
			500	16	8	32
			600	9	4	61
		-T6	70	40	23	2
			200	37	21	25
			300	28	17	35
			400	17	12	36
			500	11	8	33
			600	8	5	49
	EZ33A	-T5	70	23	16	3
			200	23	15	5
			300	22	14	10
			400	21	12	20
			500	18	10	31
			600	12	8	50

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TABLE 28 (Continued). TENSILE PROPERTIES OF SEPARATELY CAST MAGNESIUM – ALLOY TEST BARS AT ELEVATED TEMPERATURES

Govt. Spec No.	Alloy	Temper	Test Temp. °F	Ultimate Tensile Strength ksi	Tensile Yield Strength (0.2% Offset) ksi	Elongation %
SAND CAST (Cont.)						
QQ-M-56	HK31A	-T6	70	31	16	6
			200	29	16	8
			300	27	15	12
			400	24	14	17
			500	23	13	19
			600	20	12	22
			700	13	8	26
	HZ32A	-T5	70	29	15	6
			200	26	14	15
			300	22	12	23
			400	17	10	33
			500	14	9	39
			600	12	8	38
			700	10	7	29
	QE22A	-T6	70	35	25	2
			500	20	16	-
	ZK51A	-T5	70	40	26	8
			200	30	21	12
			300	23	17	14
			400	17	13	17
			500	12	9	16
			600	8	6	16
	ZH62A	-T5	70	40	27	8
			200	33	23	20
			300	26	20	24
			400	19	15	28
			500	14	10	30

TABLE 29. TENSILE PROPERTIES OF FORGED MAGNESIUM ALLOYS AT ELEVATED TEMPERATURES

Govt. Spec No.	Alloy	Temper	Test Temp. °F	Ultimate Tensile Strength ksi	Tensile Yield Strength (0.2% Offset) ksi	Elongation %
QQ-M-40	M1A	-F	70	40	19	12
			200	23	14	26
			300	20	12	31
			400	17	9	34
			500	10	6	88
			600	6	4	140
	AZ80A	-T5	70	50	34	5
			200	43	27	15
			300	30	21	30
			400	22	15	49
			500	14	8	83
			600	9	5	123

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TABLE 30. TENSILE PROPERTIES OF EXTRUDED MAGNESIUM ALLOYS AT ELEVATED TEMPERATURES

Govt. Spec No.	Alloy	Temper	Test Temp. °F	Ultimate Tensile Strength ksi	Tensile Yield Strength (0.2% Offset) ksi	Elongation %
QQ-M-31	M1A	-F	70	37	26	11
			200	27	21	15
			300	21	16	19
			400	17	12	25
			500	12	7	60
			600	9	5	93
	AZ31B	-F	70	38	29	15
			200	35	22	20
			300	25	15	38
			400	15	9	44
			500	10	5	56
			600	6	2	75
	AZ61A	-F	70	45	33	16
			200	42	26	21
			300	33	20	40
			400	21	14	42
			500	13	9	52
			600	8	5	70
	AZ80A	-T5	70	55	40	7
			200	49	32	20
			300	34	23	33
			400	23	16	52
			500	14	9	80
			600	10	5	120
	ZK60A	-T5	70	52	41	9
			200	37	28	41
			300	26	19	52
			400	15	10	84
			500	6	2	177

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TABLE 31. TENSILE PROPERTIES OF MAGNESIUM – ALLOY SHEET AT ELEVATED TEMPERATURES

Govt. Spec No.	Alloy	Temper	Test Temp. °F	Ultimate Tensile Strength ksi	Tensile Yield Strength (0.2% Offset) ksi	Elongation %
QQ-M-44	AZ31B	-O	70	38	24	23
			200	30	21	35
			300	21	14	88
			400	13	8	82
			500	8	5	92
			600	6	2	136
		-H24	70	41	32	14
			200	34	24	35
			300	22	13	58
			400	13	8	82
			500	8	5	92
			600	6	2	136
	HK31A	-O	70	33	21	23
			200	23	17	40
			300	18	15	50
			400	15	13	50
			500	13	11	34
			600	10	7	60
		-H24	70	37	29	8
			200	30	26	16
			300	26	23	20
			400	24	21	21
			500	20	17	19
			600	13	7	70
**	HM21A	-T8	70	34	25	10
			300	21	19	20
			400	18	17	30
			500	16	15	25
			600	14	12	15
			700	11	8	50
			800	5	3	100
			900	2	1	>100

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TABLE 32. ROTATING–BEAM FATIGUE STRENGTH OF SOME MAGNESIUM ALLOYS AT ELEVATED TEMPERATURES AND AS AFFECTED BY NOTCHING

Govt. Spec No.	Alloy	Temper	Temp. °F	Plain Bars, Fatigue Strength - ksi			Notched Bars, SCF^1 = 2 Fatigue Strength ksi		
				Millions of Cycles					
				0.1	10	100	0.1	10	100
Sand Cast									
QQ-M-56	AZ63A	-T6	68	18	16	15	11.5	9.5	9
			250	17	12.5	11	11.5	7	7
			300	-	9	7.5	9.5	6	5
	AZ92A	-T6	68	19.5	16	15	14	9.5	9.5
			250	14.5	11.5	10.5	13.5	8	6
			300	16	11	10	11.5	6.5	6
Forged									
QQ-M-40	AZ80A	-T5	68	25	20	18	16.5	9.5	9.5
			300	20.5	11	10.5	10.5	6.5	5

NOTES: 1. SCF, Stress Concentration Factor.

TABLE 33. AXIAL–LOADING FATIGUE STRENGTH[1] OF A SAND–CAST MAGNESIUM ALLOY AT AN ELEVATED TEMPERATURE AND AS AFFECTED BY NOTCHING

Govt. Spec No.	Alloy	Temper	Temp. °F	Plain Bars, Fatigue Strength - ksi			Notched Bars, SCF^2 = 2.4 Fatigue Strength ksi		
				Millions of Cycles					
				0.1	1	10	0.1	1	10
QQ-M-56	AZ63A	-T6	68	25	23	21	21	18	18
			275	25	24	23	18	16.5	15

NOTES: 1. For R = 0.25, R is the ratio of minimum stress in each loading cycle.
2. SCF, Stress Concentration Factor.

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TABLE 34. CREEP STRENGTHS OF VARIOUS MAGNESIUM CASTING ALLOYS AT ELEVATED TEMPERATURES—SEPARATELY SAND—CAST TEST BARS

Alloy	Temper	Temp. °F	Creep Strength - ksi									
			Hours for 0.1% Total Extension					Hours for 0.2% Total Extension				
			10 min.	1	10	100	1000	10 min	1	10	100	1000
AZ63A	-T6	200	7.9	7.1	5.9	4.6	3.5	11.7	10.7	9.2	7.4	5.7
		300	5.8	5.0	3.7	2.4	1.45	8.8	7.7	6.1	4.1	2.6
		400	3.8	3.0	1.7	-	-	5.9	4.9	3.0	-	-
		500	2.4	1.50	-	-	-	3.4	2.4	0.90	-	-
AZ92A	-T6	200	7.9	7.1	6.1	5.1	4.2	13.0	11.7	9.8	8.0	6.5
		300	5.4	4.5	3.3	2.3	1.5	9.2	7.8	5.7	3.8	2.6
		400	3.5	2.5	1.5	0.63	-	5.7	4.4	2.8	1.35	-
		500	2.2	1.25	-	-	-	3.5	2.2	0.96	-	-
EK30A	-T6	400	6.2	6.0	5.6	5.2	4.4	10.0	9.8	9.0	7.6	5.4
		500	4.5	4.1	3.4	2.15	1.5	7.9	7.0	4.9	2.85	1.8
		600	2.7	1.9	1.34	0.98	0.73	4.6	3.0	1.83	1.18	0.83
EK41A	-T5	400	-	-	-	-	-	9.5	9.2	8.6	7.8	6.9
	-T6	400	6.3	6.1	5.7	4.9	4.4	11.3	10.5	9.3	7.8	6.2
	-T6	600	3.5	2.5	1.7	1.27	1.1	4.8	3.3	2.1	1.46	1.2
EZ33A	-T5	400	6.0	5.9	5.6	5.1	4.3	9.5	9.3	8.8	8.0	6.2
		500	5.3	4.35	3.2	2.1	1.6	9.2	7.6	5.0	3.0	1.97
		600	4.1	2.6	1.5	0.93	0.66	5.7	3.5	1.97	1.18	0.82
HK31A	-T6	400	6.2	6.0	5.8	5.6	5.4	10.6	10.3	9.8	9.5	9.1
		500	5.8	5.25	4.4	3.5	3.1	10.6	10.0	8.6	6.3	4.2
		550	-	-	-	-	-	9.0	7.8	6.4	4.5	2.5
		600	4.8	4.15	3.25	2.15	0.94	7.4	6.2	4.75	2.9	1.1
		660	-	-	-	-	-	6.7	4.4	2.3	1.0	0.63
HZ32A	-T5	400	6.2	6.0	5.7	5.35	4.8	8.7	8.4	8.1	7.75	7.4
		500	5.8	5.45	4.95	4.25	3.5	8.2	7.75	6.9	5.9	5.05
		550	-	-	-	-	-	7.4	6.8	5.8	4.6	4.0
		600	5.2	4.6	3.5	2.5	1.87	6.5	6.0	5.05	3.6	2.2
		660	-	-	-	-	-	6.0	5.0	3.4	2.0	0.72
ZH62A	-T5	200	6.5	6.4	6.1	5.5	4.3	12.0	12.0	11.7	10.8	8.7
		300	5.9	5.7	5.3	4.6	3.2	10.6	9.9	8.9	7.6	5.5
		400	5.0	4.3	3.4	2.6	1.7	8.8	7.0	5.2	3.6	2.2
		500	3.8	2.7	1.9	1.4	0.9	5.9	3.7	2.4	1.7	1.2
ZK51A	-T5	200	6.3	5.9	5.3	4.5	3.5	11.8	10.7	9.5	8.0	6.5
		300	5.4	4.7	3.8	3.0	2.1	9.0	7.8	6.45	5.05	3.7
		400	4.85	3.95	2.95	2.03	1.2	7.2	5.8	4.4	3.0	1.75

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TABLE 34 (Continued).

CREEP STRENGTHS OF VARIOUS MAGNESIUM CASTING ALLOYS AT ELEVATED TEMPERATURES–SEPARATELY SAND–CAST TEST BARS

Alloy	Temper	Temp. °F	Creep Strength - ksi									
			Hours for 0.5% Total Extension					Hours for 1.0% Total Extension				
			10 min	1	10	100	1000	10 min.	1	10	100	1000
AZ63A	-T6	200	18.8	17.1	14.7	12.3	10.1	22.3	20.5	17.7	14.8	12.3
		300	13.3	11.8	9.6	6.9	4.8	15.9	14.3	11.7	8.7	6.1
		400	8.5	7.1	5.0	-	-	10.0	8.5	6.1	-	-
		500	4.7	3.4	1.65	-	-	5.6	4.2	2.3	-	-
AZ92A	-T6	200	20.5	18.8	15.9	13.0	10.5	25.0	23.2	20.2	17.0	14.2
		300	14.4	12.5	9.8	7.0	4.8	17.4	15.2	12.0	9.0	6.4
		400	9.1	7.2	4.8	2.5	0.87	11.0	8.6	5.8	3.4	1.3
		500	5.0	3.5	1.78	-	-	6.4	4.8	2.8	-	-
EK30A	-T6	400	14.0	13.2	12.0	9.8	6.7	16.0	15.3	13.8	11.3	7.3
		500	11.1	9.0	5.9	3.5	2.3	14.3	11.7	7.4	4.3	2.5
		600	5.8	3.8	2.23	1.4	0.99	7.1	4.6	2.6	1.58	1.08
EK41A	-T5	400	13.0	12.4	11.3	9.7	7.4	14.4	13.8	12.6	10.7	7.8
	-T6	400	14.9	14.0	12.3	9.8	7.4	16.4	15.4	13.8	11.2	7.9
	-T6	600	6.0	4.0	2.5	1.67	1.32	7.0	3.7	2.8	1.8	1.38
EZ33A	-T5	400	11.0	11.0	10.9	10.0	8.3	12.8	12.8	12.5	11.6	9.2
		500	10.7	9.2	6.4	4.1	2.6	12.4	10.4	7.4	4.8	3.1
		600	6.6	4.2	2.4	1.5	1.07	8.0	5.1	2.9	1.74	1.18
HK31A	-T6	400	15.0	15.0	15.0	15.0	14.0	16.1	16.0	16.0	16.0	15.8
		500	14.5	14.0	12.7	9.7	6.8	15.8	15.5	14.4	12.2	7.6
		550	13.5	12.3	9.5	6.3	3.2	-	-	-	-	-
		600	12.0	10.4	7.2	3.5	1.4	14.2	12.3	8.7	4.1	1.55
		660	9.6	6.0	3.2	1.3	0.72	-	-	-	-	-
HZ32A	-T5	400	10.3	10.2	10.0	9.9	9.8	11.0	11.0	11.0	11.0	10.4
		500	9.3	9.2	9.0	8.0	6.5	10.3	10.3	10.0	9.3	6.9
		550	8.9	8.7	8.0	6.7	4.8	-	-	-	-	-
		600	8.0	7.4	6.3	4.8	2.85	9.2	8.3	6.9	5.4	3.2
		660	7.8	6.1	4.2	2.5	1.4	-	-	-	-	-
ZH62A	-T5	200	21.0	19.9	18.2	16.3	13.6	26.3	24.8	22.3	19.9	16.9
		300	17.7	16.3	14.5	12.6	10.3	20.6	19.2	17.5	15.6	13.0
		400	13.3	10.7	7.8	5.5	3.3	17.0	13.8	10.3	7.2	4.0
		500	8.6	5.4	3.2	2.0	1.5	9.9	6.2	3.6	2.2	1.6
ZK51A	-T5	200	19.5	18.0	16.3	14.3	12.0	22.2	21.0	19.0	17.2	14.5
		300	14.3	12.8	11.0	9.2	7.1	17.0	15.7	13.7	11.5	9.0
		400	10.5	9.0	7.2	5.4	3.5	12.3	11.0	9.2	7.2	4.9

TABLE 35. CREEP STRENGTHS OF VARIOUS MAGNESIUM WROUGHT ALLOYS AT ELEVATED TEMPERATURES

Alloy	Form	Temper	Temp °F	Creep Strength - ksi							
				Hours for 0.1% Total Extension				Hours for 0.2% Total Extension			
				1	10	100	500	1	10	100	500
AZ31B	Extrusions	-F	200	6	5.0	4.0	3.5	10	9	7	6
			250	5.0	4.0	3.0	2.0	9	7	5.0	4.0
			300	4.0	3.0	1.5	1.0	7	5.0	3.0	2.0
			350	3.0	2.0	-	-	5.0	3.0	-	-
AZ61A		-F	200	6	5.0	4.0	3.0	11	10	7	6
			250	5.0	4.0	2.0	1.0	9	7	4.0	3.0
			300	3.5	2.0	-	-	7	4.0	1.0	-
			350	2.0	-	-	-	4.5	1.5	-	-
ZK60A		-T5	200	-	-	-	-	8	5.0	3.0	2.0
			250	-	-	-	-	5.0	3.0	1.5	-
			300	-	-	-	-	3.0	1.5	-	-
AZ31B	Sheet	-H24	200	6	4.0	2.5	2.0	9	6	4.0	3.0
			250	4.0	2.0	1.0	1.0	6	4.0	2.0	1.5
			300	2.0	1.0	0.5	-	4.0	2.0	1.0	0.5
			350	1.0	0.5	-	-	2.0	1.0	-	-
AZ31B		-O	200	6	5.0	4.0	4.0	10	9	8	7
			250	5.0	4.0	2.5	2.0	8	7	5.0	3.0
			300	4.0	2.5	1.0	0.5	7	4.5	2.0	1.0
			350	2.0	1.0	-	-	4.0	2.0	-	-
HK31A		-T5	300	-	-	-	-	-	-	12	-
			400	6	6	4	-	11	9	6	-
HM21A		-T5	300	-	-	-	-	-	-	12	-
			400	6	6	6	-	12	12	10	-
			500	6	6	-	-	10	9	7	-
			600	5	5	4	-	7	5	5	-
			700	3	2	-	-	4	3	3	-

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TABLE 35 (Continued).

CREEP STRENGTHS OF VARIOUS MAGNESIUM WROUGHT ALLOYS AT ELEVATED TEMPERATURES

Alloy	Form	Temper	Temp. °F	Creep Strength - ksi							
				Hours for 0.5% Total Extension				Hours for 1% Total Extension			
				1	10	100	500	1	10	100	500
AZ31B	Extrusions	-F	200	16	14	12	10	19	17	15	13
			250	13	11	9	7	16	14	11	9
			300	11	8	5.0	4.0	12	10	7	5.0
			350	8	6	-	-	10	8	-	-
AZ61A		-F	200	20	18	14	12	25	22	19	16
			250	16	13	8	6	20	16	11	8
			300	12	8	3.0	1.0	16	11	5.0	2.0
			350	9	4.0	-	-	11	6	-	-
ZK60A		-T5	200	15	10	6	4.0	20	14	8	6
			250	10	6	2.5	1.5	14	8	4	2.0
			300	5.0	2.5	1.0	-	8	4	1.5	-
AZ31B	Sheet	-H24	200	15	11	7	5.0	19	14	10	7
			250	9	6	4.0	2.0	13	8	5.0	3.0
			300	6	3.0	1.5	1.0	8	5.0	2.0	1.0
			350	4.0	1.5	-	-	6	2.0	-	-
AZ31B		-O	200	15	14	12	10	16	15	13	11
			250	12	11	9	7	14	12	10	9
			300	10	7	4.0	3.0	12	9	5.0	3.5
			350	7	4.0	-	-	8	5.0	-	-
HK31A		-T5	300	-	-	20	-	-	-	-	-
			400	16	14	10	-	18	16	11	-
HM21A		-T5	300	-	-	16	-	-	-	-	-
			400	17	16	14	-	18	17	14	-
			500	14	12	9	-	15	13	7	-
			600	9	7	6	-	10	8	-	-
			700	5	4	4	-	5	4	-	-

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TABLE 36. EFFECT OF EXPOSURE TO ELEVATED TEMPERATURES ON COMPRESSIVE YIELD STRENGTH (Tested at Room Temperature)

Alloy	Form	Temper	Exposure Temp. °F	Compressive Yield Strength After Exposure - ksi					
				Exposure Time - hrs					
				0	100	500	1000	2500	5000
AZ31B	Sheet	-H24	200	24	23	25	24	-	-
			250	24	24	25	24	-	-
			300	24	23	23	21	-	-
			400	24	18	19	18	-	-
			500	24	17	18	17	-	-
			600	24	16	16	-	-	-
HK31A	Sheet	-H24	300	25	-	-	-	-	25
			400	25	-	-	-	-	25
			500	25	-	24	22	-	21
			600	25	19	18	18	-	16
ZK60A	Extrusions	-T5	200	34	34	36	35	-	-
			250	34	35	36	35	-	-
			300	34	34	33	32	-	-
			400	34	29	30	29	-	-
			500	34	29	30	29	-	-

TABLE 37. EFFECT OF EXPOSURE TO ELEVATED TEMPERATURES ON TENSILE YIELD STRENGTH (Tested at Room Temperature)

Alloy	Form	Temper	Exposure Temp °F	Tensile Yield Strength After Exposure - ksi				
				Exposure Time - hrs				
				0	100	500	2500	5000
AZ31B	Sheet	-H24	200	32	31	32	-	-
			250	32	31	31	-	-
			300	32	29	29	-	-
			400	32	25	25	-	-
			500	32	23	24	-	-
			600	32	22	22	-	-
AZ63A	Cast	-T6	200	19	20	21	21	-
			300	19	21	21	21	21
			400	19	20	18	20	18
AZ92A	Cast	-T6	200	24	24	24	23	-
			300	24	23	24	24	-
			400	24	24	22	20	20
EK30A	Cast	-T6	400	15	15	13	12	13
			500	16	13	12	-	13
			600	16	13	12	-	12
HK31A	Sheet	-H24	300	29	-	-	-	29
			400	29	-	-	-	29
			500	29	-	24	-	28
			600	29	27	26	-	23
ZK60A	Extrusion	-T5	200	41	38	40	-	-
			250	41	41	41	-	-
			300	41	39	38	-	-
			400	41	34	32	-	-
			500	41	33	31	-	-

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TABLE 38. EFFECT OF EXPOSURE TO ELEVATED TEMPERATURES ON TENSILE STRENGTH (Tested at Room Temperature)

Alloy	Form	Temper	Exposure Temp °F	Tensile Strength After Expsoure - ksi				
				Exposure Time - Hrs				
				0	100	500	2500	5000
AZ31B	Sheet	-H24	200	41	41	41	-	-
			250	41	41	41	-	-
			300	41	40	39	-	-
			400	41	38	37	-	-
			500	41	37	37	-	-
			600	41	36	36	-	-
AZ63A	Cast	-T6	200	36	40	40	40	-
			300	36	38	40	37	38
			400	36	39	37	37	36
AZ92A	Cast	-T6	200	40	39	41	42	-
			300	40	39	39	39	39
			400	40	40	40	36	35
EK30A	Cast	-T6	400	20	20	19	-	19
			500	23	20	20	-	20
			600	23	19	19	-	19
HK31A	Sheet	-H24	300	37	-	-	-	37
			400	37	-	-	-	37
			500	37	-	37	-	36
			600	37	36	35	-	34
ZK60A	Extrusion	-T5	200	52	50	51	-	-
			250	52	50	52	-	-
			300	52	50	49	-	-
			400	52	46	45	-	-
			500	52	47	46	-	-

TABLE 39. EFFECT OF EXPOSURE AT A GIVEN TEMPERATURE ON TENSILE PROPERTIES OF AZ92A-T6 — SEPARATELY CAST TEST BARS

Property	Temperature °F		Exposure Time - Hrs						
	Exposure	Testing	0	25	100	500	1000	2500	5000
TS-ksi	200	70	40.4	40.9	38.7	41.0	40.7	41.5	-
TYS-ksi	200	70	24.0	24.9	23.5	23.6	22.1	23.2	-
Elong. - %	200	70	1.5	2.3	1.7	2.0	3.4	3.8	-
TS-ksi	200	200	40.1	40.5	39.9	-	38.6	40.0	-
TYS-ksi	200	200	20.0	19.3	19.7	-	18.5	20.9	-
Elong. -%	200	200	9.6	7.1	15.0	-	9.0	10.2	-
TS-ksi	300	70	40.4	38.0	39.1	39.1	39.1	39.4	38.7
TYS-ksi	300	70	24.0	25.2	22.8	24.4	20.8	23.7	24.0
Elong. -%	300	70	1.5	1.3	2.0	1.5	3.7	1.7	1.6
TS-ksi	300	300	28.3	27.5	26.3	-	27.2	24.8	27.0
TYS-ksi	300	300	16.9	15.8	15.8	-	15.9	14.6	14.7
Elong. -%	300	300	39.8	41.9	39.6	-	45.6	34.6	41.5
TS-ksi	400	70	40.4	40.0	39.7	40.1	39.2	35.8	35.4
TYS-ksi	400	70	24.0	25.1	23.8	22.3	22.9	19.9	19.7
Elong. -%	400	70	1.5	1.3	1.8	2.4	2.5	2.8	3.1
TS-ksi	400	400	19.6	16.9	17.2	-	17.2	17.9	17.5
TYS-ksi	400	400	12.5	11.4	11.9	-	11.3	11.2	10.2
Elong. -%	400	400	47.5	36.9	30.5	-	36.8	48.5	47.4

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TABLE 40. EFFECT OF EXPOSURE AT A GIVEN TEMPERATURE ON TENSILE PROPERTIES OF EK30A-T6 — SEPARATELY CAST TEST BARS

Property	Temperature °F		Exposure Time - Hrs					
	Exposure	Testing	0	25	100	500	1000	5000
TS-ksi	400	70	20.4	20.0	20.1	19.4	19.7	18.8
TYS-ksi	400	70	15.1	14.0	14.6	13.0	12.5	12.7
Elong. - %	400	70	1.7	1.5	2.0	2.0	-	-
TS-ksi	400	400	17.8	17.2	16.6	16.0	15.6	13.7
TYS-ksi	400	400	12.3	12.4	11.7	10.8	9.9	8.8
Elong. -%	400	400	-	10.5	15.5	27.0	-	-
TS-ksi	500	70	22.9	20.3	20.5	20.3	19.4	19.7
TYS-ksi	500	70	16.5	14.6	13.3	12.2	13.0	13.1
Elong. - %	500	70	-	2.7	3.5	3.0	2.7	2.8
TS-ksi	500	500	17.5	15.1	13.5	12.4	12.1	11.5
TYS-ksi	500	500	12.1	8.3	6.7	5.8	5.9	5.4
Elong. - %	500	500	-	38.5	46.2	50.2	38.0	43.8
TS-ksi	600	70	22.9	20.1	19.4	18.9	19.3	19.0
TYS-ksi	600	70	16.5	13.5	13.0	12.3	12.3	12.1
Elong. - %	600	70	-	3.0	3.5	3.0	3.2	3.2
TS-ksi	600	600	11.8	9.2	8.6	8.2	8.0	8.0
TYS-ksi	600	600	7.7	4.7	4.4	4.2	3.9	3.8
Elong. - %	600	600	-	79.5	115.5	84.0	110.5	106.0

TABLE 41. EFFECT OF EXPOSURE AT A GIVEN TEMPERATURE ON TENSILE PROPERTIES OF AZ63A-T6 — SEPARATELY CAST TEST BARS

Property	Temperature °F		Exposure Time - Hrs						
	Exposure	Testing	0	25	100	500	1000	2500	5000
TS-ksi	200	70	36.0	39.9	39.6	39.7	37.3	39.8	-
TYS-ksi	200	70	18.9	22.3	20.2	20.6	20.1	20.5	-
Elong. - %	200	70	3.5	4.8	5.5	5.4	4.2	5.6	-
TS-ksi	200	200	38.1	36.0	36.4	-	38.0	36.6	-
TYS-ksi	200	200	17.0	18.3	18.0	-	18.1	17.2	-
Elong. - %	200	200	15.8	10.7	17.2	-	16.1	15.5	-
TS-ksi	300	70	36.0	39.1	37.9	39.7	39.1	37.3	38.0
TYS-ksi	300	70	18.9	21.1	20.9	21.0	20.8	21.1	21.4
Elong. - %	300	70	3.5	4.3	3.8	4.0	3.7	3.7	3.7
TS-ksi	300	300	27.1	24.8	24.6	-	25.7	24.0	24.6
TYS-ksi	300	300	15.7	14.9	15.0	-	14.9	15.0	14.6
Elong. - %	300	300	35.7	39.0	35.1	-	34.3	33.8	39.1
TS-ksi	400	70	36.0	39.9	38.5	37.0	37.3	37.0	36.2
TYS-ksi	400	70	18.9	18.7	19.9	17.8	20.1	17.9	17.8
Elong. - %	400	70	3.5	6.1	5.2	4.8	4.2	5.2	5.2
TS-ksi	400	400	17.9	16.2	16.2	-	16.4	16.4	17.4
TYS-ksi	400	400	11.1	10.7	10.6	-	10.3	10.2	10.1
Elong. - %	400	400	34.5	36.2	35.9	-	38.8	36.7	37.0

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TABLE 42. EFFECT OF EXPOSURE AT A GIVEN TEMPERATURE ON TENSILE PROPERTIES OF EZ33A-T5 – SEPARATELY CAST TEST BARS

Property	Temperature °F		Exposure Time - Hrs					
	Exposure	Testing	0	312	500	1000	2500	5000
TS-ksi	400	400	21.0	-	19.1	18.9	-	-
TYS-ksi	400	400	11.2	-	13.1	14.4	-	-
TS-ksi	500	500	18.3	-	14.1	14.0	-	-
TYS-ksi	500	500	10.0	-	8.6	8.5	-	-
TS-ksi	600	600	11.1	-	11.1	11.1	-	-
TYS-ksi	600	600	7.4	-	7.4	7.4	-	-

TABLE 43. EFFECT OF EXPOSURE AT A GIVEN TEMPERATURE ON TENSILE PROPERTIES OF HK31A-T6 – SEPARATELY CAST TEST BARS

Property	Temperature °F		Exposure Time - Hrs					
	Exposure	Testing	0	25	100	500	1000	5000
TS-ksi	400	70	33.0	34.6	34.7	34.8	34.8	-
TYS-ksi	400	70	16.0	18.2	18.3	18.3	18.3	-
TS-ksi	400	400	22.0	24.9	25.3	25.8	26.0	-
TYS-ksi	400	400	14.0	14.1	14.3	14.7	14.8	-
TS-ksi	400	600	19.0	20.3	20.3	20.2	20.2	-
TYS-ksi	400	600	12.0	12.3	12.5	12.8	12.9	-
TS-ksi	500	70	33.0	34.9	35.0	34.0	33.0	-
TYS-ksi	500	70	16.0	18.0	18.0	17.6	17.0	-
TS-ksi	500	500	24.0	23.2	23.0	22.0	21.0	-
TYS-ksi	500	500	14.0	14.0	14.0	12.3	11.0	-
TS-ksi	500	600	19.0	19.7	20.0	19.0	18.0	-
TYS-ksi	500	600	12.0	12.0	12.0	11.5	11.0	-
TS-ksi	600	70	33.0	32.0	30.0	26.0	26.0	-
TYS-ksi	600	70	16.0	15.8	14.0	11.0	11.0	-
TS-ksi	600	600	19.0	17.5	15.0	10.0	9.0	-
TYS-ksi	600	600	12.0	10.5	8.0	4.0	3.0	-

TABLE 44. EFFECT OF EXPOSURE AT A GIVEN TEMPERATURE ON TENSILE PROPERTIES OF HZ32A-T5 – SEPARATELY CAST TEST BARS

Property	Temperature °F		Exposure Time - Hrs					
	Exposure	Testing	0	25	100	500	1000	5000
TS-ksi	400	70	-	-	-	-	31.5	-
TYS-ksi	400	70	-	-	-	-	15.6	-
TS-ksi	400	400	16.5	16.7	16.7	16.8	16.8	-
TYS-ksi	400	400	10.0	10.0	10.0	10.0	10.0	-
TS-ksi	500	70	-	-	-	-	32.6	-
TYS-ksi	500	70	-	-	-	-	15.2	-
TS-ksi	500	500	13.0	13.2	13.3	13.4	13.5	-
TYS-ksi	500	500	7.7	8.3	8.5	8.7	8.8	-
TS-ksi	600	70	-	-	-	-	33.0	-
TYS-ksi	600	70	-	-	-	-	15.0	-
TS-ksi	600	600	11.5	11.5	11.5	11.5	11.5	-
TYS-ksi	600	600	7.4	7.55	7.6	7.7	7.7	-

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TABLE 45. MECHANICAL PROPERTIES OF WROUGHT MAGNESIUM ALLOYS AT ROOM AND ELEVATED TEMPERATURES AFTER EXPOSURE AT ELEVATED TEMPERATURE

Alloy	Temper	Temperature °F		Exposure Time - Hrs	Tensile Strength ksi	Tensile Yield Strength ksi	Elong. %	Compressive Yield Strength ksi
		Exposure	Testing					
				Sheet				
HK31A	-H24	-	70	0	37	29	8	25
		300	70	5000	37	29	8	25
		400	70	5000	37	29	8	25
		500	70	500	37	29	11	24
		-	70	1000	37	29	15	22
		-	70	5000	36	28	16	21
		600	70	1/2	37	29	9	25
		-	70	1	36	29	11	24
		-	70	100	36	27	20	19
		-	70	500	35	26	21	18
		-	70	1000	35	25	21	18
		-	70	5000	34	23	20	16
		300	300	0	26	23	20	23
		-	300	500	26	23	20	23
		400	400	0	24	21	21	22
		-	400	500	21	19	31	22
		-	400	1000	20	18	35	20
		-	400	5000	24	20	20	-
		-	70	0	34	25	10	15
HM21A	-T8	300	70	5000	34	25	10	15
		400	70	5000	34	25	10	15
		500	70	5000	34	25	10	15
		600	70	5000	34	25	10	15
		700	70	100	32	23	12	-
		800	70	100	30	18	15	-
		300	300	0	21	19	20	-
		-	-	5000	21	19	20	-
		400	400	0	18	17	30	-
		-	-	5000	18	17	30	-
		500	500	0	16	15	25	-
		-	-	5000	16	15	25	-
		600	600	0	14	12	15	-
		-	-	5000	14	12	15	-
		700	700	0	11	8	50	-
		-	-	100	11	8	50	-
		-	-	1000	7	5	112	-
		-	-	2000	7	5	55	-
		800	800	0	5	3	100	-
		-	-	100	5	3	100	-
		900	900	0	2	1	>100	-
		-	-	100	2	1	>100	-

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TABLE 46. MECHANICAL PROPERTIES OF WROUGHT MAGNESIUM ALLOYS AT ROOM TEMPERATURES AFTER EXPOSURE AT ELEVATED TEMPERATURE

Alloy	Temper	Property	Temperature °F		Exposure Time - Hrs					
			Exposure	Testing	0	16	48	192	500	1000
Sheet										
AZ31B	-H24	TS-ksi	200	70	40.7	40.5	40.7	41.0	40.8	40.8
		TYS-ksi			32.2	32.0	31.4	30.0	31.8	32.9
		Elong. - %			13.8	13.7	14.5	12.5	11.0	13.0
		CYS-ksi			24.1	24.0	23.1	24.0	24.9	23.6
		TS-ksi	250	70	40.7	40.3	40.9	40.6	40.5	40.4
		TYS-ksi			32.2	31.9	31.8	29.5	30.8	31.7
		Elong. - %			13.8	11.2	14.2	13.7	13.2	17.0
		CYS-ksi			24.1	24.3	23.8	23.4	24.7	23.5
		TS-ksi	300	70	40.7	39.9	39.9	39.6	39.4	39.3
		TYS-ksi			32.2	30.9	29.2	28.4	28.7	29.6
		Elong. - %			13.8	15.2	16.0	18.5	17.5	20.0
		CYS-ksi			24.1	23.1	23.7	21.6	22.6	21.1
		TS-ksi	400	70	40.7	37.2	37.9	38.1	37.4	37.4
		TYS-ksi			32.2	25.4	24.9	25.0	25.0	25.3
		Elong. - %			13.8	21.5	21.2	22.5	19.0	21.2
		CYS-ksi			24.1	19.2	18.6	17.2	18.8	17.9
		TS-ksi	500	70	40.7	36.6	37.0	36.0	37.0	38.0
		TYS-ksi			32.2	23.7	22.6	22.5	23.8	23.9
		Elong. - %			13.8	21.2	22.5	22.2	21.2	22.5
		CYS-ksi			24.1	17.0	18.0	16.4	17.5	16.7
		TS-ksi	600	70	40.7	35.7	-	36.2	36.5	-
		TYS-ksi			32.2	22.5	-	22.3	22.2	-
		Elong. - %			13.8	18.5	-	23.0	21.5	-
		CYS-ksi			24.1	16.9	-	15.7	16.0	-
Extrusions										
ZK60A	-T5	TS-ksi	200	70	51.8	50.0	50.1	50.3	51.3	52.1
		TYS-ksi			40.9	39.8	40.9	35.3	40.2	43.8
		Elong. - %			8.8	10.0	11.2	11.0	9.2	10.0
		CYS-ksi			34.4	34.4	35.1	33.8	36.1	34.8
		TS-ksi	250	70	51.8	49.7	50.6	50.3	52.5	51.8
		TYS-ksi			40.9	39.2	42.4	39.8	40.7	42.4
		Elong. - %			8.8	10.0	10.5	10.5	8.0	9.5
		CYS-ksi			34.4	34.5	35.1	35.3	35.8	35.2
		TS-ksi	300	70	51.8	50.3	50.2	49.0	48.8	47.1
		TYS-ksi			40.9	40.6	41.9	34.6	38.1	37.0
		Elong. - %			8.8	10.2	10.0	12.0	10.2	12.0
		CYS-ksi			34.4	34.7	34.3	33.1	33.1	31.9
		TS-ksi	400	70	51.8	46.0	45.9	45.3	45.3	45.6
		TYS-ksi			40.9	34.7	34.8	31.1	32.4	32.9
		Elong. - %			8.8	12.5	13.7	14.0	12.7	12.7
		CYS-ksi			34.4	29.9	28.8	29.2	30.4	28.9
		TS-ksi	500	70	51.8	45.6	47.2	45.7	45.9	46.4
		TYS-ksi			40.9	31.8	33.7	31.1	31.3	32.7
		Elong. - %			8.8	14.5	14.5	15.2	14.5	13.7
		CYS-ksi			34.4	29.1	29.8	28.1	29.5	28.9

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TABLE 47. PROPERTIES AND CHARACTERISTICS OF MAGNESIUM CASTING ALLOYS — ALLOY COMPOSITION AND SPECIFICATIONS

Alloy	Nominal Composition,[1] %							ASTM Alloy Specification	Similar Specifications For Comparative Purposes		
	Al	Zn	Th	R. E.[3]	Zr	Ag	Mn		Federal or Military	SAE	AMS[2]
Sand Cast											
AM100A	10.0	-	-	-	-	-	0.10 min	B80	-	-	-
AZ63A	6.0	3.0	-	-	-	-	0.15 min	B80	QQ-M-56	50	4420G, 4422H, 4424F
AZ81A	7.6	0.7	-	-	-	-	0.13 min	B80	QQ-M-56	-	-
AZ91C	8.7	0.7	-	-	-	-	0.13 min	B80	QQ-M-56	504	4437
AZ92A	9.0	2.0	-	-	-	-	0.10 min	B80	QQ-M-56	500	4434F
EK30A	-	-	-	3.0	0.20 min	-	-	B80	-	-	-
EK41A	-	-	-	4.0	0.6	-	-	B80	-	-	4440A, 4441A
EZ33A	-	2.7	-	3.0	0.7	-	-	B80	QQ-M-56	-	4442A
HK31A	-	0.30 max.	3.2	-	0.7	-	-	B80	QQ-M-56	-	4445A
HZ32A	-	2.1	3.2	0.10 max.	0.7	-	-	B80	QQ-M-56	-	4447
K1A	-	-	-	-	0.7	-	-	B80	-	-	-
QE22A	-	-	-	2.1	0.7	2.5	-	B80	-	-	-
ZE41A	-	4.2	-	1.25	0.7	-	0.15 max.	B80	-	-	-
ZH62A	-	5.7	1.8	-	0.7	-	-	B80	QQ-M-56	-	-
ZK51A	-	4.6	-	-	0.7	-	-	B80	QQ-M-56	-	4443A
ZK61A	-	6.0	-	-	0.7	-	-	B80	QQ-M-56	-	4444
Permanent Mold											
AM100A	10.0	-	-	-	-	-	0.10 min	B199	QQ-M-55	502	4483
AZ63A	6.0	3.0	-	-	-	-	0.15 min	-	QQ-M-55	-	-
AZ81A	7.6	0.7	-	-	-	-	0.13 min	-	QQ-M-55	-	-
AZ91C	8.7	0.7	-	-	-	-	0.13 min	B199	QQ-M-55	-	-
AZ92A	9.0	2.0	-	-	-	-	0.10 min	B199	QQ-M-55	503	4484E
EK41A	-	0.30 max.	-	4.0	0.6	-	-	B199	-	-	-
EZ33A	-	2.7	-	3.0	0.7	-	-	B199	QQ-M-55	-	-
Die Casting											
AZ91A	9.0	0.7	-	-	-	-	0.13 min	B94	QQ-M-38	501	4490D
AZ91B	9.0	0.7	-	-	-	-	0.13 min	B94	-	501A	-
Investment Casting											
AM100A	10.0	-	-	-	-	-	0.10 min	-	-	-	-
AZ92A	9.0	2.0	-	-	-	-	0.10 min	-	-	-	4453

NOTES: 1. For specific compositions refer to applicable specifications.

2. Aerospace Materials Specifications.

3. R. E. stands for total rare earth content -- usually signifying mischmetal content. In QEZ2A it stands for didymium -- essentially 85% neodymium and 15% praseodymium.

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TABLE 48. PROPERTIES AND CHARACTERISTICS OF MAGNESIUM CASTING ALLOYS – OUTSTANDING CHARACTERISTICS

Alloy	Typical Green Sand Pattern Shrinkage Allowance[1] In. /Ft.	Melting Range Approx °F			Foundry Characteristics[3]					Other Characteristics				
		Non-Equilibrium Solidus[2]	Solidus	Liquidus	Resistance to Hot Cracking[4]	Pressure Tightness	Fluidity[5]	Microporosity Tendency[6]	Normally Heat Treated	Machining[7]	Electroplating[8]	Surface Treatment[9]	Suitability to Brazing[10]	Suitability to Welding[11]
Sand Cast														
AM100A	9/64	810	867	1100	1	2	1	2	Yes	1	2	2	No	1
AZ81A	9/64	790	882	1115	2	2	1	2	Yes	1	2	2	No	1
AZ63A	9/64	685	850	1130	2	3	1	3	Yes	1	1	1	No	3
AZ91C	9/64	785	875	1105	2	3	1	3	Yes	1	2	2	No	1
AZ92A	9/64	770	830	1100	2	3	1	3	Yes	1	2	2	No	2
EK30A	3/16	-	1100	1184	2	2	2	2	Yes	1	2	2	*	2
EK41A	3/16	-	1095	1193	2	1	2	2	Yes	1	2	1	*	2
EZ33A	5/32	-	1010	1189	2	2	2	1	Yes	1	1	1	*	1
HK31A	3/16	-	1092	1204	2	2	3	2	Yes	1	1	2	*	1
HZ32A	3/16	-	1026	1198	2	2	3	2	Yes	1	2	1	*	3
K1A	3/16	-	-	1205	4	2	2	2	No	1	3	2	*	1
QE22A	5/32	-	1020	1190	2	2	2	2	Yes	1	*	2	*	*
ZE41A	5/32	-	975	1190	*	8	2	*	Yes	1	1	1	*	3
ZH62A	5/32	-	960	1169	2	3	3	3	Yes	1	1	1	*	4
ZK51A	5/32	-	1020	1186	4	4	2	4	Yes	1	2	1	*	4
ZK61A	5/32	-	985	1175	4	4	2	4	Yes	1	2	1	*	4
Permanent Mold														
AM100A		810	867	1100	1	2	1	2	Yes	1	1	2	No	1
AZ63A		685	850	1130	3	4	1	4	Yes	1	1	1	No	3
AZ81A		790	882	1115	2	3	1	3	Yes	1	2	2	No	1
AZ91C		785	875	1105	2	3	1	3	Yes	1	1	2	No	1
AZ92A		770	830	1100	2	3	1	3	Yes	1	1	2	No	2
EK41A		-	1095	1193	2	2	2	2	Yes	1	1	1	*	2
EZ33A		-	1010	1189	2	1	2	1	Yes	1	1	1	*	1
Die Casting														
AZ91A		785	875	1105	2	1	2	1	No	1	2	2	No	No
AZ91B		785	875	1105	2	1	2	1	No	1	2	2	No	No
Investment Casting														
AM100A		810	867	1100	*	*	*	*	Yes	1	*	*	No	No
AZ92A		770	830	1100	*	*	*	*	Yes	1	*	*	No	No

NOTES:
1. Allowance for average castings. Shrinkage requirements will vary with intricacy of design and dimensions and with mold materials.
2. As measured on metal solidified under normal casting conditions.
3. Rating of 1 indicates best of group. 3 or 4 indicates poorest of group.
4. Ability of alloy to withstand contraction stresses while cooling through hot-short or brittle temperature range.
5. Ability of liquid alloy to flow readily in mold and fill thin sections.
6. Based on radiographic evidence.
7. Composite rating based on ease of cutting, chip characteristics, quality of finish, and tool life. Ratings, in the heat-treatable alloys, based on -T6 type temper. Other tempers, particularly the annealed temper, may have lower rating.
8. Ability of casting to take and hold an electroplate applied by present standard methods.
9. Ability of castings to be cleaned in standard pickle solution and to be conditioned for best paint adhesion.
10. Refers to suitability of alloy to withstand brazing temperature without excessive distortion or melting.
11. Based on ability of material to be fusion-welded with filler rod of same alloy or of an alloy whose composition is recommended.

* Inexperience with these alloys under wide production conditions makes it undesirable to supply ratings at this time.

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TABLE 49. PROPERTIES AND CHARACTERISTICS OF MAGNESIUM CASTING ALLOYS – MECHANICAL PROPERTIES

A.S.T.M. Alloy Specification	Specified Minimum Properties (1)				Typical Properties (2)				
					Tensile Strength		Yield Strength (0.2 percent offset), ksi		Brinell Hardness 500 Kg - 10 mm
Sand Cast Alloys B80-70	Temper	Yield Strength ksi	Tensile Strength ksi	Elong. (2 In.) %	KSI	min (3)	avg (4)	min (5)	
AM100A	-T6	17	35	Not Req.	-	-	-	-	-
AZ63A	-F	10	26	4	-	-	-	-	50
	-T4	11	34	7	25.5	17.0	10.0	9.0	55
	-T5	11	26	2					55
	-T6	16	34	3	25.5	17.0	14.5	12.0	73
AZ81A	-T4	10	34	7	25.5	17.0	10.0	9.0	55
AZ91C	-F	10	23	Not Req.					60
	-T4	11	34	7	25.5	17.0	10.0	9.0	55
	-T5	11	23	2					62
	-T6	16	34	3	25.5	17.0	14.5	12.0	70
AZ92A	-F	10	23	Not Req.					65
	-T4	11	34	6	25.5	17.0	10.0	9.0	63
	-T5	11	23	Not Req.					69
	-T6	18	34	1	25.5	17.0	16.0	13.5	81
EZ33A	-T5	14	20	2	15.0	13.0	12.5	11.0	50
HK31A	-T6	13	27	4	23.0	19.0	11.7	10.5	66
HZ32A	-T5	13	27	4	23.0	19.0	11.7	10.5	55
K1A	-F	6	24	14	-	-	-	-	-
QE22A[2]	-T6	25	35	2	32.0	28.0	23.0	20.0	78
ZE41A	-T5	19	29	2.5	28.0	26.0	19.5	17.5	62
ZH62A	-T5	22	35	4	31.5	26.5	17.5	15.5	70
ZK51A	-T5	20	34	5	29.0	24.0	17.0	14.0	65
ZK61A	-T6	25	40	5	34.0	30.0	24.0	21.0	70

NOTES:
(1) Separately cast test bar
(2) *Specimens cut from castings*
(3) The average value of all specimens tested from each casting shall conform to the minimum average. Not less than three specimens shall be tested from each casting: one from the thinnest section, one from the thickest section, and one from an intermediate section.
(4) Any specimen cut from a casting shall conform to the minimum value specified.
(5) Where a value is not listed the minimum average percent elongation of specimens cut from a casting shall not be less than 25 percent of the value specified for separately cast test bars.

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TABLE 49 (Continued). PROPERTIES AND CHARACTERISTICS OF MAGNESIUM CASTING ALLOYS – MECHANICAL PROPERTIES OF SEPARATELY CAST TEST BARS

A.S.T.M. Alloy Specification		Specified Minimum Properties			Typical Properties					
Permanent Mold Cast Alloys B199-59	Temper	Yield Strength ksi	Tensile Strength ksi	Elong. (2 In.) %	Yield Strength ksi	Tensile Strength ksi	Elong. (2 In.) %	Brinell Hardness 500 Kg - 10 mm	Shear Strength ksi	Endurance Limit -ksi 10^8 Cycles R. R. Moore
AM100A	-F	10	20	Not Req.	12	22	2	53	18	10
	-T4	10	34	6	13	40	10	52	20	11
	-T6	15	34	2	16	40	4	60	21	10
	-T61	17	34	Not Req.	22	40	1	70	22	10
AZ81A	-T4	11	34	Not Req.	-	-	-	-	-	-
AZ91C	-F	11	23	Not Req.	14	24	2	52	18	-
	-T4	11	34	7	12	40	14	53	17	12
	-T5	12	23	2	17	26	3	-	-	-
	-T6	16	34	3	19	40	5	66	20	12
AZ92A	-F	11	23	Not Req.	14	24	2	65	18	12
	-T4	11	34	6	14	40	10	63	20	14
	-T5	12	23	Not Req.	16	25	1	69	19	11
	-T6	18	34	Not Req.	21	40	2	80	22	12
EZ33A	-T5	14	20	2	15	23	3	50	-	10
Die Cast Alloys B94-58										
AZ91A	-F	Not Req.	Not Req.	Not Req.	23	34	3	60	20	14
AZ91B	-F	Not Req.	Not Req.	Not Req.	23	34	3	60	-	-
Investment Casting Alloys										
AM100[6]	-T6	15	34	1	-	-	-	-	-	-
AZ92A[7]	-T6	20	37	1	-	-	-	-	-	-

NOTES: (6) Minimum properties from AMS.
(7) Minimum properties from AMS4453.

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TABLE 50. PROPERTIES AND CHARACTERISTICS OF MAGNESIUM CASTING ALLOYS – PHYSICAL PROPERTIES

Alloys	Specific Gravity 75°F	Density Lbs./cu. in.	Thermal Conductivity at 68°F C.G.S. Units	Coefficient of Thermal Expansion Per °F x 10-6 68° - 212°F	Temper	Electrical Conductivity % I.A.C.S.
			Sand Cast			
AM100A	1.81	0.065	0.12	14.5	-F	12
			0.10	-	-T4	10
			0.14	-	-T6	14
			0.15	-	-T61	15
AZ63A	1.82	0.066	0.14	14.5	-F	14
			0.12	-	-T4	12
			0.15	-	-T6	15
AZ81A	1.80	0.065	0.12	14.5	-T4	12
AZ91C	1.81	0.065	0.13	14.5	-F	13
			0.11	14.5	-T4	11
AZ92A	1.83	0.066	0.12	14.5	-F	12
			0.11	-	-T4	10
			0.14	-	-T6	14
EK30A	1.79	0.065	0.26	14.5	-T6	27
EK41A	1.81	0.065	0.23	14.5	-T6	24
EZ33A	1.83	0.066	0.24	14.5	-T5	25
HK31A	1.79	0.065	0.22	14.5	-T6	22
HZ32A	1.83	0.066	0.26	14.5	-T5	27
K1A	1.75	0.063	0.30	14.5	-F	31
QE22A	1.82	0.066	0.25	14.5	-T6	25
ZE41A	1.84	0.066	0.30	14.5	-T5	31
ZH62A	1.86	0.067	0.26	14.5	-T5	27
ZK51A	1.81	0.065	0.26	14.5	-T5	27
ZK61A	1.83	0.066	0.26	14.5	-T6	27
			Permanent Mold			
AM100A	1.81	0.065	0.12	14.5	-F	12
			0.10	-	-T4	10
			0.15	-	-T6	15
AZ63A	1.82	0.066	0.14	14.5	-F	14
			0.12	-	-T4	12
			0.16	-	-T5	16
			0.15	-	-T6	15
AZ81A	1.80	0.065	0.12	14.5	-T4	12
AZ91C	1.81	0.065	0.13	14.5	-F	13
			0.11	-	-T4	11
			-	-	-T5	-
			0.13	-	-T6	13
AZ92A	1.83	0.066	0.12	14.5	-F	12
			0.11	-	-T4	10
			0.14	-	-T6	14
EK41A	1.81	0.065	0.24	14.5	-T5	25
			0.23	-	-T6	24
EZ33A	1.83	0.066	0.24	14.5	-T5	25
			Die Casting			
AZ91A	1.81	0.065	0.13	14.5	-F	13
AZ91B	1.81	0.065	0.13	14.5	-F	13
			Investment Casting			
AM100A	1.81	0.065	0.14	14.5	-T6	14
AZ92A	1.83	0.066	0.14	14.5	-T6	14

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TABLE 51. ALLOY COMPOSITION AND SPECIFICATIONS OF WROUGHT MAGNESIUM ALLOYS

Alloy	Nominal Composition, %							ASTM Spec	Federal or Military Spec.	AMS Spec	SAE No.
	Al	Mn	Zn	RE	Sn	Th	Zr				
Extruded Bar, Rod and Special Shaped Sections											
AZ31B	3.0	0.20	1.0	-	-	-	-	B107-59	QQ-M-31	-	510
AZ61A	6.5	0.15	1.0	-	-	-	-	B107-59	QQ-M-31	4350	520
AZ80A	8.5	0.12	0.5	-	-	-	-	B107-59	QQ-M-31	-	523
HM31A	-	1.2	-	-	-	-	-	-	MIL-M-8916	4388, 4389	-
M1A	-	1.2	-	-	-	-	-	B107-59	QQ-M-31	-	522
ZK21A	-	-	2.3	-	-	-	0.6	-	MIL-M-46039	-	-
ZK60A	-	-	5.5	-	-	-	0.45	B107-59	QQ-M-31	4352	524
ZK60B	-	-	5.8	-	-	-	0.45	-	MIL-M-26696	-	-
Tube											
AZ31B	3.0	0.20	1.0	-	-	-	-	B217-59	WW-T-825	-	510
AZ61A	6.5	0.15	1.0	-	-	-	-	B217-59	WW-T-825	4350	520
M1A	-	1.2	-	-	-	-	-	B217-59	WW-T-825	-	522
ZK60A	-	-	5.5	-	-	-	0.45	B217-59	WW-T-825	4352	524
Forgings											
AZ31B	3.0	0.20	1.0	-	-	-	-	B91-59	QQ-M-40	-	-
AZ61A	6.5	0.15	1.0	-	-	-	-	B91-59	QQ-M-40	4358	520
AZ80A	8.5	0.12	0.5	-	-	-	-	B91-59	QQ-M-40	4360	523
M1A	-	1.2	-	-	-	-	-	-	QQ-M-40	-	522
TA54A	3.5	0.20	-	-	5.0	-	-	B91-59	QQ-M-40	-	53
ZK60A	-	-	5.5	-	-	-	0.45	B91-59	QQ-M-40	4362	-
Sheet and Plate											
AZ31B	3.0	0.20	1.0	-	-	-	-	B90-59	QQ-M-44	4375, 4376, 4377	510
HK31A	-	-	-	-	-	3.2	0.7	B90-59	MIL-M-26075	4384, 4385	-
HM21A	-	0.8	-	-	-	2.0	-	B90-59	MIL-M-8917	4390	-
M1A	-	1.2	-	-	-	-	-	-	QQ-M-54	-	522
ZE10A	-	-	1.2	0.17	-	-	-	B90-59	MIL-M-46037	-	534

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TABLE 52. MINIMUM TENSILE PROPERTIES OF WROUGHT MAGNESIUM ALLOYS – SHEET AND PLATE

Alloy	Temper	Thickness in.	Tensile Strength ksi	Tensile Yield Strength ksi	Elong. (2 In.) %
AZ31B	-O	0.016 to 0.500	*32	-	12
		0.501 to 2.000	*30	-	10
		2.001 to 3.000	*30	-	9
AZ31B	-H24	0.016 to 0.249	39	29	6
		0.250 to 0.374	38	26	8
		0.375 to 0.500	37	24	8
		0.501 to 1.000	36	22	8
		1.001 to 2.000	34	20	8
		2.001 to 3.000	34	18	8
AZ31B	-H26	0.250 to 0.374	39	27	6
		0.375 to 0.500	38	26	6
		0.501 to 0.750	37	25	6
		0.751 to 1.000	37	23	6
		1.000 to 1.500	35	22	6
		1.501 to 2.000	35	21	6
HK31A	-O	0.016 to 0.250	**30	18	12
		0.251 to 0.500	30	16	12
		0.501 to 1.000	30	15	12
		1.001 to 3.000	29	14	12
HK31A	-H24	0.016 to 0.125	34	26	4
		0.126 to 0.250	34	23	4
		0.251 to 0.500	34	21	4
		0.501 to 1.000	34	18	10
		1.001 to 2.000	33	15	10
		2.001 to 3.000	33	15	8
HM21A	-T8	0.016 to 0.250	31	18	4
		0.251 to 0.500	32	21	6
		0.501 to 1.000	30	19	6
		1.001 to 2.000	29	18	6
M1A	-O	0.020 to 0.249	**	-	12
M1A	-H24	0.020 to 0.249	32	22	3
ZE10A	-O	0.016 to 0.250	30	-	15
		0.251 to 0.500	29	-	12
ZE10A	-H24	0.016 to 0.125	36	-	4
		0.126 to 0.188	34	-	4
		0.189 to 0.250	31	-	4

NOTES: * Tensile strength shall be 40 ksi max.
** Tensile strength shall be 38 ksi max.

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TABLE 53. MINIMUM TENSILE PROPERTIES OF WROUGHT MAGNESIUM ALLOYS — BARS, ROD AND SPECIAL SHAPED SECTIONS

Alloy	Temper	Form	Diameter or Thickness, In.	Cross-Sectional Area, Sq. In.	Tensile Strength ksi	Tensile Yield Strength ksi	Elong. (2 In.) %
AZ31B	-F	Bars, Rods and Shapes	0.249 and under	All	35	21	7
			0.250 to 1.499	All	35	22	7
			1.500 to 2.499	All	34	22	7
			2.500 to 3.999	All	32	20	7
		Hollow Shapes	All	All	32	16	8
AZ61A	-F	Bars, Rods, and Shapes	0.249 and under	All	38	21	8
			0.250 to 2.499	All	39	24	9
			2.500 to 4.999	All	40	22	7
		Hollow Shapes	All	All	36	16	7
AZ80A	-F	Bars, Rods, and Shapes	0.249 and under	All	43	28	9
			0.250 to 1.499	All	43	28	8
			1.500 to 2.499	All	43	28	6
			2.500 to 4.999	All	42	27	4
AZ80A	-T51	Bars, Rods, and Shapes	0.249 and under	All	47	30	4
			0.250 to 2.499	All	48	33	4
			2.500 to 4.999	All	45	30	2
HM31A	-T5	Bars, Rods, and Shapes		Under 1.000	37	26	4
				1.000 - 3.999	37	26	4
M1A	-F	Bars, Rods, and Shapes	0.249 and under	All	30	-	2
			0.250 to 1.499	All	32	-	3
			1.500 to 2.499	All	32	-	2
			2.500 to 4.999	All	29	-	2
		Hollow Shapes	All	All	28	-	2
ZK21A	-F	Bars, Rods, and Shapes	-	Under 5.000	38	28	4
		Hollow Shapes	-	All	34	26	4
ZK60A	-F	Bars, Rods, and Shapes	All	4.999 & under	43	31	5
			All	5.000 to 39.999	43	31	4
		Hollow Shapes	All	All	40	28	5
ZK60A	-T5	Bars, Rods, and Shapes	All	4.999 & Under	45	36	4
		Hollow Shapes	All	All	46	38	4
ZK60B	-T5	Bars, Rods, and Shapes	Up to 2.999	Under 20.000	45	35	4

TABLE 54. MINIMIM TENSILE PROPERTIES OF WROUGHT MAGNESIUM ALLOYS — TUBE AND FORGINGS

Alloy	Temper	Outside Diameter in.	Wall Thickness in.	Tensile Strength ksi	Tensile Yield Strength ksi	Elongation in 2" %
			Tube			
AZ31B	-F	6.000 max.	0.028 - 0.250	32	16	8
		6.000 max.	0.251 - 0.750	32	16	4
AZ61A	-F	6.000 max.	0.028 - 0.750	36	16	7
M1A	-F	6.000 max.	0.028 - 0.750	28	-	2
ZK21A	-F	3.000 max.	all	34	26	4
ZK60A	-F	3.000 max.	0.028 - 0.250	40	28	5
ZK60A	-T5	Up thru 3.000	0.028 - 0.250	46	38	4
		3.001 - 8.500	0.094 - 1.188	44	33	4
			Forgings			
AZ31B	-F	-	-	34	19	6
AZ61A	-F	-	-	38	22	6
AZ80A	-F	-	-	42	26	5
AZ80A	-T5	-	-	42	28	2
M1A	-F	-	-	30	18	3
TA54A	-F	-	-	36	22	7
ZK60A	-T5	-	-	42	26	7

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TABLE 55. MECHANICAL PROPERTIES OF CAST MAGNESIUM ALLOYS

Govt Spec No.	Alloy	Temper	Minimum			Typical						
			US ksi	YS (0.2% offset) ksi	Elong %	CYS (0.2% offset) ksi	Elong %	SS ksi	Bearing Strength ksi		Brinell Hard. No. 500 kg 10 mm	Fatigue Str ksi
									US	YS		
Die Castings												
QQ-M-38	AZ91A	-	-	-	-	23	3	20	-	-	60	12
Permanent Mold Castings												
QQ-M-55	AM100A	-F	20	10	-	12	2	-	-	-	54	-
		-T4	34	10	6	13	10	-	-	-	52	-
		-T6	34	15	2	16	4	-	-	-	69	-
		-T61	34	17	-	22	1	-	-	-	-	-
	AZ81A	-T4	34	10	7	12	15	17	60	44	55	12
	AZ91C	-F	18	10	-	14	2	16	60	40	52	12
		-T4	34	10	7	12	14	17	60	44	53	13
		-T5	20	11	2	17	2	-	-	-	-	-
		-T6	34	16	3	19	5	19	75	52	66	11
	AZ92A	-F	20	10	-	14	2	16	50	46	65	13
		-T4	34	10	6	14	9	17	68	46	63	13
		-T5	20	11	-	16	2	16	50	46	80	-
		-T6	34	18	-	21	2	20	80	65	84	13
	EZ33A	-T5 Rm Temp	20	14	2	15	3	20	57	40	50	10
		-T5 500°F	13	8	-	10	31	14	38	28	-	-
	AZ63A	-F	24	10	4	14	6	16	60	40	50	10
		-T4	34	10	7	13	12	17	60	44	55	14
		-T5	24	11	2	14	4	17	60	40	55	-
		-T6	34	16	3	19	5	19	75	52	73	13
Sand Castings												
QQ-M-56	AZ63A	-F	24	10	4	14	6	16	60	40	50	10
		-T4	34	10	7	13	12	17	60	44	55	14
		-T5	24	11	2	14	4	17	60	40	55	-
		-T6	34	16	3	19	5	19	75	52	73	13
	AZ81A	-T4	34	10	7	12	15	17	60	44	55	12
	AZ91C	-F	18	10	-	14	2	16	60	44	52	12
		-T4	34	10	7	12	14	17	60	40	53	13
		-T5	20	11	2	17	2	-	-	-	-	-
		-T6	34	16	3	19	5	19	75	52	66	11
	AZ92A	-F	20	10	-	14	2	16	50	46	65	13
		-T4	34	10	6	14	9	17	68	46	63	13
		-T5	20	11	-	15	2	16	50	46	80	-
		-T6	34	18	-	21	2	20	80	65	84	13
	EZ33A	-T5 Rm Temp	20	14	2	15	3	20	57	40	50	10
		-T5 500°F	13	8	-	10	31	14	38	28	-	-
	HK31A	-T6 Rm Temp	27	13	4	15	8	21	61	40	55	10
		-T6 500°F	13	8	-	14	20	15	44	28	-	-
	HZ32A	-T5 Rm Temp	27	13	4	14	7	20	60	37	57	10
		-T5 500°F	13	8	-	9	40	12	39	23	-	-
	QE22A	-T6										
	ZK51A	-T5	34	20	5	24	8	22	72	47	65	9
	ZH62A	-T5	35	22	4	25	6	23	72	49	75	-

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TABLE 56. MECHANICAL PROPERTIES OF MAGNESIUM – ALLOY SHEET

Govt Spec No.	Alloy	Temper	Thickness	Minimum			Typical				
				US ksi	YS (0.2% offset) ksi	Elong %	CYS (0.2% offset) ksi	SS ksi	Bearing Strength ksi		Brinell Hard. No. 500 kg 10 mm
									US	YS	
QQ-M-44	AZ31B	-0	0.016-0.500	32	-	12	16	21	70	42	55
			0.501-2.000	30	-	10	14	-	-	-	-
		-H24	0.016-0.249	39	29	6	26	23	72	47	73
			0.250-0.374	38	26	8	23	-	-	-	-
			0.375-0.500	37	24	8	21	-	-	-	-
			0.501-1.000	36	22	8	16	-	-	-	-
			1.001-2.000	34	20	8	14	-	-	-	-
		-H26	0.250-0.374	39	27	6	24	-	-	-	-
			0.375-0.500	38	26	6	21	-	-	-	-
			0.501-0.750	37	25	6	19	-	-	-	-
			0.751-1.000	37	23	6	18	-	-	-	-
			1.001-1.500	35	22	6	16	-	-	-	-
			1.501-2.000	35	21	6	15	-	-	-	-
--	ZE10A	-0	0.016-0.060	30	23 Typ[1]	15	16	-	-	-	-
			0.061-0.250	30	20 Typ[1]	15	16	-	-	-	-
			0.251-0.500	29	16 Typ[1]	12	12	-	-	-	-
		-H24	0.016-0.125	36	28 Typ[1]	4	26	-	-	-	-
			0.126-0.188	34	25 Typ[1]	4	24	-	-	-	-
			0.189-0.250	31	19 Typ[1]	4	16	-	-	-	-
MIL-M-26075	HK31A	-H24	0.016-0.125	34 Min	26 Min	4	23	27	61	41	-
			0.126-0.250	34 Min	23 Min	4	23	-	-	-	-
			0.251-0.500	34 Min	21 Min	4	21	-	-	-	-
			0.501-1.000	34 Min	18 Min	10	17	-	-	-	-
			1.001-2.000	33 Min	15 Min	10	14	-	-	-	-
			2.001-3.000	33 Min	15 Min	8	-	-	-	-	-
		-0	0.016-0.250	30 Min 38 Max	18 Min	12	15	23	-	-	-
			0.251-0.500	30 Min	16 Min	12	-	-	-	-	-
			0.501-1.000	30 Min	15 Min	12	-	-	-	-	-
			1.001-2.000	29 Min	14 Min	12	-	-	-	-	-
			2.001-3.000	29 Min	14 Min	12	-	-	-	-	-
	HM21A	-T8	0.016-0.250	31	18	4	19	19	60	40	-
			0.251-0.500	32	21	6	24	-	-	-	-
			0.501-1.000	30	19	6	22	-	-	-	-
			1.001-2.000	29	18	6	-	-	-	-	-

NOTE: 1. Specified Minimums

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TABLE 57. MECHANICAL PROPERTIES OF MAGNESIUM – ALLOY EXTRUSIONS

					Minimum			Typical					
										Bearing Strength ksi		Brinell Hard. No. 500kg 10mm	
Govt Spec No.	Alloy	Temp	Form	Diam or Thickness Inches	US ksi	YS (0.2% offset) ksi	Elong %	CYS (0.2% offset) ksi	SS ksi	US	YS		Fatigue Str ksi
QQ-M-31	AZ31B	-F	Bars, Rods, and Shapes	0.249 and under	35	21	7	15	19	56	33	-	19
				0.250-1.499	35	22	7	14	19	56	33	49	-
				1.500-2.499	34	22	7	14	19	56	33	-	-
			Hollow Shapes	-	32	16	8	12	-	-	-	-	-
	AZ61A		Bars, Rods, and Shapes	0.249 and under	38	21	8	-	-	65	38	-	20
				0.250-2.499	39	24	9	19	20	68	41	60	-
			Hollow Shapes	-	36	16	7	16	-	-	-	50	-
	AZ80A	-F	Bars, Rods and Shapes	0.249 and under	43	28	9	-	22	68	48	60	22
				0.250-1.499	43	28	8	-	22	68	48	60	-
				1.500-2.499	43	28	6	-	22	68	48	60	-
		-T5	Bars, Rods and Shapes	0.249 and under	47	30	4	34	-	-	-	82	-
				0.250-1.499	48	33	4	35	24	-	-	82	-
				1.500-2.499	48	33	4	32	24	-	-	82	-
	M1A		Bars, Rods, and Shapes	0.249 and under	30	-	2	12	-	51	25	44	11
				0.250-1.499	32	-	3	12	18	51	25	44	-
				1.500-2.499	32	-	2	-	18	51	25	44	-
			Hollow Shapes	-	28	-	2	9	-	-	-	42	-
				Cross-Sect Area Sq in.									
--	ZK21A	-F	Bars, Rods and Shapes	0.249 and under	38	28	4	25	-	-	-	-	-
			Hollow Shapes	-	34	26	4	17	-	-	-	-	-
QQ-M-31	ZK60A	-F	Bars, Rods, and Shapes	Up to 5 incl	43	31	5	29	27	80	55	75	20
			Hollow Shapes	-	40	28	5	25	-	-	-	75	-
		-T5	Bars, Rods, and Shapes	Up to 5 incl	45	36	4	32	26	85	59	82	20
			Hollow Shapes	-	46	38	4	31	-	-	-	82	-
--	ZK60B	-T5	Bars, Rods, and Shapes	Up to 20 incl	45	35	4	40	26	90	70	-	20
--	HM31A	-T5	Bars, Rods, and Shapes	0.999 and under	37	26	4	25	23	58	44	-	12

TABLE 58. MECHANICAL PROPERTIES OF MAGNESIUM – ALLOY FORGINGS

Govt Spec No.	Alloy	Temper	Minimum			Typical			
			US ksi	YS (0.2% offset) ksi	Elong %	CYS (0.2% offset) ksi	SS ksi	Brinell Hard. No. 500 kg 10 mm	Fatigue Str ksi
QQ-M-40	AZ61A	-F	38	22	6.0	17	20	55	19
	AZ80A	-F	42	26	5.0	25	22	69	19
		-T5	42	28	2.0	28	23	72	17
	M1A	-F	30	18	3.0	16	18	47	-
	TA54A	-F	36	22	7.0	-	-	-	-
	ZK60A	-T5	42	26	7.0	28	23	72	17

TABLE 59. MECHANICAL PROPERTIES OF MAGNESIUM – ALLOY TUBE

Govt Spec No.	Alloy	Temper	Tubing Wall Thickness Inches	Minimum		Typical	
				US ksi	YS (0.2% offset) ksi	CYS (0.2% offset) ksi	Brinell Hard. No. 500kg 10 mm
WW-T-825	AZ31B		0.028-0.250	32	16	12	46
			0.251-0.750	32	16	-	46
	AZ61A		0.028-0.750	36	16	16	50
	M1A		0.028-0.750	28	-	9	42
--	ZK21A	-F	All	34	26	17	-
WW-T-825	ZK60A	-F	0.028-0.250	40	28	25	75
		-T5	0.028-0.250	46	38	29	82

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TABLE 60. STRESS–RUPTURE VALUES FOR SOME SAND–CAST MAGNESIUM ALLOYS

Govt Spec No.	Alloy	Temper	Temp °F	Stress ksi	Hours to Third Stage	Extension, Third Stage, %	Hours to Rupture	Extension, at Rupture, %
QQ-M-56	AZ63A	-T6	300	9	330	2.4	728	10.8
				10	90	1.4	310	11.0
	AZ92A	-T6	95	34	480	7.4	1,156	28.8
				40	115	9.0	233	26.8
			250	12	495	2.7	1,080	30.5
				14	120	5.7	290	30.9
			300	8	405	2.5	1,181	32.9
				10	84	2.1	288	35.7

TABLE 61. STRESS–RUPTURE VALUES FOR SOME WROUGHT MAGNESIUM ALLOYS

Govt Spec No.	Alloy	Temper	Temp °F	Stress ksi	Hours to Third Stage	Extension, Third Stage, %	Hours to Rupture	Extension, at Rupture, %
Extruded								
QQ-M-31	M1A	-F	95	24	1,300	16.6	1,420	21.1
				25	340	13.5	477	22.1
			300	12	680	14.7	1,115	38.5
				14	-	-	232	41.8
	AZ31B	-F	95	31	570	15.1	866	30.4
				32	330	15.9	418	28.9
			300	7	390	10.5	900	55.0
				8	180	9.5	410	54.0
	AZ61A	-F	95	42	660	8.4	1,158	24.7
				44	210	9.1	299	21.9
			300	6	360	4.2	1,174	71.0
				10	114	3.2	312	44.4

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TABLE 62. EFFECT OF HEATING ON PROPERTIES OF WROUGHT AND CAST MAGNESIUM ALLOYS

Alloy	Composition %	Condition	Temper	Density lb/cu in.	Elect Cond (IACS) %	YS (0.2% offset) ksi	TS ksi	Elong (2 in.) %	Brinell Hard.	Fatigue Str (5×10^8 cycles) ksi	Shear Str ksi
			Wrought								
AZ80A	8.5 Al, 0.5 Zn, 0.2 Mn	Press Forged	-	0.065	10.8	30	45	7	64	17	22
		Press Forged, aged 350 F, 16-24 hr	-T5	0.065	14.6	34	50	6	72	16	22
		As Extruded	-	0.065	10.8	36	48	14	62	18	21.5
		Extruded, Sol H.T. at 750 F, aged 350 F, 16-24 hr	-T51	0.065	14.6	39	54	6	80	-	-
			Sand or Permanent Mold Cast								
AM100A	10 Al, 0.2 Mn	As cast	-F	0.066	11.5	12	22	2	53	10	18
		Sol H.T. at 780-800 F, 18-24 hr	-T4	0.066	9.9	13	40	10	52	11	20
		Sol H.T. at 780-800 F, 18-24 hr, aged 325-400 F, 10-12 hr	-T6	0.066	11.0	20	40	4	60	10	21
AZ92A	9 Al, 2Zn, 0.2 Mn	As cast	-F	0.066	12.3	14	25	2	65	12	18
		Sol H.T. at 760-775 F, 18-22 hr	-T4	0.066	10.5	14	40	10	63	13	20
		Sol H.T. at 760-775 F, 18-22 hr, aged 400-425 F, 5-15 hr	-T6	0.066	12.3	22	40	3	81	12	22
AZ63A	6 Al, 3 Zn, 0.2 Mn	As cast	-F	0.066	15.0	11	27	6	51	11	18
		Sol H.T. at 720-740 F, 10-15 hr	-T4	0.066	12.3	13	40	15	53	12	18
		Sol H.T. at 720-740 F, 10-15 hr, aged 400-425 F, 5-15 hr	-T6	0.066	13.8	19	40	5	71	11	20
AZ91C	8.7 Al, 0.7 Zn, 0.2 Mn	As cast	-F	0.066	10.2	14	24	25	60	-	-
		Sol H.T. at 775-790 F, 18-20 hr	-T4	0.066	9.9	13	40	15	55	-	-
		Sol H.T. at 775-790 F, 18-20 hr, aged 400-425 F, 3-5 hr	-T6	0.066	11.2	19	40	5	70	-	-

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TABLE 63. EFFECT OF HEATING ON PROPERTIES OF HARD--ROLLED MAGNESIUM – ALLOY SHEET

Heating Time and Temperature	ROOM TEMPERATURE PROPERTIES AFTER HEATING							
	QQ-M-44-Cond-H				QQ-M-54-Cond-H			
	Sheet Thickness Inch	TS ksi	YS ksi	Elong %	Sheet Thickness Inch	TS ksi	YS ksi	Elong %
None	0.064	43	35	6	0.064	36	30	5
5 min at 300°F.	0.064	41	31.5	10	0.064	37	30.5	5
40 min at 300°F.	0.064	40.5	31	14	0.064	38	30.5	4
5 min at 350°F.	0.064	40	29.5	17	0.064	37.5	30	5
40 min at 350°F.	0.064	39	27	18	0.064	37	29	5
5 min at 400°F.	0.064	38	25.5	19	0.064	36	27	6
40 min at 400°F.	0.064	37.5	23.5	21	0.064	36	26.5	7
5 min at 450°F.	0.064	37.5	23	21	0.064	35.5	26.5	7
40 min at 450°F.	0.064	36	21.5	22	0.064	34.5	25	8
None	0.080	43	37.5	8	0.064	35.5	28	5
2 hrs at 300°F.	0.080	40.5	31.5	18	0.064	35	28.5	6
2 hrs at 400°F.	0.080	38	25	27	0.064	34.5	26.5	7
2 hrs at 450°F.	0.080	37	23	25	0.064	32.5	25	10
None	0.051	43	33	8	0.064	35.5	28	5
16 days at 200°F	0.051	41	29	13	0.064	35.5	28	5
10 hrs at 300°F	0.051	41	29	16	0.064	35.5	28	6
16 days at 300°F	0.051	39	25	21	0.064	34.5	26	6
None	0.064	45	39	7	0.064	35.5	28	5
4 hrs at 500°F.	0.064	37	24.5	24	0.064	35	23.5	16
Heated to 600°F.	0.064	38	24.5	24	0.064	34	24.5	18
4 hrs at 600°F.	0.064	37	24	23	0.064	33.5	22	19
4 hrs at 700°F.	0.064	37	22	23	0.064	33	20.5	19

NOTE: All properties were determined on longitudinal specimens. QQ-M-44-Cond-H has approximately the same properties in longitudinal and transverse directions. QQ-M-54-Cond-H in transverse direction has from 3 to 5 ksi lower yield strength, but higher elongation. Heating at the same temperature for one hour results in practically the same properties as heating for 4 hours.

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TABLE 64. FORMING CHARACTERISTICS—MAGNESIUM—ALLOY SHEET

Alloy	Temper	Approx Minimum 90° Bend Radius[1]			Maximum Reduction % per Draw[2]			Hot Working Temp[3] °F	Approx Temp °F of Beginning Hot Shortness	Workability Rating[4] (Relative)
		70°F	400°F	700°F	70°F	400°F	700°F			
AZ31B	-O	5.5t	3t	-	15-20	65	-	550	900	A
	-H24	8t	3t	-	-	65	-	375	900	C
HK31A	-O	6t	5t	2t	-	-	-	750	1000	B
	-H24	12t	9.5t	3t	-	-	70	650	1000	D
HM21A	-T8	9t	9t	6t	-	-	60	750	1000	D
ZE10A	-O	5.5t	3t	-	20-25	65	-	550	1000	A
	-H24	8t	3t	-	-	65	-	375	1000	C

NOTES: 1. t = sheet thickness. Values apply to 0.040 to 0.072 inch thickness. They decrease for thinner and increase for thicker material. Many gages of magnesium-alloy sheet can be bent cold to a radius closer by approximately one t if the material is bent with-grain rather than cross-grain.

2. Percent reduction = $\frac{\text{Blank diameter minus punch diameter}}{\text{Blank diameter}} \times 100$

3. ± 50 deg F.

4. Applies to bending, forming, and drawing of wrought products in general. Best, A. Worst, D.

TABLE 65. FORMING CHARACTERISTICS—MAGNESIUM—ALLOY FORGINGS

Alloy	Characteristics
M1A	Low cost, easy formability; fair strength; suitable for press and hammer forgings.
AZ80A	Hot press forgings of relatively simple design requiring maximum mechanical properties.
TA54A	Best hammer forging alloy.
ZK60A	Excellent press forging characteristics and the best combination of strength and ductility of the magnesium forging alloys.
AZ61A	Fairly good formability; medium mechanical properties; usually press forged.

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TABLE 66. COMPARATIVE STRENGTH AND STIFFNESS IN BENDING OF SOME STRUCTURAL METALS AT ROOM TEMPERATURE[1]

Material	Thickness	Bending Strength	Stiffness	Weight
For Equal Thickness				
1025 steel	100	100.0	100.0	100.0
6061-T6 aluminum sheet and extrusions	100	97.2	34.5	34.5
AZ31B magnesium extrusions	100	47.2	22.4	22.5
ZK60A-T5 magnesium extrusions	100	88.9	22.4	22.5
AZ31B-H24 magnesium sheet	100	73.4	22.4	22.5
For Equal Strength				
1025 steel	100	100.0	100.0	100.0
6061-T6 aluminum sheet and extrusions	101	100.0	35.8	34.8
AZ31B magnesium extrusions	146	100.0	69.2	32.9
ZK60A-T5 magnesium extrusions	106	100.0	26.7	23.9
AZ31B-H24 magnesium sheet	117	100.0	35.6	26.3
For Equal Stiffness				
1025 steel	100	100.0	100.0	100.0
6061-T6 aluminum sheet and extrusions	143	199.0	100.0	49.3
AZ31B magnesium extrusions	165	129.0	100.0	37.2
ZK60A-T5 magnesium extrusions	165	242.0	100.0	37.2
AZ31B-H24 magnesium sheet	165	200.0	100.0	37.2
For Equal Weight				
1025 steel	100	100.0	100.0	100.0
6061-T6 aluminum sheet and extrusions	290	817.0	841.0	100.0
AZ31B magnesium extrusions	444	930.0	1962.0	100.0
ZK60A-T5 magnesium extrusions	444	1753.0	1962.0	100.0
AZ31B-H24 magnesium sheet	444	1451.0	1962.0	100.0

NOTE: 1. Rectangular beams of constant width with the following minimum yield strengths: 1025 steel, 36 ksi; 6061-T6 aluminum, 35 ksi; magnesium alloys, average of minimum tensile yield and compressive yield strengths.

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TABLE 67. MINIMUM BEND RADII FOR SOME MAGNESIUM ALLOYS[1]

Govt Spec No.	Composition	Temper	Thickness Range Inch	Forming Temp[2] °F	Incidence of Breakage at Various Bend Radii[3]				
					1 in 2 (Typical)	1 in 20	1 in 100	1 in 1,000	1 in 10,000
QQ-M-54	*	-O	0.016 to 0.064	70	3.7t	4.8t	5.2t	5.7t	6.1t
				600	0.5t	-	1.2t	-	1.6t
			0.072 to 0.250	70	4.3t	5.1t	5.4t	5.8t	6.1t
				600	0.5t	-	1.2t	-	1.6t
		-H24	0.016 to 0.064	70	6.2t	9.0t	10.1t	11.4t	12.4t
				450	5.0t	-	8.3t	-	10.3t
			0.064 to 0.250	70	8.0t	10.8t	11.9t	13.2t	14.2t
				450	6.4t	-	9.4t	-	11.3t
QQ-M-44	**	-O	0.020 to 0.250	70	3.0t	4.1t	4.5t	5.0t	5.4t
				500	(3.2t)†	(4.9t)	(5.6t)	(6.4t)	(7.0t)
					0.7t	-	1.6t	-	2.1t
					(0.7t)	-	(1.7t)	-	(2.3t)
		-H24	0.020 to 0.064	70	4.6t	6.2t	6.9t	7.6t	8.2t
				325	2.4t	-	4.1t	-	5.0t
			0.064 to 0.250	70	5.3t	6.9t	7.6t	8.3t	8.9t
				325	2.4t	-	4.1t	-	5.0t

NOTES: 1. 90-degree press-brake bends on 6-inch wide specimens in rectangular rubber-retaining dies.
2. Sheet and dies at temperature indicated. Elevated temperatures listed are maxima for which original room-temperature properties will be essentially retained.
3. Bend radii are in terms of sheet thicknesses (t).
* Nominal composition: 1.5 percent manganese; same as ASTM alloy M1A.
** Nominal composition: 3 percent aluminum, 1 percent zinc; same as ASTM alloy AZ31A.
† Process changes for sheet material AZ31A, in -O temper, have been introduced since collection of data reported. Values in parentheses are those actually obtained from data. Other values are estimates for AZ31A-O sheet presently being produced.

TABLE 68. WELDABILITY

Alloy	Relative Rating of Weldability[1]	
	Elect Resist.	Inert Gas Shield Arc
AZ31B	1	A
AZ61A	1	B
AZ63A	-	D
AZ80A	1	B
AZ81A	-	B
AZ91C	-	C
AZ92A	-	C
EZ33A	-	A
HK31A	-	A
HZ32A	-	D
M1A	1	A
ZE10A	1	A
ZH62A	-	D
ZK21A	1	B
ZK51A	-	D
ZK60A	1	D

NOTE: 1. Ratings A to D indicate weldability from good to poor.

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TABLE 69. TYPICAL STRENGTH PROPERTIES OF INERT–GAS, TUNGSTEN–ARC–WELDED, BUTT JOINTS IN MAGNESIUM

Alloy	Temper	Welding Rod	Weld TS ksi	Elong (2 in.) %	Joint Efficiency %
			Sheet and Plate		
AZ31B	-H24	AZ61A	37	10.0	88
	-O	AZ61A	36	12.0	97
HK31A	-H24*	HK31A	30	2.0	-
			Extrusions		
AZ31B	-	AZ61A	35	12.0	92
AZ61A	-	AZ61A	40	8.0	89
AZ80A	-F	AZ92A	42	4.0	86
M1A	-	M1A	20	2.0	54
			Castings		
AZ63A	-T6	AZ63A	28	3.0	80
AZ92A	-T6	AZ92A	28	3.0	80
HK31A	-T6*	HK31A	31	9.5	-
HZ32A	-T5*	HK31A	28	4.0	-
EK41A	-T5	EZ33A	18.3	1.0	90
EK41A	-T6	EZ33A	18.8	1.5	91
EZ33A	-T5	EZ33A	20	2.0	100

NOTE: *Room temperature properties. At elevated temperatures (400-600°F) the weld efficiencies of HK31A and HZ32A approach 100%.

TABLE 70. TYPICAL STRENGTH PROPERTIES OF WELDS MADE BY THE CONSUMABLE ELECTRODE–ARC WELDING PROCESS

Alloy	Temper	Thickness Inch	Wire Dia Inch	Weld Str ksi	Elong (2 in.) %	Weld Efficiency %
			Sheet			
AZ31B	-H24	0.1875	0.051	36.8	6.6	88.5
	-H24	0.1875	0.051	38.5	7.8	92.5
	-H24	0.1875	0.051	40.1	8.8	96.4
	-O	0.032	0.050	33.7	8.3	92.7
	-O	0.040	0.050	35.8	7.0	95.5
	-O	0.050	0.050	35.5	8.3	94.6
	-O	0.064	0.050	35.7	10.6	99.5
	-O	0.040	0.064	36.3	9.3	96.8
	-O	0.250	0.098	35.1	8.8	95.9
			Plate			
AZ31B	-O	0.375	0.098	33.0	6.0	89.9
	-O	0.375	0.098	36.6	13.4	99.7
	-O	0.375	0.098	33.9	7.1	92.4
	-O	0.500	0.098	36.1	17.5	98.3
	-O	0.500	0.098	36.3	17.4	98.9
	-O	0.500	0.098	36.2	17.6	98.6

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TABLE 71. CHARACTERISTICS OF ADHESIVES USED FOR BONDING MAGNESIUM[1]

General Type of Composition	Curing Conditions			Adhesive Thick Inch	Shear Strength ksi
	Temp °F	Time Minutes	Pressure psi		
Phenol formaldehyde plus polyvinyl formal powder[2]	270	32	50 to 500	0.000 to 0.006	1.6 to 2.6
Phenolic rubber-base resin[2]	325	20	200	0.003 to 0.006	2.2 to 2.6
Phenolic synthetic rubber base plus thermosetting resin	350	10	7 to 45	0.005 to 0.020	1.0 to 2.5
	350	60	100	0.005 to 0.020	2.1 to 2.9
Ethoxyline resin-liquid, powder or stick used like solder	390	60	Contact	0.003 to 0.006	1.5 to 2.2
Ethoxyline resin (two liquids)	Room	24 hr	Contact	0.001 to 0.006	-
Epoxy type resin paste plus liquid activator	200	60	Contact	0.003 to 0.005	3.0 max
Rubber base	400	8	200	0.010 to 0.015 (tape)	1.7 to 2.3
Vinyl phenolic	300	8 preheat	200	0.004 to 0.012	1.0 to 1.7
	275 to 400	70 to 4			
Epoxy-type resin	200	45	Contact	0.010 to 0.030	1.2 to 1.8
	200	45	14	(tape)	1.4 to 1.7
Phenolic	300	15	28	0.002 to 0.004	2.4 max

NOTES: 1. These adhesives are for service at temperatures up to 185°F.
2. Known to meet USAF specifications.

TABLE 72. CLEANING AND PICKLING METHODS USED ON MAGNESIUM

Type of Cleaning	Method of Application	Uses
Mechanical	Wire brushing, sanding, sand shot and grit blasting.	A rapid method of removing heavy layers of oxide and dirt. Sand, shot and grit blasting should always be followed by acid pickle.
Solvent	By rinsing, by vapor degreasing or as an emulsion type cleaner.	Used to remove superfluous amounts of oily matter prior to alkaline cleaning and as a pre-cleaning step prior to painting.
Alkaline	Applied in hot solution as a soak cleaner or an electrolytic cleaner.	The most satisfactory method of degreasing magnesium prior to chemical treating. Will remove old chemical finishes and thus insure uniform coatings.
Acid Pickle	By immersion.	Several acid pickling treatments are available for use on magnesium usually as cleaning steps in application of chemical coatings.

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TABLE 73. CHEMICAL TREATMENTS FOR MAGNESIUM

Treatment	Type	Specification	Alloys used on	Appearance	Uses	Remarks
Chrome-pickle, Dow 1	Chemical	MIL-M-3171 Type 1	All alloys	Matte gray to yellow red	For protection during shipment and storage.	Simple dip treatment. 0.006" per surface dimensional loss - good paint base.
Chrom-Alum Dow 4	Chemical	-	AZ91A, AZ91B	Black	For black finish on die castings.	No dimensional loss. Not recommended for outdoor service unless painted.
Dichromate Dow 7	Chemical	MIL-M-3171 Type 3	All aluminum containing alloys and ZK60A	Brassy to dark brown	For general longtime protection of large castings where close dimensional tolerances are required.	Accepts inserts of brass, bronze, cadmium, aluminum, (with special treatment), steel. Does not materially affect dimensions.
Galvanic Anodize Dow 9	Electro-chemical	MIL-M-3171 Type 4	All alloys	Dark brown to black	Used especially on those alloys which will not take No. 7 or where a black coating is needed.	Requires galvanic couple between work and steel cathode. No external current required. Does not materially affect dimensions.
Sealed chrome-pickle Dow 10	Chemical	MIL-M-3171 Type 2	All alloys	Matte gray to yellow red brown	For wrought products but may also be applied to castings.	Slight dimensional loss. Not for parts containing steel or brass inserts; not for temperature above 500°F.
Caustic anodize Dow 12	Electro-chemical	MIL-M-3171 Type 5	All alloys	Light gray to light tan	For solution annealed castings.	Specialty treatment - can be dyed. When sealed gives good paint-base qualities. Good dielectric strength.
Anodize Dow 17	Electro-chemical, medium to high voltage	MIL-M-45202 Type I - light Type II - heavy	All alloys	Clear, light gray to green, dark green	For rigid parts; dissimilar metal inserts must be insulated. Can also be used as casting impregnant.	Clear: 0.001" build up per surface. Thin: 0.001" to 0.003" per surface. Full: 0.001 to 0.0015. Coating is stable above melting point of metal. Good abrasion resistance-good paint base.

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TABLE 73 (Continued). CHEMICAL TREATMENTS FOR MAGNESIUM

Treatment	Type	Specification	Alloys used on	Appearance	Uses	Remarks
Anodize HAE	Electro-chemical, high-and-low-voltage	MIL-M-45202 Type I - light Type II - heavy	All alloys except Mg-Li alloys	Light tan to brown	For rigid parts; dissimilar metal inserts must be insulated.	Exhibits exceptionally high hardness, good dielectric properties, high corrosion, abrasion and heat resistance. Dimensional increase of 0.3 to 1.0 mil occurs during treatment. Low-voltage coatings used for moderate environment.
Anodize CR-22	Electro-chemical, high-voltage	MIL-M-45202	All alloys	Green	Used on all forms of Mg alloys including those having metallic inserts.	Parts treated do not require special treatment. Only degreasing is needed. The hardness of CR-22 is such that it will abrade copper and mild steel. It requires high voltage - to 320 volts.
Iridite #15	Chemical	-	All alloys	Dark brown to light brown	Used on all forms for electronic equipment, cameras, etc.	Metal loss 0.001 - 0.003. Coating buildup very thin. Fair abrasion resistance. Fair paint base. Low electrical resistance.
Dilute chromic acid Dow 19	Chemical	-	All alloys	Brassy to brown	Touch up, paint base. Used in refinishing procedures or on parts that are too large to be immersed.	No appreciable change in dimensions. Has adverse effect on anodic coatings such as HAE if allowed to remain in contact too long. Can be applied by spray, brush or dipped.
Ferric Nitrate	Chemical	-	All alloys	Silvery	General clean-up pickle for removal of mill scale, oxidation and corrosion products.	Bright finish that protects magnesium from tarnish for short periods.
Stannate	Chemical	-	All except ZK60A	Medium to dark gray	Used with dissimilar metal inserts or in assemblies.	Retards galvanic corrosion. Good paint base. Does not require electrical equipment. Requires neutralization before painting.

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TABLE 74. PAINT SCHEDULES

Condition of Exposure	Primer	Finish
Class I Normal indoor	One coat of metal primer of zinc chromate type optional	One or two coats of paint, lacquer or enamel compatible with primer
Class II Outdoor, inland area	One coat of metal of zinc chromate type	Two coats of spar varnish or synthetic resin enamels compatible with primer
Class III Marine or corrosive atmosphere	One or two coats of zinc chromate primer	Three or more coats of synthetic resin enamel, preferably carrying aluminum pigment

TABLE 75. PLATING ON MAGNESIUM WROUGHT ALLOYS

Process	Appearance	Uses	Remarks
Electroplating Decorative Chromium Hard Chrom um Copper + Tin (hot flowed)	Bright metallic Dull to semi-bright Semi-bright to bright	On all alloys. For appearance, wear resistance protection, for grounding and as soldering base.	Any metal which can be electrodeposited, can be plated onto magnesium.
Electroless Nickel	Dull to semi-bright metallic	On all alloys. For protection wear resistance grounding and as soldering base.	Complex shapes, internal bores, etc. can be uniformly coated.

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APPENDIX

This appendix provides the reader with a ready reference of specifications, physical properties and characteristics, chemical composition, and mechanical properties of magnesium alloys.

Specifications are provided in table A, and physical properties and characteristics in tables B and C. Chemical composition and mechanical properties are provided in individual tables, classified by alloys in an alpha-numerical order. Chemical composition values are maximum unless otherwise specified. Mechanical properties of cast magnesium alloys are for separately cast specimens, unless otherwise noted.

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TABLE A. CURRENT PRODUCT SPECIFICATIONS FOR MAGNESIUM AND MAGNESIUM ALLOYS

Form	ASTM			GOVERNMENT	AMS	SAE
	Alloy	Temper	Spec. No.	Spec. No.	Spec. No.	No. 2
Ingot	9980A	-	B 92	MIL-M-20161B	-	-
	9990A	-	B 92	-	-	-
	9998A	-	B 92	-	-	-
	99995A	-	B 92	-	-	-
	AM100A	-	B 93	-	-	-
	AZ63A	-	B 93	-	-	-
	AZ81A	-	B 93	-	-	-
	AZ91A	-	B 93	-	-	-
	AZ91B	-	B 93	-	-	-
	AZ91C	-	B 93	-	-	-
	AZ92A	-	B 93	-	-	-
Castings	AZ91C	-T6	-	MIL-M-46062*	-	-
	AZ92A	-T6	-	MIL-M-46062*	-	-
	HK31A	-T6	-	MIL-M-46062*	-	-
	ZK51A	-T5	-	MIL-M-46062*	-	-
	ZK61A	-T6	-	MIL-M-46062*	-	-
	ZH62A	-T5	-	MIL-M-46062*	-	-
	QE22A	-T6	-	MIL-M-46062*	-	-
	LS141A	-T7	Not Current	MIL-M-46143	-	-
	LZ145A	-T7	Not Current	MIL-M-46143	-	-
	(*This specification covers high strength magnesium alloy castings produced by any casting method.)					
Sand Castings	AM100A	-T6	B 80	-	-	-
	AZ63A	-F	B 80	QQ-M-56	4420	50
	AZ63A	-T4	B 80	QQ-M-56	4422	50
	AZ63A	-T5	B 80	QQ-M-56	-	50
	AZ63A	-T6	B 80	QQ-M-56	4424	50
	AZ81A	-T4	B 80	QQ-M-56	-	505
	AZ91C	-F	B 80	QQ-M-56	-	504
	AZ91C	-T4	B 80	QQ-M-56	-	504
	AZ91C	-T5	B 80	QQ-M-56	-	504
	AZ91C	-T6	B 80	QQ-M-56	4437	504
	AZ92A	-F	B 80	QQ-M-56	-	500
	AZ92A	-T4	B 80	QQ-M-56	-	500
	AZ92A	-T5	B 80	QQ-M-56	-	500
	AZ92A	-T6	B 80	QQ-M-56	4434	500
	EZ33A	-T5	B 80	QQ-M-56	4442	506
	HK31A	-T6	B 80	QQ-M-56	4445	507
	HZ32A	-T5	B 80	QQ-M-56	4447	-
	K1A	-F	B 80	MIL-M-45207	-	-
	QE22A	-T6	B 80	QQ-M-56	4418	-
	ZE41A	-T5	B 80	-	4439	-
	ZH62A	-T5	B 80	QQ-M-56	4438	508
	ZK51A	-T5	B 80	QQ-M-56	4443	509
	ZK61A	-T6	B 80	QQ-M-56	-	513
Permanent Mold Castings	AM100A	-F	B 199	QQ-M-55	-	502
	AM100A	-T4	B 199	QQ-M-55	-	502
	AM100A	-T6	B 199	QQ-M-55, MIL-M-46062	4483	502
	AM100A	-T61	B 199	QQ-M-55	-	502
	AZ63A	-F	-	QQ-M-55	-	-
	AZ63A	-T4	-	QQ-M-55	-	-
	AZ63A	-T5	-	QQ-M-55	-	-
	AZ63A	-T6	-	QQ-M-55	-	-
	AZ81A	-T4	B 199	QQ-M-55	-	-
	AZ91C	-F	B 199	QQ-M-55	-	504
	AZ91C	-T4	B 199	QQ-M-55	-	504
	AZ91C	-T5	B 199	QQ-M-55	-	504
	AZ91C	-T6	B 199	QQ-M-55	-	504
	AZ92A	-F	B 199	QQ-M-55	-	500
	AZ92A	-T4	B 199	QQ-M-55	-	500
	AZ92A	-T5	B 199	QQ-M-55	-	500
	AZ92A	-T6	B 199	QQ-M-55	4484	500
	EZ33A	-T5	B 199	QQ-M-55	-	506
	HK31A	-T6	B 199	QQ-M-55	-	-
	HZ32A	-T5		QQ-M-55	-	-
	QE22A	-T6	B 199	QQ-M-55	-	-

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TABLE A (Continued). CURRENT PRODUCT SPECIFICATIONS FOR MAGNESIUM AND MAGNESIUM ALLOYS

Form	ASTM			GOVERNMENT	AMS	SAE
	Alloy	Temper	Spec. No.	Spec. No.	Spec. No.	No. 2
Die Castings	AZ91A	-F	B 94	QQ-M-38	4490	501
	AZ91B	-	B 94	-	-	501A
Investment Castings	AM100A	-F	B 403	-	-	-
	AM100A	-T4	B 403	-	-	-
	AM100A	-T6	B 403	-	4455	-
	AM100A	-T7	B 403	-	-	-
	Z81A	-T4	B 403	-	-	-
	AZ91C	-F	B 403	-	-	-
	AZ91C	-T4	B 403	-	-	-
	AZ91C	-T5	B 403	-	-	-
	AZ91C	-T6	B 403	-	-	-
	AZ92A	-F	B 403	-	-	-
	AZ92A	-T4	B 403	-	-	-
	AZ92A	-T5	B 403	-	-	-
	AZ92A	-T6	B 403	-	4453	-
	EZ33A	-T5	B 403	-	-	-
	HK31A	-T6	B 403	-	-	-
	K1A	-F	B 403	-	-	-
	QE22A	-T6	B 403	-	-	-
	ZK61A	-T6	B 403	-	-	-
Extruded Bar, Rod, and Shapes	AZ31B	-F	B 107	QQ-M-31	-	510
	AZ31C	-F	B 107	-	-	-
	AZ61A	-F	B 107	QQ-M-31	4350	520
	AZ80A	-F	B 107	QQ-M-31	-	523
	AZ80A	-T5	B 107	QQ-M-31	-	523
	*HM31A	-T5	-	MIL-M-8916	4389	-
	M1A	-F	B 107	QQ-M-31	-	522
	*ZK21A	-F	-	MIL-M-46039	4387	-
	ZK60A	-F	B 107	QQ-M-31	-	524
	ZK60A	-T5	B 107	QQ-M-31	4352	524
	ZK60B	-T5	-	MIL-M-26696	-	-
Tube	AZ31B	-	B 217	WW-T-825	-	510
	AZ31C	-	B 217	-	-	-
	AZ61A	-	B 217	WW-T-825	4350	520
	M1A	-	B 217	WW-T-825	-	522
	ZK60A	-F	B 217	WW-T-825	-	524
	ZK60A	-T5	B 217	WW-T-825	4352	524
Forgings	AZ31B	-F	B 91	-	-	510
	AZ61A	-F	B 91	QQ-M-40	4358	520
	AZ80A	-F	B 91	QQ-M-40	-	523
	AZ80A	-T5	B 91	QQ-M-40	-	523
	*HM21A	-T5	-	QQ-M-40	4360	-
	*ZK60A	-T6	B 91	QQ-M-40	4362	524
	M1A	-F	-	QQ-M-40	-	522
	HM21A	-T5	-	QQ-M-40	-	522
Sheet and Plate	AZ31B	-O	B 90	QQ-M-44	4375	510
	AZ31B	-H24	B 90	QQ-M-44	4377	510
	AZ31B	-H26	B 90	MM-M-44	4376	510
	AZ31B	-O	-	MIL-F-46048	-	-
	AZ31B	-	B 90	MIL-T-38749	4382	-
	AZ31C	-	B 90	-	-	-
	HK31A	-O	B 90	MIL-M-26075	4384	507
	HK31A	-H24	B 90	MIL-M-26075	4385	507
	HM21A	-T81	-	-	4383	-
	HM21A	-T8	B 90	MIL-M-8917	4390	-
	LA141A	-T7	B 90	MIL-R-6944	4386	-
	ZE10A	-O	B 90	MIL-M-46037	-	534
	ZE10A	-H24	B 90	MIL-M-46037	-	534
Welding Rod	*AZ61A	-	B 90	MIL-R-6944	-	-
	AZ63A	-	B 90	MIL-R-6944	-	-
	AZ92A	-	B 90	MIL-R-6944	4395	-
	EZ33A	-	B 90	MIL-R-6944	4396	-
	HK31A	-	B 90	MIL-R-6944	-	-
	LA141A	-	-	-	4397	-
	M1A	-	B 90	MIL-R-6944	-	-

*NOT A,S,T,M.

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TABLE B. PHYSICAL AND FABRICATING PROPERTIES OF CAST MAGNESIUM ALLOYS

Characteristics	ASTM Type					
	AZ63A	AZ81A	AZ91 + AZ291B	AZ91C	AX92A	AM100A
PHYSICAL PROPERTIES						
Density, lb/cu in.	0.066	0.065	0.065	0.065	0.065	0.065
Melting Temp Range °F	850-1130	882-1115	875-1120	875-1105	830-1110	885-1100
Ther Cond (68F), Btu/hr/sq ft/°F/ft	29-39	29	31	27-31	27-34	24-34
Coef of Ther Exp, per °F						
68°F	14×10^{-6}	14×10^{-6}	14×10^{-6}	14×10^{-6}	14×10^{-6}	14×10^{-6}
68-750°F	16×10^{-6}	16×10^{-6}	16×10^{-6}	16×10^{-6}	16×10^{-6}	16×10^{-6}
Spec Ht (68°F), Btu/lb/°F	0.245	0.245	0.245	0.245	0.245	0.245
Elec Res (68°F), Microhm-cm	11-15	12	14	13-16	12-17	10-15
THERMAL TREATMENT						
Solution Temp, °F	730	780	---	780	770	780
Aging Temp, °F	425	---	---	420	500	425
FABRICATING PROPERTIES						
Weldability[1]						
Inert-Gas Arc	D	B	---	C	C	A
Stress Relief Temp, °F	500	500	---	500	500	500
Machinability Index (free-cutting brass = 100)	500	500	500	500	500	500
CORROSION RESISTANCE	Good resistance to atmosphere, attacked by salt water unless finished.					
AVAILABLE FORMS	Sand and permanent mold castings.		Die Castings	Sand and permanent mold castings		Permanent mold castings
USES	Wheels, aircraft and missile parts, ordnance equipment, automotive and truck parts, housings, office machines, appliances, reciprocating machinery parts, cameras and optical equipment, hand tools, material handling equipment, sporting goods, griddles.					

NOTE: 1. Letter A indicates most favorable, B less favorable, etc. Relative to magnesium alloys only.

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TABLE B (Continued). PHYSICAL AND FABRICATING PROPERTIES OF CAST MAGNESIUM ALLOYS

Characteristics	ASTM Type			
	QE22A-T6	EZ33A-T5	HK31A-T6	HZ32A-T5
PHYSICAL PROPERTIES				
Density, lb/cu in	0.065	0.066	0.065	0.066
Melting Temp Range, °F	1020-1190	1110-1189	1092-1195	1026-1198
Ther Cond, Btu/hr/sq ft/°F/ft				
68°F	---	58	52	62
600°F	---	69	68	74
Coef of Ther Exp, per °F				
68°F	14×10^{-6}	14×10^{-6}	14×10^{-6}	14×10^{-6}
68-750°F	16×10^{-6}	16×10^{-6}	16×10^{-6}	16×10^{-6}
Spec Ht(68F), Btu/lb/°F	0.245	0.245	0.245	0.245
Elec Res, microhm-cm				
68°F	---	7.0	7.7	6.5
600°F	---	12.1	12.4	11.3
FABRICATING PROPERTIES				
Weldability (inert-gas arc)[1]	---	A	A	A
Machinability Index (free-cutting brass=100)	500	500	500	500
CORROSION RESISTANCE	Good resistance to atmosphere; attacked by salt water unless finished.			
AVAILABLE FORMS	Sand and permanent mold castings			
SERVICE TEMPERATURE, MAXIMUM, F	500	500	700	700
USES	Aircraft and missile components			

NOTE:
1. Letter A indicates most favorable, B less favorable, etc. Relative to magnesium alloys only.

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TABLE B (continued). PHYSICAL AND FABRICATING PROPERTIES OF CAST MAGNESIUM ALLOYS

Characteristics	ASTM Type				
	ZE41A-T5	ZK51A-T5	ZH62A-T5	K1-A	ZK61A-T6
PHYSICAL PROPERTIES					
Density, lb/cu in.	0.066	0.065	0.067	0.063	0.066
Melting Temp Range, °F	975-1190	1020-1185	960-1169	1202[1]	985-1175
Ther Cond(68F), Btu/hr/sq ft/°F/ft	---	48	63	60	---
Coef of Ther Exp, per °F					
68°F	14×10^{-6}	14×10^{-6}	14×10^{-6}	---	14×10^{-6}
68-750°F	16×10^{-6}	16×10^{-6}	16×10^{-6}	15×10^{-6}	16×10^{-6}
Spec Ht(68°F), Btu/lb/°F	0.245	0.245	0.245	---	0.245
Elec Res(68°F), Microhm-cm	---	8.4	6.5	5.7	---
THERMAL TREATMENT					
Aging Temp, °F	350	350	350		
FABRICATING PROPERTIES					
Weldability (inert-gas arc)[2]	B	D	D	B	
Machinability Index (free-cutting brass-100)	500	500	500	500	
CORROSION RESISTANCE	Good resistance to atmosphere; attacked by salt water unless finished.				
AVAILABLE FORMS	Sand and permanent mold castings			Sand and die castings	Sand and perm mold
USES	Aircraft and missile parts, ordnance equipment				

NOTES:
1. Melting point.
2. Letter A indicates most favorable, B less favorable, etc. Relative to magnesium alloys only.

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TABLE C. PHYSICAL AND FABRICATING PROPERTIES OF WROUGHT MAGNESIUM ALLOYS

Characteristics	ASTM Types			
	AZ31B-F	AZ61A-F	AZ80A-T5	ZK60A-T5
PHYSICAL PROPERTIES				
Density, lb/cu in.	0.064	0.065	0.065	0.066
Melting Temp Range °F	1050-1170	950-1140	900-1115	968-1175
Ther Cond(68°F), Btu/hr/sq ft/°F/ft	44	34	29	68-70
Coef of Ther Exp, per °F				
68°F	14 x 10-6	14 x 10-6	14 x 10-6	14 x 10-6
68 to 750°F	16 x 10-6	16 x 10-6	16 x 10-6	16 x 10-6
Spec Ht(68°F), Btu/lb/°F	0.245	0.245	0.245	0.245
Elec Res(68°F), Microhm-cm	9.2	12.5	14.5	6.0-5.7
FABRICATING PROPERTIES				
Hot Working Temp Range, °F[2]	450-550	450-650	350-375	300-500
Weldability[1]				
Inert-Gas Arc	A	B	B	D
Elec Res	A	A	A	A
Stress Relief Temp, °F	500	500	400	---
Machinability Index (free-cutting brass=100)	500	500	500	500
Hot Formability[1]	A	B	C	A
Cold Formability[1]	B	C	D	C
CORROSION RESISTANCE	Good resistance to atmosphere; attacted by salt water unless finished.			
AVAILABLE FORMS	Rod, bar, extruded shapes, tubing (except AZ80A), forgings			
USES	Aircraft and missile parts, ordinance vehicles, automotive and truck parts, electronic equipment wave guides, office machines, reciprocating machinery parts, drillable oil well casing tools, hand tools, levels, material handling equipment, concrete forms, plastering tools, jigs and fixtures, luggage and furniture, boats, sporting goods, wheels			

NOTES:
1. Letter A indicates most favorable, B less favorable, etc., Relative to magnesium alloys only.
2. Higher working temperatures can be used if they do not require heating aged alloys above the aging temperature.

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TABLE C (Continued). PHYSICAL AND FABRICATING PROPERTIES OF WROUGHT MAGNESIUM ALLOYS

Characteristics	ASTM Types			
	ZE10A-H24[1]	AZ31B-H24[2]	HK31A-H24[1]	HM21A-T8
PHYSICAL PROPERTIES				
Density, lb/cu in.	0.063	0.064	0.065	0.064
Melting Temp Range °F	1100-1195	1050-1170	1092-1195	1121-1202
Ther Cond(68°F), Btu/hr/sq ft/°F/ft	77	44	66	79
Coef of Ther Exp(68 to 750°F) per °F	16 x 10-6	16 x 10-6	16 x 10-6	16 x 10-6
Spec Ht(68°F), Btu/lb/°F	0.245	0.245	0.245	0.245
Elec Res(68°F), microhm-cm	5.2	9.2	6.1	5.0
FABRICATING PROPERTIES				
Hot Working Temp Range, °F	325-425	325-425	550-700	650-800
Weldability[3]				
Inert-Gas Arc	A	A	A	A
Elect Res	A	A	A	A
Stress Relief Temp, °F	---	500	---	---
Machinability Index (free-cutting brass=100)	500	500	500	500
Hot Formability[3]	A	A	A	A
Cold Formability[3]	B+	B	C	C
CORROSION RESISTANCE	Good resistance to atmosphere; attacked by salt water unless finished.			
AVAILABLE FORMS	Sheet and plate			Extrusions
USES	Aircraft and missile parts, ordnance vehicles, truck bodies, electronic cabinets, chassis, covers and reflectors, office machines, appliances materials handling equipment, concrete forms, jigs and fixtures, templates, foundry pattern plate, deck plate, photo-engraving plate, luggage, furniture, sporting goods		Aircraft and Missile components	

NOTES:

1. ZE10A-H24 and HK31A also available in -0 temper.
2. AZ31B-H24 sheet and plate also available in -0 temper. AZ31B plate also available in -H26 temper. AZ31B also available as special bending sheet in the -0 temper and 0.040-0.190-in. thickness for cold bending around a mandrel radius of 2-3 or more times the sheet thickness.
3. Letter A indicates most favorable, B less favorable, etc., Relative to magnesium alloys only.

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AM100A

Chemical Composition - %										
Spec. No.	Al	Zn	Mn	Si	Cu	Ni	R.E.	Zr	Th	Others
QQ-M-55	9.3-10.7	0.30	0.10	0.30	0.10	0.01	-	-	-	0.30
MIL-M-46062	9.3-10.7	0.30	Min	0.30	0.10	0.01	-	-	-	0.30

Minimum Mechanical Properties								
Specification Number	Form	Temper	Size Thick. (In.)	Size Area ($In.^2$)	TS ksi	YS ksi	YS Ext	Elong %
QQ-M-55	Castings PM and SPM	-F[1]			20	10	0.0035	-
		-T4[1]			34	10	0.0035	6
		-T6[1]			34	15	0.0043	2
		-T61[1]			34	17	0.0046	-
		-T4[2]			17	9	0.0034	-
		-T6[2]			17	10	0.0034	-
		-T61[2]			17	11	0.0038	-
MIL-M-46061	PM	-T6			17	10		0.75

NOTES: 1. Separately Cast Specimens
2. Specimens Cut from Castings

AZ31B

Chemical Composition - %										
Spec No.	Al	Zn	Mn	Ca	Zr	Si	Fe	Cu	Ni	Others
QQ-M-31	2.5-3.5	0.6-1.4	0.2	-	-	0.3	0.005	0.05	0.005	0.30
QQ-M-40	2.5-3.5	0.6-1.4	0.2 Min	0.04	-	0.1	0.005	0.05	0.005	0.30
QQ-M-44	2.5-3.5	0.7-1.3	0.2	0.04	-	0.1	0.005	0.05	0.005	0.30
WW-T-825	2.5-3.5	0.6-1.4	0.2	0.04	-	0.1	0.005	0.05	0.005	0.30
MIL-F-46068	2.5-3.5	0.6-1.4	0.2 Min	0.04	-	0.1	0.005	0.05	0.005	0.30

Minimum Mechanical Properties								
Specification Number	Form	Temper	Size Thick. (In.)	Size Area ($In.^2$)	TS ksi	YS ksi	YS Ext.	Elong %
QQ-M-31	Extrusions							
	Bars		Up thru 0.249		35	21	0.0105	7
	Rods		0.250-1.499		35	22	0.0108	7
	Shapes		1.500-2.499		34	22	0.0108	7
	Hollow Shape		All		32	16	0.0089	8
			All					
QQ-M-40	Forgings	-F	All		34	19	0.0049	6
QQ-M-44	Plate Sheet	-O	0.016-0.500		32	--	--	12
			0.501-2.000		32	--	--	10
			2.001-3.000		32	--	--	9
		-H24	0.016-0.249		39	29	0.0129	6
			0.250-0.374		38	26	0.0120	8
			0.375-0.500		37	24	0.0114	8
			0.501-1.000		36	22	0.0108	8
			1.001-2.000		34	20	0.0102	8
			2.001-3.000		34	18	0.0095	8
		-H26	0.250-0.374		39	27	0.0123	6
			0.375-0.500		38	26	0.0120	6
			0.501-0.750		37	25	0.0117	6
			0.751-1.000		37	23	0.0111	6
			1.001-1.500		35	22	0.0108	6
			1.501-2000		35	21	0.0105	6
WW-T-825	Extrusion		0.028-0.250		32	16	0.0089	8
	Tube		0.251-0.750		32	16		4
MIL-F-46048	Plate	-F	0.125-1.000		32	15	0.0043	8

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AZ61A

Chemical Composition _ %										
Spec. No.	Al	Zn	Mn	Ca	Zr	Si	Fe	Cu	Ni	Others
QQ-M-31	5.8-7.2	0.4-1.5	0.15	-	-	0.30	0.005	0.05	0.005	0.30
QQ-M-40	5.8-7.2	0.4-1.5	0.15	-	-	0.10	0.005	0.05	0.005	0.30
WW-T-825	5.8-7.2	0.4-1.5	0.15 Min	-	-	0.10	0.005	0.05	0.005	0.30
MIL-R-6944	5.8-7.2	0.4-1.5	0.15 Min	-	-	0.05	0.005	0.05	0.005	0.30

Minimum Mechanical Properties								
Specification Number	Form	Temper	Size Thick (In.)	Size Area (In.^{2})	TS ksi	YS ksi	YS Ext	Elong %
QQ-M-31	Extrusions		Up thru 0.249		38	21	0.0101	8
	Bars		0.250-2.499		39	24	0.0114	9
	Rods							
	Shapes							
	Hollow Shapes		All		36	16	0.0089	7
QQ-M-40	Forgings	-F			38	22	0.0054	6
WW-T-825	Extrusions							
	Tubing		0.050-0.250		36	16	0.0089	7
MIL-R-6944	Welding				No Mechanical Requirements			
	Rod							
	Wire							

AZ63A

Chemical Composition - %										
Spec. No.	Al	Zn	Mn	R.E.	Zr	Si	Fe	Cu	Ni	Others
QQ-M-55	5.3-6.7	2.5-3.5	0.15	-	-	0.30	-	0.10	0.01	0.30
QQ-M-56	5.3-6.7	2.5-3.5	0.15	-	-	0.30	-	0.10	0.01	0.30
MIL-R-6944	5.3-6.7	2.5-3.5	0.15 Min	-	-	0.30	-	0.10	0.01	0.30

Minimum Mechanical Properties								
Specification Number	Form	Temper	Size Thick, (In.)	Size Area (In.^{2})	TS ksi	YS ksi	YS Ext	Elong %
QQ-M-55	Castings PM & SFM	-F [1]			26	11	0.0037	4
		-T4[1]			34	11	0.0037	7
		-T5[1]			26	12	0.0038	2
		-T6[1]			34	16	0.0045	3
		-T4[2]			17	9	0.0034	-
		-T6[2]			17	12	0.0039	-
QQ-M-56	Castings Sand	-F [1]			26	11	0.0037	4
		-T4[1]			34	11	0.0037	7
		-T5[1]			26	12	0.0038	2
		-T6[1]			34	16	0.0045	3
		-T4[2]			17	9	0.0034	-
		-T6[2]			17	12	0.0038	-
MIL-T-6944	Welding				No Mechanical Property Requirements			
	Rod							
	Wire							

NOTE: 1. Separately Cast Specimens.
2. Specimens Cut from Castings.

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AZ80A

Chemical Composition - %										
Spec No.	Al	Zn	Mn	Ca	Zr	Si	Fe	Cu	Ni	Others
QQ-M-31	7.8-9.2	0.2-0.8	0.12	-	-	0.30	0.005	0.05	0.005	0.30
QQ-M-40	7.8-9.2	0.2-0.8	0.12 Min	-	-	0.10	0.005	0.05	0.005	0.30

Minimum Mechanical Properties								
Specification Number	Form	Temper	Size Thick. (In.)	Size Area(In.^{2})	TS ksi	YS ksi	YS Ext	Elong %
QQ-M-31	Extrusions Bars Rods Shapes	-F	Up thru 0.249		43	28	0.0126	9
			0.250-1.499		43	28	0.0126	8
			1.500-2.499		43	28	0.0126	6
		-T5	Up thru 0.249		47	30	0.0132	4
			0.250-1.499		48	33	0.0142	4
			1.500-2.499		48	33	0.0142	4
QQ-M-40	Forgings	-F			42	26	0.0060	5
		-T5			42	28	0.0063	2

AZ81A

Chemical Composition - %										
Spec No.	Al	Zn	Mn	Ca	Zr	Si	Fe	Cu	Ni	Others
QQ-M-55	7.0-8.1	0.4-1.0	0.13 Min	-	-	0.30	-	0.10	0.01	0.30
QQ-M-56	7.0-8.1	0.4-1.0	0.13 Min	-	-	0.30	-	0.10	0.01	0.30

Minimum Mechanical Properties								
Specification Number	Form	Temper	Size Thick.(In.)	Size Area (In.^{2})	TS ksi	YS ksi	YS Ext	Elong %
QQ-M-55	Castings PM and SPM	-T4[1]			34	11	0.0037	7
		-T4[2]			17	9	0.0034	-
QQ-M-56	Castings Sand	-T4[1]			34	11	0.0037	7
		-T4[2]			17	9	0.0034	-

NOTES: 1. Separately Cast Specimens
2. Specimens Cut from Castings

AZ91A

Chemical Composition - %										
Spec No.	Al	Zn	Mn	Ca	Zr	Si	Fe	Cu	Ni	Others
QQ-M-38	8.3-9.7	0.35-1.0	0.13	-	-	0.50	-	0.10	0.03	0.30

Typical Mechanical Properties								
Specification Number	Form	Temper	Size Thick.(In.)	Size Area (In.^{2})	TS ksi	YS ksi	YS Ext	Elong %
QQ-M-38	Casting Die				29-34	22-24	-	2-5

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AZ91C

Chemical Composition - %										
Spec. No.	Al	Zn	Mn	Ca	Zr	Si	Fe	Cu	Ni	Others
QQ-M-55	8.1-9.3	0.4-1.0	0.13	-	-	0.30	-	0.10	0.01	0.30
QQ-M-56	8.1-9.3	0.4-1.0	0.13	-	-	0.30	-	0.10	0.01	0.30
MIL-M-46062	8.1-9.3	0.4-1.0	0.13	-	-	0.30	-	0.10	0.01	0.30

Minimum Mechanical Properties								
Specification Number	Form	Temper	Size Thick (In.)	Size Area ($In.^2$)	TS ksi	YS ksi	YS Ext	Elong %
QQ-M-55	Castings PM & SPM	-F [1]			23	11	0.0037	-
		-T4 [1]			34	11	0.0037	7
		-T5 [1]			23	12	0.0038	2
		-T6 [1]			34	16	0.0045	3
		-T4 [2]			17	9	0.0034	-
		-T6 [2]			17	12	0.0039	-
QQ-M-56	Castings Sand	-F [1]			23	11	0.0037	-
		-T4 [1]			34	11	0.0037	7
		-T5 [1]			23	12	0.0038	2
		-T6 [1]			34	16	0.0045	3
		-T4 [2]			17	9	0.0034	-
		-T6 [2]			17	12	0.0038	-
MIL-R-46062	Castings	-T6			17	12	0.0039	0.75

NOTE 1. Separately Cast Specimens
2. Specimens Cut from Castings
3. Extremely Sensitive to Variations from Good Foundry Practice and Heat Treatment Procedure.

AZ92A

Chemical Composition - %										
Spec. No.	Al	Zn	Mn	Ca	Zr	Si	Fe	Cu	Ni	Others
QQ-M-55	8.3 - 9.7	1.6-2.4	0.13	-	-	0.30	-	0.10	0.01	0.30
QQ-M-56	8.3 - 9.7	1.6-2.4	0.10	-	-	0.30	-	0.10	0.01	0.30
MIL-M-46062	8.3 - 9.7	1.6-2.4	0.10	-	-	0.30	-	0.10	0.01	0.30

Minimum Mechanical Properties								
Specification Number	Form	Temper	Size Thick (In.)	Size Area ($In.^2$)	TS ksi	YS ksi	YS Ext	Elong %
QQ-M-55	Castings PM & SPM	-F [1]			23	11	0.0037	-
		-T4 [1]			34	11	0.0037	6
		-T5 [1]			23	12	0.0038	-
		-T6 [1]			34	18	0.0048	-
		-T4 [2]			17	9	0.0034	-
		-T6 [2]			17	13.5	0.0041	-
QQ-M-56	Castings Sand	-F [1]			23	11	0.0037	-
		-T4 [1]			34	11	0.0037	6
		-T5 [1]			23	12	0.0038	-
		-T6 [1]			34	18	0.0045	-
		-T4 [2]			17	9	0.0034	-
MIL-R-46062		-T6 [2]			17	13.5	0.0041	0.50

NOTE 1. Separately Cast Specimens
2. Specimens Cut from Castings
3. Extremely Sensitive to Variations from Good Foundry Practice and Heat Treatment Procedure.

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EZ33A

Chemical Composition - %										
Spec No.	Al	Zn	Mn	R. E.	Zr	Si	Fe	Cu	Ni	Others
QQ-M-55	-	2.0-3.0	-	2.5-4.0	0.50	-	-	0.10	0.01	0.30
QQ-M-56	-	2.0-3.1	-	2.5-4.0	0.5-1.0	-	-	0.10	0.01	0.30

Minimum Mechanical Properties								
Specification Number	Form	Temper	Size Thick.(In.)	Size Area (In.^{2})	TS ksi	YS ksi	YS Ext	Elong %
QQ-M-55	Castings PM & SPM	-T5[1]			20	14	0.0042	2
		-T5/500°F[1]			13	8	-	-
		-T5[2]			13	11	0.0037	-
		-T5/500°F[2]			10	6	-	-
QQ-M-56	Castings Sand	-T5[1]			20	14	0.0042	2
		-T5/500°F[1]			13	8	-	-
		-T5[2]			13	11	0.0037	-
		-T5/500°F[2]			10	6	-	-

NOTE: 1. Separately Cast Specimens
2. Specimens Cut from Castings

HK31A

Chemical Composition - %											
Spec. No.	Al	Zn	Mn	R.E.	Zr	Si	Fe	Cu	Ni	Th	Others
QQ-M-55	-	0.30	-	-	0.5-1.0	-	-	0.10	0.01	2.5-4.0	0.30
QQ-M-56	-	0.30	-	-	0.5-1.0	-	-	0.10	0.01	2.5-4.0	0.30
MIL-M-26075	-	-	-	-	0.45-1.0	-	-	-	-	2.5-4.0	0.30
MIL-R-6944 MIL-M-46062	-	0.30	-	-	0.5-1.0	-	-	-	-	2.5-4.0	0.30

Minimum Mech Properties								
Specification Number	Form	Temper	Size Thick (In.)	Size Area (In.^{2})	TS ksi	YS ksi	YS Ext	Elong %
QQ-M-55	Castings PM & SPM	-T6[1]			27	13	.0040	4
		-T6[1] @ 500°F			21	13	-	-
		-T6[2]			19	10.5	.0036	-
		-T6 @ 500°F			14	10.5	-	-
QQ-M-56	Castings Sand	-T6[1]			27	13	.0040	4
		-T6[1] @ 500°F			21	13	-	2
		-T6[2]			19	10.5	.0036	-
		-T6[2] @ 500°F			14	10.5	-	-
MIL-M-26075	Sheet Plate	-H24	.016-.125		34	26	-	4
			.126-.250		34	24	-	4
			.251-1.000		34	25	-	4
			1.001-3.000		33	25	-	4
		-O	.016-.250		30	18	-	12
			.251-.500		30	16	-	12
			.501-1.000		30	15	-	12
			1.001-3.000		29	14	-	12
		-H24/600°F	0.016-0.125		10	-	-	20
MIL-R-6944	Welding Rod Wire	No Mechanical Requirements						
MIL-M-46062	Castings	-T6			19	10.5	-	1.0

NOTE: 1. Separately Cast Specimens
2. Specimens Cut from Casting

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HM21

Chemical Composition - %										
Spec. No.	Al	Zn	Mn	Si	Cu	Ni	R. E.	Zr	Th	Others
QQ-M-40	-	-	0.45-1.1	-	-	-	-	-	1.5-2.5	0.10
MIL-M-8917 (ASG)	-	-	0.45-1.1	-	-	-	-	-	1.5-2.5	0.10 0.30

Minimum Mechanical Properties								
Specification Number	Form	Temper	Size Thick. (In.)	Size Area ($In.^2$)	TS ksi	YS ksi	YS Ext	Elong %
MIL-M-8917 (ASG)	Sheet & Plate at 600°F	-T8	0.016-0.250		33	18		6
			0.251-0.500		32	21		6
			0.500-2.0		30	21		6
	Sheet & Plate at 600°F	-T81	0.125-0.312		11			8
			0.125-0.312		34	25		4
					15			
QQ-M-40	Forgings	-T5	up to 4.0		33	25	.0059	3.0

HM31A-T5

Chemical Composition - %										
Spec. No.	Al	Zn	Mn	Si	Cu	Ni	R. E.	Zr	Th	Others
MIL-M-8916 (ASG)	-	-	1.2	-	-	-	-	-	2.5-3.5	0.10 (Max.) (0.30 Total impurities)

Minimum Mechanical Properties								
Specification Number	Form	Temper	Size Thick. (In.)	Size Area ($In.^2$)	TS ksi	YS ksi	YS Ext	Elong %
MIL-M-8916 (ASG)	Extrusions Bar Rods Shapes (Solid At 600°F	-T5	under 4		37	26		4
					12	15		10

HZ32A

Chemical Composition - %											
Spec. No.	Al	Zn	Mn	R. E.	Zr	Si	Fe	Cu	Ni	Th	Others
QQ-M-55	-	1.7-2.5	-	0.10	0.5-1.0	-	-	0.10	0.01	2.5-4.0	0.30
QQ-M-56	-	1.7-2.5	-	0.10	0.5-1.0	-	-	0.10	0.01	2.5-4.0	0.30

Minimum Mechanical Properties								
Specification Number	Form	Temper	Size Thick (In.)	Size Area ($In.^2$)	TS ksi	YS ksi	YS Ext	Elong %
QQ-M-55	Castings PM & SPM	-T5[1]			27	13	0.0040	4
		-T5/500°F[1]			13	8	-	-
		-T5[2]			19	10.5	0.0036	-
		-T5/500°F[2]			10	6	-	-
QQ-M-56	Castings Sand	-T5[1]			27	13	0.0040	4
		-T5/500°F[1]			13	8	-	-
		-T5[2]			19	10.5	0.0036	-
		-T5/500°F[2]			10	6	-	-

NOTE: 1. Separately Cast Specimens
2. Specimens Cut from Castings

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K1A

Chemical Composition - %											
Spec No.	Al	Zn	Mn	Ca	Zr	Si	Fe	Cu	Ni	Sn	Others
MIL-M-45207	-	-	-	-	0.40-1.0	-	-	-	-	-	0.30

Minimum Mechanical Properties								
Specification Number	Form	Temper	Size Thick. (In.)	Size Area ($In.^2$)	TS ksi	YS ksi	YS Ext	Elong %
MIL-M-45207	Castings Sand	-F			24	6	0.0029	14

LS141A

Chemical Composition - %										
Spec. No.	Al	Zn	Mn	Si	Fe	Cu	Ni	Na	Li	Other
MIL-M-46143	0.05	-	0.15	0.5-0.6	0.005	0.05	0.005	0.005	12.0-15.0	0.20

Minimum Mechanical Properties								
Specification Number	Form	Temper	Size Thick (In.)	Size Area ($In.^2$)	TS ksi	KS ksi	KS Ext	Elong %
MIL-M-46143	Castings	-T7			18.5	13.5		18

LS145A

Chemical Composition - %											
Spec. No.	Al	Zn	Mn	Si	Fe	Cu	Ni	Na	Ag	Li	Other
MIL-M-46143	0.04	4.5-5.0	0.15	1.5-2.0	0.005	0.05	0.005	0.005	2.0-3.0	12.0-15.0	0.20

Minimum Mechanical Properties								
Specification Number	Form	Temper	Size Thick (In.)	Size Area ($In.^2$)	TS ksi	YS ksi	YS Ext	Elong %
MIL-M-46143	Castings	-T7			28	22.5		4

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M1A

Chemical Composition - %										
Spec No.	Al	Zn	Mn	Ca	Zr	Si	Fe	Cu	Ni	Others
QQ-M-31	-	-	1.2	0.30	-	0.30	-	0.05	0.01	0.30
QQ-M-40	-	-	1.2 Min	0.30	-	0.10	-	0.05	0.01	0.30
WW-T-825 MIL-R-6944	-	-	1.2	0.10-0.14	-	0.05	-	0.05	0.01	0.30

Minimum Mechanical Properties								
Specification Number	Form	Temper	Size Thick.(In.)	Size Area (In.^{2})	TS ksi	YS ksi	YS Ext	Elong %
QQ-M-31	Extrusions Bars		Up thru 0.249		30	-	-	2
	Rods		0.250-1.499		32	-	-	3
	Shapes		1.500-2.499		32	-	-	2
	Hollow Shapes		All		28	-	-	2
QQ-M-40	Forgings	-F			30	18	0.0048	3
WW-T-825	Extrusions Tubing		0.050-0.250		28	-	-	2
MIL-R-6944	Welding Rod Wire				No Mechanical Requirements			

QE22A

Chemical Composition - %						
Spec. No.	Cu	Ni	Ag	R.E.	Zk	Others (Max.)
QQ-M-55	0.10	0.01	2.0-3.0	1.80-2.5	0.40-1.0	0.30
QQ-M-56	0.10	0.01	2.0-3.0	1.80-2.5	0.40-1.0	0.30
MIL-M-46062	0.10	0.01	2.0-3.0	1.80-2.5	0.40-1.0	0.30

Minimum Mechanical Properties								
Specification Number	Form	Temper	Size Thick. (In.	Size Area (In.^{2})	TS ksi	YS ksi	YS Ext	Elong %
QQ-M-56	Castings Sand	-T6			35	25	0.0058	2
		-T6/500°F			20	16	0.0045	-
		-T6			32	23	0.0055	-
		-T6/500°F			23	20	-	-
QQ-M-55	Castings PM & SPM	-T6[1]			35	25	0.0062	2
		T6/600°F[1]			13	10	0.0035	-
		-T6[2]			28	20	0.0051	-
		-T6/600°F[2]			10	8	-	-
		-T6[1]			35	25	0.0058	2
		-T6/600°F[1]			13	10	-	-
		-T6[2]			28	20	0.0051	-
		-T6/600°F[2]			10	8	-	-
MIL-M-46062		-T6			28	20	-	1

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TA54A

Chemical Composition - %											
Spec No.	Al	Zn	Mn	Ca	Zr	Si	Fe	Cu	Ni	Sn	Others
QQ-M-40	3.0-4.0	0.30	0.20	-	-	0.30	-	0.05	0.030	4.0-6.0	0.30

Minimum Mechanical Properties								
Specification Number	Form	Temper	Size Thick. (In.)	Size Area (In.^{2})	TS ksi	YS ksi	YS Ext	Elong %
QQ-M-40	Forgings	-F			36	22	0.0054	7

ZE10A

Chemical Composition - %										
Spec No.	Al	Zn	Mn	R. E.	Zr	Si	Fe	Cu	Ni	Others
MIL-M-46037	-	1.0-1.5	-	0.12-0.22	-	-	-	-	-	0.30

Minimum Mechanical Properties								
Specification Number	Form	Temper	Size Thick (In.)	Size Area (In.^{2})	TS ksi	YS ksi	YS Ext	Elong %
MIL-M-46037	Sheet	-O	0.016-0.060		30	18		15
	Plate		0.061-0.250		30	15		15
			0.251-0.500		29	12		12
		-H24	0.016-0.125		36	25		6
			0.126-0.188		34	22		6
			0.189-0.250		31	20		6

ZH62A

Chemical Composition - %											
Spec. No.	Al	Zn	Mn	Ca	Zr	Si	Fe	Cu	Ni	Th	Others
QQ-M-56	-	5.2-6.2	-	-	0.5-1.0	-	-	0.10	0.01	1.4-2.2	0.30
MIL-M-46062	-	5.2-6.2	-	-	0.5-1.0	-	-	0.10	0.01	1.4-2.2	0.30

Minimum Mechanical Properties								
Specification Number	Form	Temper	Size Thick (In.)	Size Area (In.^{2})	TS ksi	YS ksi	YS Ext	Elong %
QQ-M-56	Castings	-T5[1]	-	-	35	22	0.0054	5
	Sand	-T5[2]	-	-	28.5	17.5	0.0047	
MIL-M-46062	Castings	-T5	-	-	28.5	17.5		1.25

NOTE: 1. Separately Cast Specimens
2. Specimens Cut from Castings

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ZK21A

Chemical Composition - %										
Spec No.	Al	Zn	Mn	Ca	Zr	Si	Fe	Cu	Ni	Others
MIL-M-46039	-	2.0-2.6	-	-	0.45-0.80	-	-	-	-	0.30

Minimum Mechanical Properties								
Specification			Size		TS	YS		Elong
Number	Form	Temper	Thick. (In.)	Area (In.^{2})	ksi	ksi	Ext	%
MIL-M-46039	Extrusions Bars Rods Shapes			Less than 5	38	28	0.0126	4

ZK51A

Chemical Composition - %										
Spec. No.	Al	Zn	Mn	Ca	Zr	Si	Fe	Cu	Ni	Others
QQ-M-56	-	3.6-5.5	-	-	0.5-1.0	-	-	0.10	0.01	0.30
MIL-M-46062	-	3.6-5.5	-	-	0.5-1.0	-	-	0.10	0.01	0.30

Minimum Mechanical Properties								
Specification			Size		TS	YS		Elong
Number	Form	Temper	Thick. (In.)	Area (In.^{2})	ksi	ksi	Ext	%
QQ-M-56	Castings	-T5[1]			34	20	0.005	5
	Sand	-T5[2]			24	14	0.0042	-
MIL-M-46062	Castings	-T5			24	14	-	1.25

NOTE: 1. Separately Cast Specimens
2. Specimen Cut from Casting

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ZK60A

Chemical Composition - %										
Spec No.	Al	Zn	Mn	Ca	Zr	Si	Fe	Cu	Ni	Others
QQ-M-31	-	4.8-6.2	-	-	0.45	-	-	-	-	0.30
QQ-M-40	-	4.8-6.2	-	-	0.45	-	-	-	-	0.30
WW-T-825	-	4.8-6.2	-	-	0.45	-	-	-	-	0.30

Minimum Mechanical Properties									
Specification Number	Form	Temper	Size Thick.(In.)	Size Area (In.^{2})	TS ksi	YS ksi	YS Ext	Elong %
QQ-M-31	Extrusion Bars Rods Shapes	-F		Up thru 5	43	31	0.0068	5
		-T5		Up thru 5	45	36	0.0075	4
	Hollow Shapes	-F	All		40	28	0.0068	5
		-T5	All		46	38	0.0078	4
QQ-M-40	Forgings	-T5	Up thru 3		42	26	0.0060	7
WW-T-825	Extrusions Tubing		0.050-0.250		40	28	0.0063	5

ZK60B

Chemical Composition - %										
Spec No.	Al	Zn	Mn	Ca	Zr	Si	Fe	Cu	Ni	Others
MIL-M-26696 (USAF)	-	4.8-6.8	-	-	0.45	-	-	-	-	0.30

Minimum Mechanical Properties									
Specification Number	Form	Temper	Size Thick.(In.)	Size Area (In.2)	TS ksi	YS ksi	YS Ext	Elong %	CYS ksi
MIL-M-26696	Extruded Bars Rods Special Shaped Sections	-T5	Up thru 2.999	Under 20.000	45	35		4	35

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ZK61A

Chemical Composition - %										
Spec. No.	Al	Zn	Mn	Si	Cu	Ni	R.E.	Ze	Th	Others
QQ-M-56	-	5.5-6.5	-	-	0.10	0.01	-	0.60-1.0	-	0.30
MIL-M-46062	-	5.5-6.5	-	-	0.10	0.01	-	0.60-1.0	-	0.30

Minimum Mechanical Properties								
Specification			Size		TS	YS		Elong
Number	Form	Temper	Thick. (In.)	Area ($In.^2$)	ksi	ksi	Ext	%
QQ-M-56	Castings	-T5			39	26	0.0060	5
	Sand	-T6			39	26	0.0060	5
		-T5			30	21	0.0053	-
		-T6			30	21	0.0053	-
MIL-M-46062	Castings	-T6			30	21	0.0053	1.25

Custodians:
Army - MR
Navy - AS
Air Force - 84
DSA - IS

Preparing activity:
Army - MR
(Project 95GP-0025)

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